

Engineering classification of rocks and rock masses

- Classification systems in rock engineering
- Classification of intact rocks
- Classification of rockmass
 - Terzaghi's rock load
 - RQD
 - Rock Structure Rating
 - Bieniawski's RMR
 - Barton's Q-System
 - Laubscher's-MRMR
 - Hoek's-GSI
 - Palmstrom's RMi
 - CMRI-ISM Rock mass classification and
- Recent developments; correlations between different classification systems; Applications of Rockmass Classification in rock engineering (being discussed in respective topics).

- Rock mass classification systems are used for various engineering design and stability analysis.
- These are based on empirical relations between rock mass parameters and engineering applications, such as tunnels, slopes, foundations, and excavatability.
- rock mass classification, methods are extensively used for feasibility and pre-design studies, and often also for the final design
- We cannot change the ground conditions, but we can understand the conditions, and according to that we can select excavation methodology, that offer optimum performance in the given conditions and we can improve the ability to cope with adverse geological conditions.

The objectives of rock mass classifications are (after Bieniawski 1989):

1. Identify the most significant parameters influencing the behaviour of a rock mass.
2. Divide a particular rock mass formulation into groups of similar behaviour – rock mass classes of varying quality.
3. Provide a basis of understanding the characteristics of each rock mass class
4. Relate the experience of rock conditions at one site to the conditions and experience encountered at others
5. Derive quantitative data and guidelines for engineering design
6. Provide common basis for communication between engineers and geologists

The main benefits of rock mass classifications:

1. Improving the quality of site investigations by calling for the minimum input data as classification parameters.
2. Providing quantitative information for design purposes.
3. Enabling better engineering judgement and more effective communication on a project.
4. provide a basis for understanding the characteristics of each rock mass

Engineering Classification of Rock Mass are usually applied for the evaluation of

- **stability of excavations,**
- **rock support requirements, and**
- **determination of rock mass deformation and strength parameters.**

Rock mass classification puts a number to the quality of the rock mass. It divides the rock mass into groups of similar behavior

Disadvantages of rock mass classification

According to Bieniawski (1993), the major pitfalls of rock mass classification systems arise when:

- 1) using rock mass classifications as the ultimate empirical 'cook book', i.e. ignoring analytical and observational design methods;
- 2) using one rock mass classification system only, i.e. without cross-checking the results with at least one other system;
- 3) using rock mass classifications without enough input data;
- 4) using rock mass classifications without full realization of their conservative nature and their limits arising from the database on which they were developed.

Some people are of the opinion that

- a) natural materials cannot be described by a single number,
- b) other important (often dominating) factors are not considered,
- c) results of rock mass classification are prone to misuse (e.g., claims for changed conditions) (Bieniawski, 1989).

Classification of Intact rocks

CLASSIFICATION OF ROCK MATERIAL BASED ON UNCONFINED COMPRESSIVE STRENGTH

Manual Indent Tests

Grade	Description	Field identification	Approx. range of UCS (MPa)
S1	Very soft clay	Easily penetrated several inches by fist	< 0.025
S2	Soft clay	Easily penetrated several inches by thumb	0.025-0.05
S3	Firm clay	Can be penetrated several inches by thumb with moderate strength	0.05-0.10
S4	Stiff clay	Readily indented by thumb but penetrated only with thumb with great effort	0.10-0.25
S5	Very stiff clay	Readily indented by thumbnail with	0.25-0.50
S6	Hard clay	Indented by with difficulty by thumbnail	>0.50
R0	Extremely weak rock	Indented by thumbnail	0.25-1.0
R1	Very weak rock	Crumbles under firm blow with point of geological hammer, can be peeled by pocket knife	1.0-5.0
R2	Weak rock	Can be peeled by a pocket knife with difficulty, shallow indentation made by firm blow with point of geological hammer	5.0-25
R3	Medium strong rock	Cannot be scraped or peeled by pocket knife, specimen can be fractured with single blow of geological hammer	25-50
R4	Strong rock	Specimen requires more than one blow of geological hammer to fracture it	50-100
R5	Very strong rock	Specimen requires many blows of geological hammer to fracture it	100-250
R6	Extremely strong rock	Specime can only be chipped with geological hammer	>250

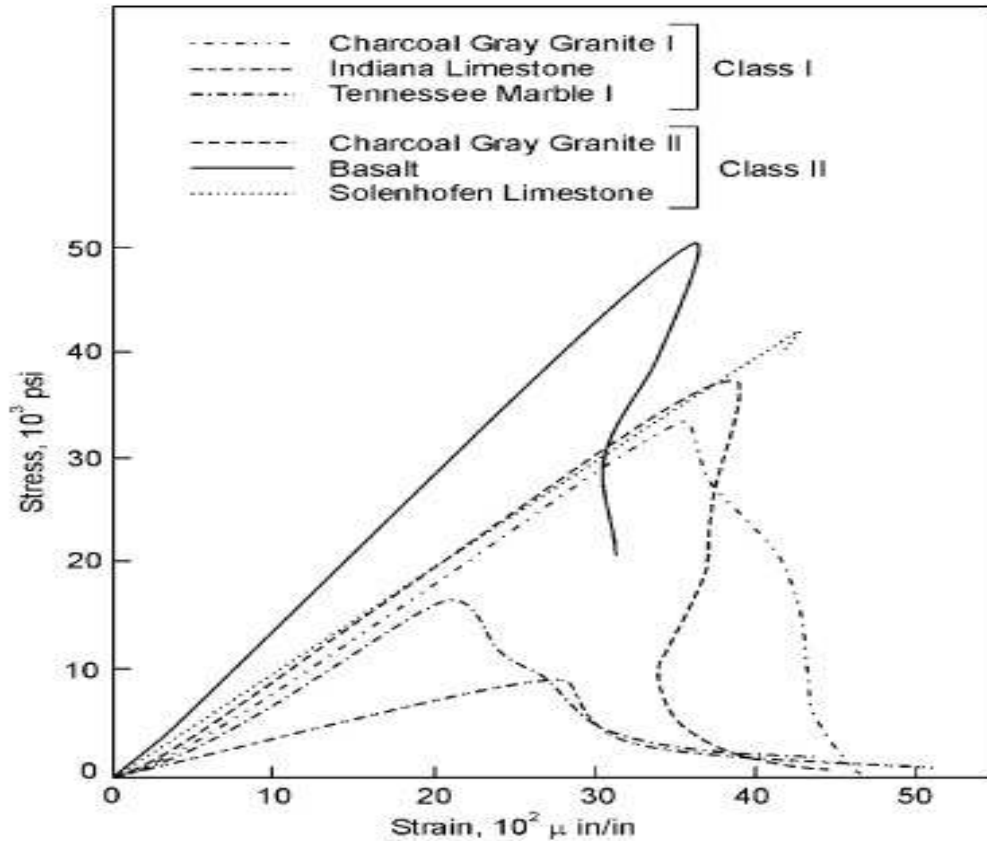
TABLE 3.1 Classification of Rock Material Based on Unconfined Compressive Strength

Term for uniaxial compressive strength	Symbol	Strength (MPa)	Ranges for common rock materials				
			<i>Granite, basalt, gneiss, quartzite, marble</i>	<i>Schist, sandstone</i>	<i>Limestone, siltstone</i>	<i>Slate</i>	<i>Concrete</i>
Extremely weak*	EW	<1		**	**		
Very weak	VW	1–5		**	**	**	**
Weak	W	5–25		**	**	**	**
Medium strong	MS	25–50	**		**	**	
Strong	S	50–100	**				
Very strong	VS	100–250	**				
Extremely Strong	ES	>250	**				

*Some extremely weak rocks behave as soils and should be described as soils.

**Indicates the range of strength of rock material.

Source: ISO 14689-1, 2003.



Stress-strain curve for six representative rocks in uniaxial compression (Wawersik, 1968)

- Typical S-shape stress-strain curves may be obtained for rocks with micro-fractures.
- Post-peak curve for Class II rocks shows reduction of strain after failure. The lateral strain increases rapidly after peak stress in Class II rocks. Brittle rocks, therefore, may be kept in the Class II category.
- A deep tunnel within dry, massive, hard Class II and laminated rocks may fail because of rock bursts due to uncontrolled fracturing where tangential stress exceeds the strength of the rock material. Hence, it is necessary to test rock material in a Servo-controlled closed loop testing machine to get the post-peak curve.

TABLE 3.2 Rock Material Stability in Water

Term	Description (after 24 h in water)	Grade
Stable	No changes	1
Fairly stable	A few fissures are formed or specimen surface crumbles slightly	2
	Many fissures are formed and broken into small lumps or specimen surface crumbles	3
Unstable	Specimen disintegrates or nearly the whole specimen surface crumbles	4
	The whole specimen becomes muddy or disintegrates into sand	5

Source: ISO 14689-1, 2003.

TABLE 3.3 Slake Durability Classification

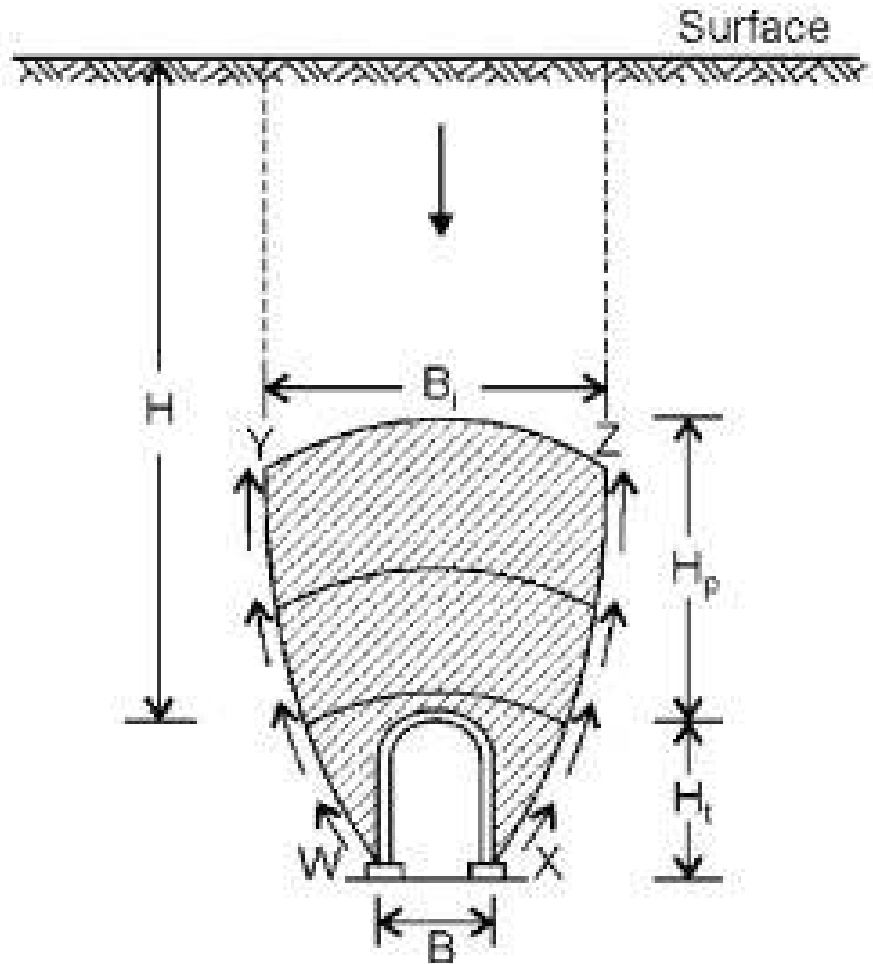
Group name	% retained after one 10-minute cycle (dry weight basis)	% retained after two 10-minute cycles (dry weight basis)
Very high durability	>99	>98
High durability	98–99	95–98
Medium high durability	95–98	85–95
Medium durability	85–95	60–85
Low durability	60–85	30–60
Very low durability	<60	<30

Source: Gamble, 1971, 2003.

Terzaghi's Rock Load Classification

The geotechnical engineer should apply theory and experimentation but temper them by putting them into the context of the uncertainty of nature. Judgement enters through engineering geology.

Karl Terzaghi



- probably the first successful attempt at classifying rock masses for engineering purposes
- The rock load factors (H_p) were estimated by Terzaghi from a 5.5-m-wide steel-arch supported railroad tunnel in the Alps during the late 1920s.
- **wooden blocks of known strengths** were used for blocking the steel arches to the surrounding rock masses.
- Rock loads were **estimated from the known strengths of the failed wooden blocks**
- Subsequently, he conducted **“trap-door” experiments on different sands** and found that the height of loosened arch above the roof increased directly with the opening width in the sand

Definitions of Rock Classes of Terzaghi's Rock Load Theory

Rock class	Type of rock	Definition
I	Hard and intact	The rock is unweathered. It contains neither joints nor hair cracks. If fractured, it breaks across intact rock. After excavation the rock may have some popping and spalling failures from the roof. At high stresses spontaneous and violent spalling of rock slabs may occur from the sides or the roof. The unconfined compressive strength is equal to or more than 100 MPa.
II	Hard stratified and schistose	The rock is hard and layered. The layers are usually widely separated. The rock may or may not have planes of weakness. In this type of rock, spalling is quite common.
III	Massive moderately jointed	A jointed rock. The joints are widely spaced. The joints may or may not be cemented. It may also contain hair cracks, but the huge blocks between the joints are intimately interlocked so that vertical walls do not require lateral support. Spalling may occur.
IV	Moderately blocky and seamy	Joints are less spaced. Blocks are about 1 m in size. The rock may or may not be hard. The joints may or may not be healed, but the interlocking is so intimate that no side pressure is exerted or expected.
V	Very blocky and seamy	Closely spaced joints. Block size is less than 1 m. It consists of almost chemically intact rock fragments that are entirely separated from each other and imperfectly interlocked. Some side pressure of low magnitude is expected. Vertical walls may require supports.

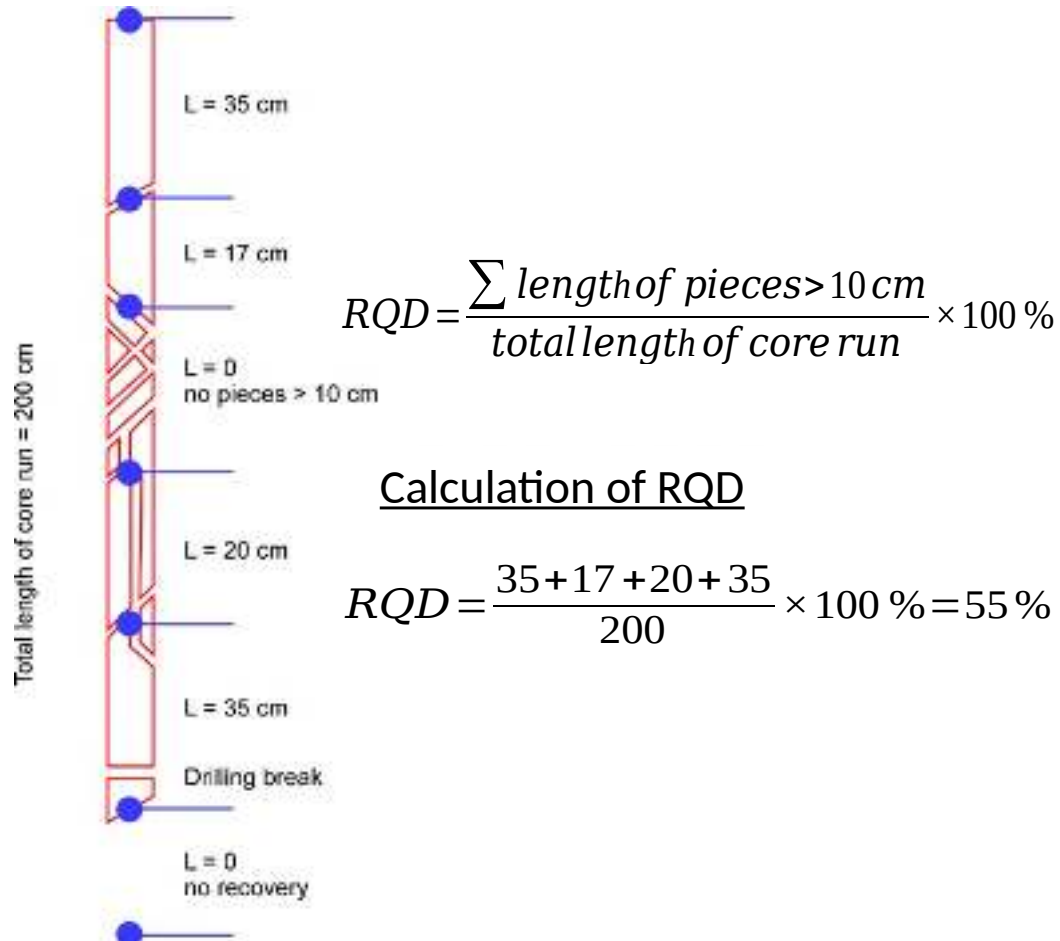
Rock class	Type of rock	Definition
VI	Completely crushed but chemically intact	Comprises chemically intact rock having the character of a crusher-run aggregate. There is no interlocking. Considerable side pressure is expected on tunnel supports. The block size could be a few centimeters to 30 cm.
VII	Squeezing rock — moderate depth	Squeezing is a mechanical process in which the rock advances into the tunnel opening without perceptible increase in volume. Moderate depth is a relative term and could be 150 to 1000 m.
VIII	Squeezing rock — great depth	The depth may be more than 150 m. The maximum recommended tunnel depth is 1000 m.
IX	Swelling rock	Swelling is associated with volume change and is due to chemical change of the rock usually in the presence of moisture or water. Some shales absorb moisture from air and swell. Rocks containing swelling minerals such as montmorillonite, illite, kaolinite, etc., can swell and exert heavy pressure on rock supports.

Rock Condition ^c	Rock load H_p (ft)	Remarks
1. Hard and intact	Zero	Light lining required only if spalling or popping occurs.
2. Hard stratified or schistose	0-0.5B	Light support, mainly for protection against spalls. Load may change erratically from point to point.
3. Massive, moderately jointed	0-0.25B	
4. Moderately blocky and seamy	0.25B-0.35 (B + H_l)	No side pressure.
5. Very blocky and seamy	(0.35-1.10) (B + H_l)	Little or no side pressure.
6. Completely crushed	1.10 (B + H_l)	Considerable side pressure. Softening effects of seepage toward bottom of tunnel require either continuous support for lower ends of ribs or circular ribs.
7. Squeezing rock, moderate depth	(1.10-2.10) (B + H_l)	Heavy side pressure, invert struts required. Circular ribs are recommended.
8. Squeezing rock, great depth.	(2.10-4.50) (B + H_l)	
9. Swelling rock	Up to 250 ft, irrespective of the value of (B + H_l)	Circular ribs are required in extreme cases, use yielding support.

$$P_v = \gamma H_p$$

Rock Quality Designation (RQD)

Rock in field is generally jointed. It was classified by core recovery in the past and latter in sixties by modified core recovery (RQD)



Rock Quality Designation (RQD) is a modified core recovery percentage in which all the pieces of core >10 cm long are counted as recovery and is expressed as percentage of length drilled

Artificial fractures can be identified by close fitting cores and unstained surfaces. All of the artificial fractures should be ignored while counting the core length for RQD. A slow rate of drilling will also give better RQD.

CORRELATION BETWEEN RQD AND ROCK MASS QUALITY

S. No.	RQD (%)	Rock Quality
1	<25	Very poor
2	25-50	Poor
3	50-75	Fair
4	75-90	Good
5	90-100	Excellent

Indirect Methods for Determination of RQD

$$RQD = (V_F/V_L)^2 \cdot 100$$

V_F = in situ compressional wave velocity

V_L = compressional wave velocity in intact rock core

RQD for rock masses (Palmstrom, 2005)

$$RQD = 110 - 2.5 J_v$$

J_v = total number of joints per cubic meter or the volumetric joint count.

$$J_v = \sum_{i=1}^J \left(\frac{1}{S_i} \right) + \frac{N_r}{5\sqrt{A}}$$

S_i = average joint spacing in meters for the i^{th} joint set
 J = total number of joint sets except the random joint set
 N_r = number of random joints in the actual location
 A = the area in m^2

TABLE 4.2 Classification of Volumetric Joint Count (J_v)

S. No.	Degree of jointing	J_v
2	Very low	< 1.0
3	Low	1–3
4	Moderately	3–10
5	High	10–30
6	Very high	30–60
7	Crushed	> 60

Source: Palmstrom, 2005.

APPLICATION OF RQD

Used to estimate the deformation modulus of the rock mass
(Zhang and Einstein 2004)

$$\frac{E_d}{E_r} = 10^{0.0186 \text{ RQD} - 1.91}$$

E_d and E_r are the deformation moduli of the rock mass and the intact rock, respectively

Rock Structure Rating (RSR)

Rock Structure Rating (RSR)

- Wickham et al (1972) described a quantitative method for describing the quality of a rock mass and for selecting appropriate support on the basis of their Rock Structure Rating (RSR) classification.
- Most of the case histories, used in the development of this system, were for relatively small tunnels supported by means of steel sets, although historically this system was the first to make reference to shotcrete support.
- In spite of this limitation, it is worth examining the RSR system in some detail since it demonstrates the logic involved in developing a quasi-quantitative rock mass classification system.

Rock Structure Rating (RSR)

- it introduced the concept of rating each of the components listed below to arrive at a numerical value of

$$RSR = A+B+C$$

- where A , B and C are parameters related to Geology, Geometry and Groundwater.

Rock Structure Rating (RSR)

i. Parameter A : Geology

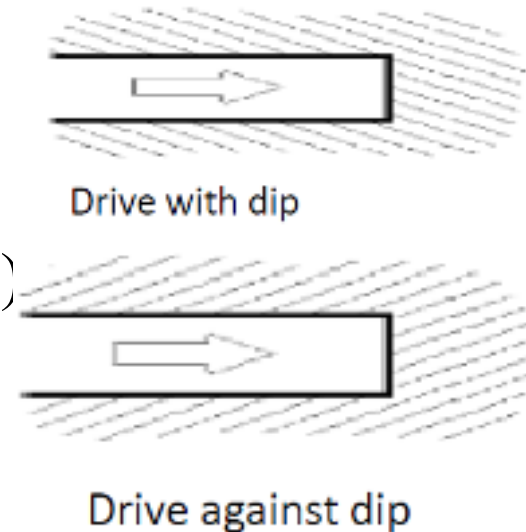
- General appraisal of geological structure on the basis of:
 - a) Rock type origin (igneous, metamorphic, sedimentary)
 - b) Rock hardness (hard, medium, soft, decomposed)
 - c) Geological structure (massive, slightly, faulted/folded, moderately)

Rock Structure Rating (RSR)

ii. Parameter B: Geometry

- Effect of discontinuity pattern with respect to the direction of the tunnel drive on the basis of:

- a) Joint spacing
- b) Joint orientation (strike and dip)
- c) Direction of tunnel drive



Rock Structure Rating (RSR)

iii. Parameter C : Effect of Groundwater

- Effect of ground water inflow and joint condition on the basis of:
 - a) Overall rock mass quality on the basis of combined A and B
 - b) Joint condition (good, fair, poor)
 - c) Amount of water inflow (in gallons per minute per 1000 feet of tunnel)
- There are three tables (Tables 1, 2 and 3) which can be used to evaluate the rating of each of these parameters to arrive at the RSR value (maximum RSR value = 100)

Rock Structure Rating (RSR)

Example:

In a hard metamorphic rock which is slightly folded or faulted has a rating of A = 22 (from Table 1). The rock mass is moderately jointed, with joints striking perpendicular to the tunnel axis which is being driven east-west, and dipping at between 20° and 50°.

	Basic Rock Type				Geological Structure			
	Hard	Medium	Soft	Decomposed				
Igneous	1	2	3	4		Slightly	Moderately	Intensively
Metamorphic	1	2	3	4		Folded or	Folded or	Folded or
Sedimentary	2	3	4	4	Massive	Faulted	Faulted	Faulted
Type 1					30	22	15	9
Type 2					27	20	13	8
Type 3					24	18	12	7
Type 4					19	15	10	6

Rock Structure Rating (RSR)

Example:

- The rock mass is moderately jointed, with joints striking perpendicular to the tunnel axis which is being driven east-west, are dipping at between 20° and 50°.
- The rating for B = 24 driving with dip.

Average joint spacing	Strike ⊥ to Axis					Strike to Axis		
	Direction of Drive					Direction of Drive		
	Both	With Dip		Against Dip		Either direction		
	Dip of Prominent Joints ^a					Dip of Prominent Joints		
	Flat	Dipping	Vertical	Dipping	Vertical	Flat	Dipping	Vertical
1. Very closely jointed, < 2 in	9	11	13	10	12	9	9	7
2. Closely jointed, 2-6 in	13	16	19	15	17	14	14	11
3. Moderately jointed, 6-12 in	23	24	28	19	22	23	23	19
4. Moderate to blocky, 1-2 ft	30	32	36	25	28	30	28	24
5. Blocky to massive, 2-4 ft	36	38	40	33	35	36	24	28
6. Massive, > 4 ft	40	43	45	37	40	40	38	34

Rock Structure Rating (RSR)

Example:

- The value of $A + B = 46$ and this means that, for joints of fair condition (slightly weathered and altered) and a moderate water inflow of between 200 and 1000 gallons per minute. The rating gives $C = 16$.
- Hence, the final value of the rock structure rating

$$\text{RSR} = A + B + C = 62$$

Anticipated water inflow gpm/1000 ft of tunnel	Sum of Parameters A + B					
	13 - 44			45 - 75		
	Joint Condition ^b					
	Good	Fair	Poor	Good	Fair	Poor
None	22	18	12	25	22	18
Slight, < 200 gpm	19	15	9	23	19	14
Moderate, 200-1000 gpm	15	22	7	21	16	12
Heavy, > 1000 gp	10	8	6	18	14	10

^a Dip: flat: 0-20°; dipping: 20-50°; and vertical: 50-90°

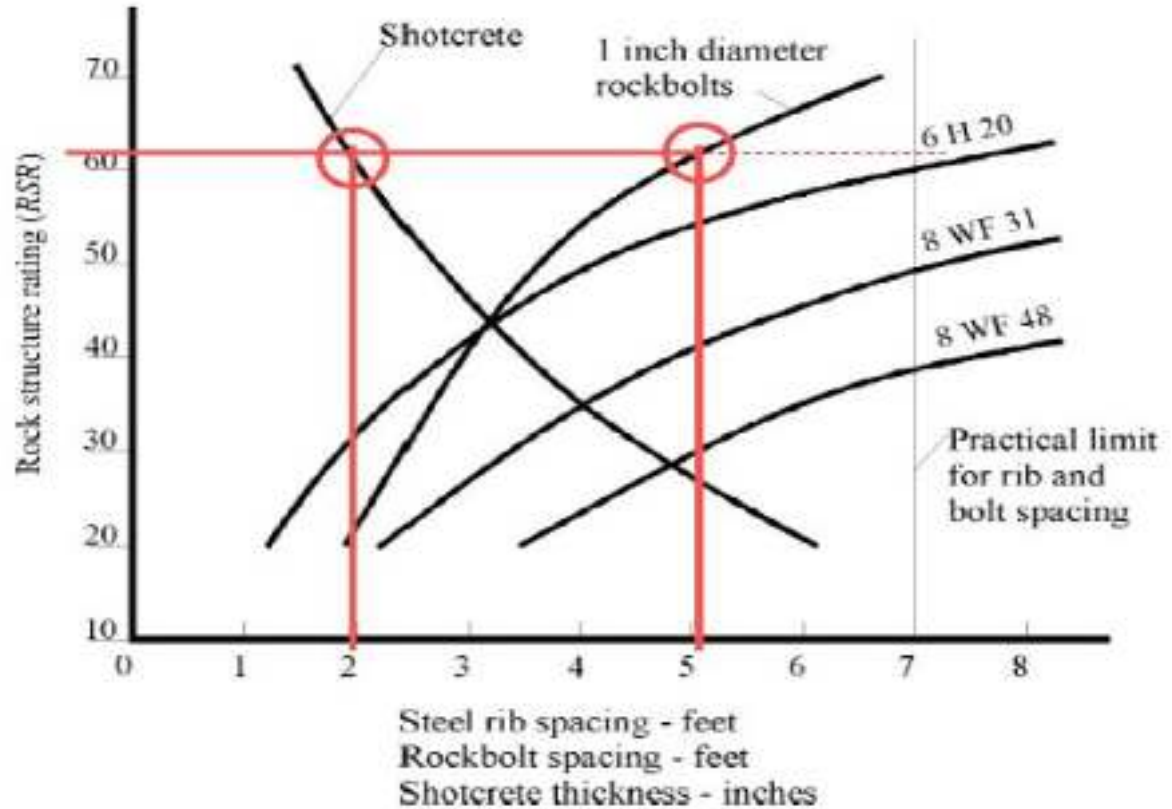
^b Joint condition: good = tight or cemented; fair = slightly weathered or altered; poor = severely weathered, altered or open

Rock Structure Rating (RSR)

Example:

- **Remedial Measure:**
- RSR support estimates for a 24 ft (7.3m) diameter circular tunnel.
- Note that rock bolts and shotcrete are generally used together.

8 WF 31 steel sets
= 8 inch deep wide flange I section
weighing 31 lb per foot



Bieniawski's Rock Mass Rating (RMR)

Rock Mass Rating (RMR)

- Bieniawski (1973) developed RMR method popularly known as geo-mechanics classification system at the South African Council of Scientific and Industrial Research (CSIR).
- Over the years, it has been modified several times by Bieniawski (1979, 1989, 1993).
- The most significant use of RMR is in various geo-engineering fields such as mining operations, tunnels, hydro-power projects and stability of rock slopes.

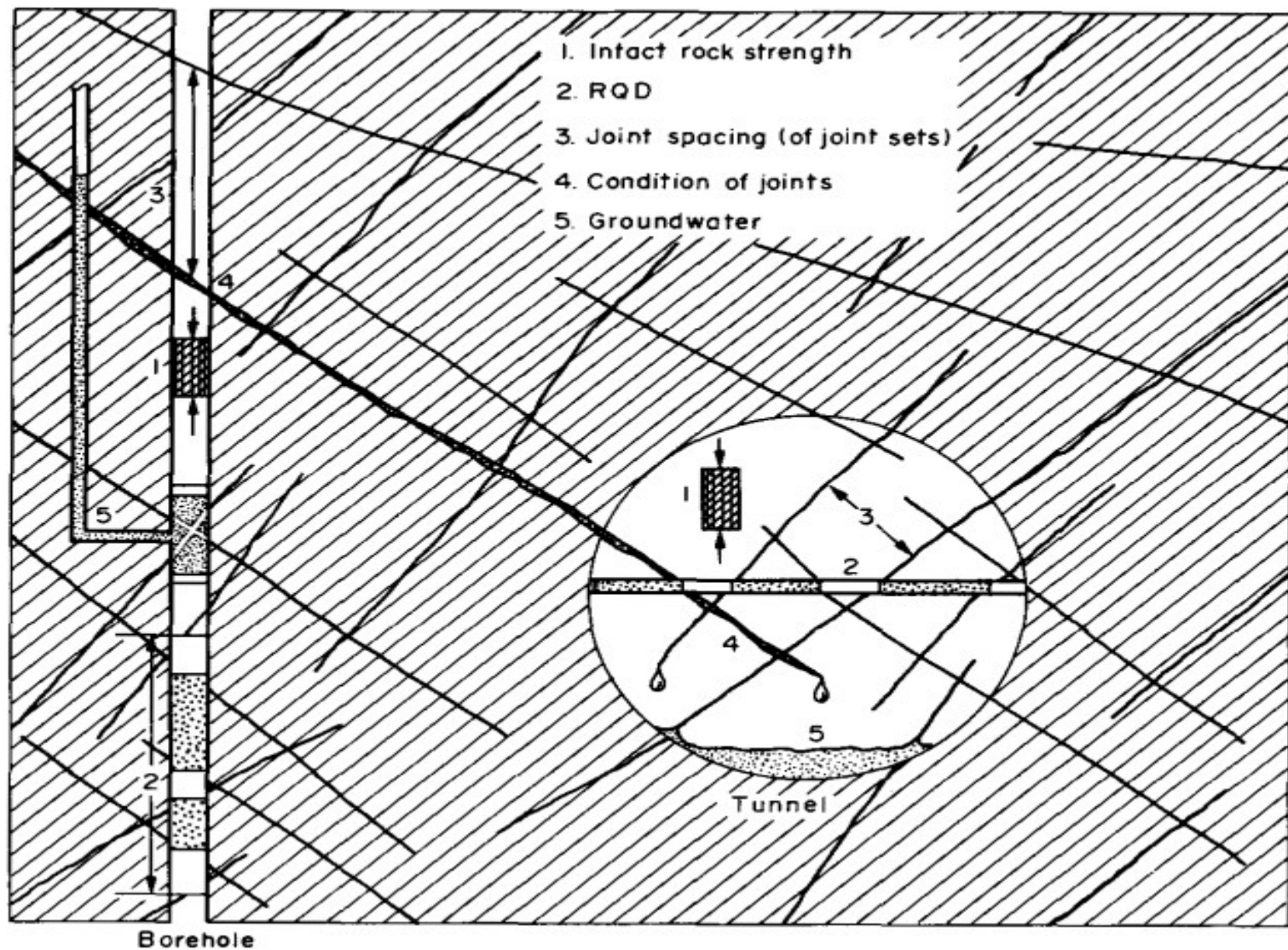
Rock Mass Rating (RMR)

- RMR is estimated by adding the rating of following five parameters.

$$\text{RMR} = R_{\text{RS}} + R_{\text{RQD}} + R_{\text{JS}} + R_{\text{JC}} + R_{\text{GC}}$$

where

- R_{RS} is rating of Rock Strength and it can be estimated by using UCS or PLSI values.
- R_{RQD} is rating of Rock Quality Designation and it can be estimated by field procedure or empirical formulae
- R_{JS} is rating of Joint Spacing among the discontinuities and it can be estimated in the field
- R_{JC} is rating of Joint Conditions and it can be estimated in the field
- R_{GC} is rating of Groundwater Condition and it can be observed in the field



Rock Mass Rating (RMR)

1. Rock Strength:

- The strength of intact rock is determined either by finding the Uniaxial Compressive Strength (UCS) or Point Load Strength Index (PLSI) in laboratory.

Rating based on intact rock strength							
Qualitative description	Strong				Weak		
	Extreme	Very	Strong	Medium	Weak	Very	Extreme
Compressive strength	> 250	250-100	100-50	50-25	25-5	5-1	< 1
Point load strength	8	8-4	4-2	2-1	For this low range uniaxial compressive test is preferred		
Rating	15	12	7	4	2	1	0

Rock Mass Rating (RMR)

2. RQD:

- It should be determined either from the core logs or from the volumetric joint count (J_v).
- The direct method involves the percentage of rock recovery in 1 m of drill run (Deere, 1963).
- The indirect method includes an empirical formula based on J_v (Palmstrom, 1982) which can be estimated in the field survey. J_v is defined as the number of joints occurring in one of volume.
- $RQD = 115 - 3.3 \times J_v$

Rating based on RQD					
Qualitative description	Excellent	Good	Fair	Poor	Very Poor
	100-90	90-75	75-50	50-25	< 25
Rating	20	17	13	8	3

Rock Mass Rating (RMR)

3. Joint Spacing:

- It can be measured as a perpendicular distance between two neighbouring discontinuities in the field.
- The spacing should be measured for all sets of discontinuities to account for the maximum vulnerability of slope failure. The vulnerability of slope failure is higher for closely spaced discontinuities.

Rating based on spacing of discontinuities					
Description	Very wide	Wide	Moderate	Close	Very Close
Spacing	> 2	2-0.6	0.6-0.2	0.2-0.06	< 0.06
Rating	20	15	10	8	5

Rock Mass Rating (RMR)

4. Joint Conditions:

- The condition of discontinuities includes length of discontinuity, their separation, roughness of its surfaces, weathering of rock surface or the weak planes and infilling material.

Rating based on the condition of discontinuities					
Persistence	< 1	1-3	3-10	10-20	> 20
Rating	6	4	2	1	0
Aperture	None	< 0.1	0.1-1	1-5	> 5
Rating	6	5	4	1	0
Roughness	Very Rough	Rough	Slightly Rough	Smooth	Slickensided
Rating	6	5	3	1	0
Weathering	Unweathered	Slightly	Moderately	Highly	Decomposed
Rating	6	5	3	1	0
Infilling	-	Hard filling		Soft filling	
	None	< 5	> 5	< 5	> 5
Rating	6	4	2	2	0

Rock Mass Rating (RMR)

5. Groundwater Conditions:

- The rating for the condition of groundwater is assigned on the basis of the rate of inflow of groundwater or availability of actual data of water pressure or a general description.

Rating based on the condition of ground-water					
Inflow per 10 m tunnel length	None	< 10	10-25	25-125	> 125
Ratio of joint water pressure to major principal stress	0	0-0.1	0.1-0.2	0.2-05	> 0.5
General description	Completely dry	Damp	Wet	Dripping	Flowing
Rating	15	10	7	4	0

Rock Mass Rating (RMR): Example



Parameter		Condition
UCS		41-80 MPa
RQD		92-100%
Joint Spacing		0.25 - 6.0 m
Joint Condition	Persistence	up to 20 m
	Aperture	up to 5 mm
	Roughness	Slightly
	Infilling	None
	Weathering	Moderate
Ground Water Condition		Dry

Rock Mass Rating (RMR): Example

TABLE 6.1 Strength of Intact Rock Material

Qualitative description	Compressive strength (MPa)	Point load strength (MPa)	Rating
Extremely strong*	>250	8	15
Very strong	100–250	4–8	12
Strong	50–100	2–4	7
Medium strong*	25–50	1–2	4
Weak	5–25	Use of UCS is preferred	2
Very weak	1–5	<0.5	0
Extremely weak	<1	<0.1	0

*At compressive strength of rock material less than 1.0 MPa, many rock materials would be regarded as soft.
*Terms redefined according to ISO 7469.

Sources: Bieniawski, 1979, 1986; ISRI 1989-1, 2003.

TABLE 6.5 The RMR System: Guidelines for Classification of Discontinuity Conditions

Parameter*	Ratings				
Discontinuity length (see natural continuity)	<1 m	1–3 m	3–10 m	10–20 m	>20 m
	0	4	2	1	0
Separation (apart)	None	<0.1 mm	0.1–1.0 mm	1–5 mm	>5 mm
	0	5	4	1	0
Roughness of discontinuity surface	Very rough	Rough	Slightly rough	Smooth	Slip-sidled
	6	5	3	1	0
Irregularity (gouge)	Hard filling			Soft filling	
	None	<5 mm	>5 mm	<5 mm	>5 mm
	6	4	2	2	0
Weathering discontinuity surface	Unweathered	Slightly weathered	Moderately weathered	Highly weathered	Decomposed
	6	5	3	1	0

*Some conditions are mutually exclusive. For example, if filling is present, it is irrelevant what the roughness may be, since its effect will be overshadowed by the influence of the gouge. In such cases see Table 6.4 directly.

Sources: Bieniawski, 1963.

TABLE 6.2 Rock Quality Designation

Qualitative description	RQD (%)	Rating
Excellent	90–100	20
Good	75–90	17
Fair	50–75	13
Poor	25–50	8
Very poor	<25	5

Sources: Bieniawski, 1979.

TABLE 6.3 Spacing of Discontinuities

Description	Spacing (m)	Rating
Very wide	>2	20
Wide	0.6–2	15
Moderate	0.2–0.6	10
Close	0.06–0.2	8
Very close	<0.06	5

If more than one discontinuity set is present and the spacing of discontinuities of each set varies, consider the unfavorably oriented set with lowest rating. ISCI 146-89 uses the term "extremely close" for joint spacing less than 0.02 m.

Sources: Bieniawski, 1979; ISCI 1989-1, 2003.

TABLE 6.6 Groundwater Condition

Inflow per 10 m tunnel length (L/min)	None	<10	10–25	25–125	>125
Ratio of joint water pressure to major principal stress	0	0–0.1	0.1–0.2	0.2–0.5	>0.5
General description	Completely dry	Damp	Wet	Dripping	Flowing
Rating	15	10	7	4	0

Sources: Bieniawski, 1979.

Parameter		Condition	Rating
UCS		41 MPa	4
RQD		92 %	20
Joint Spacing		0.256 m	10
Joint Condition	Persistence	20 m	1
	Aperture	5 mm	1
	Roughness	Slightly	3
	Infilling	None	6
	Weathering	Moderate	3
Ground Water Condition		Dry	15
RMR			63

Rock Mass Rating (RMR)

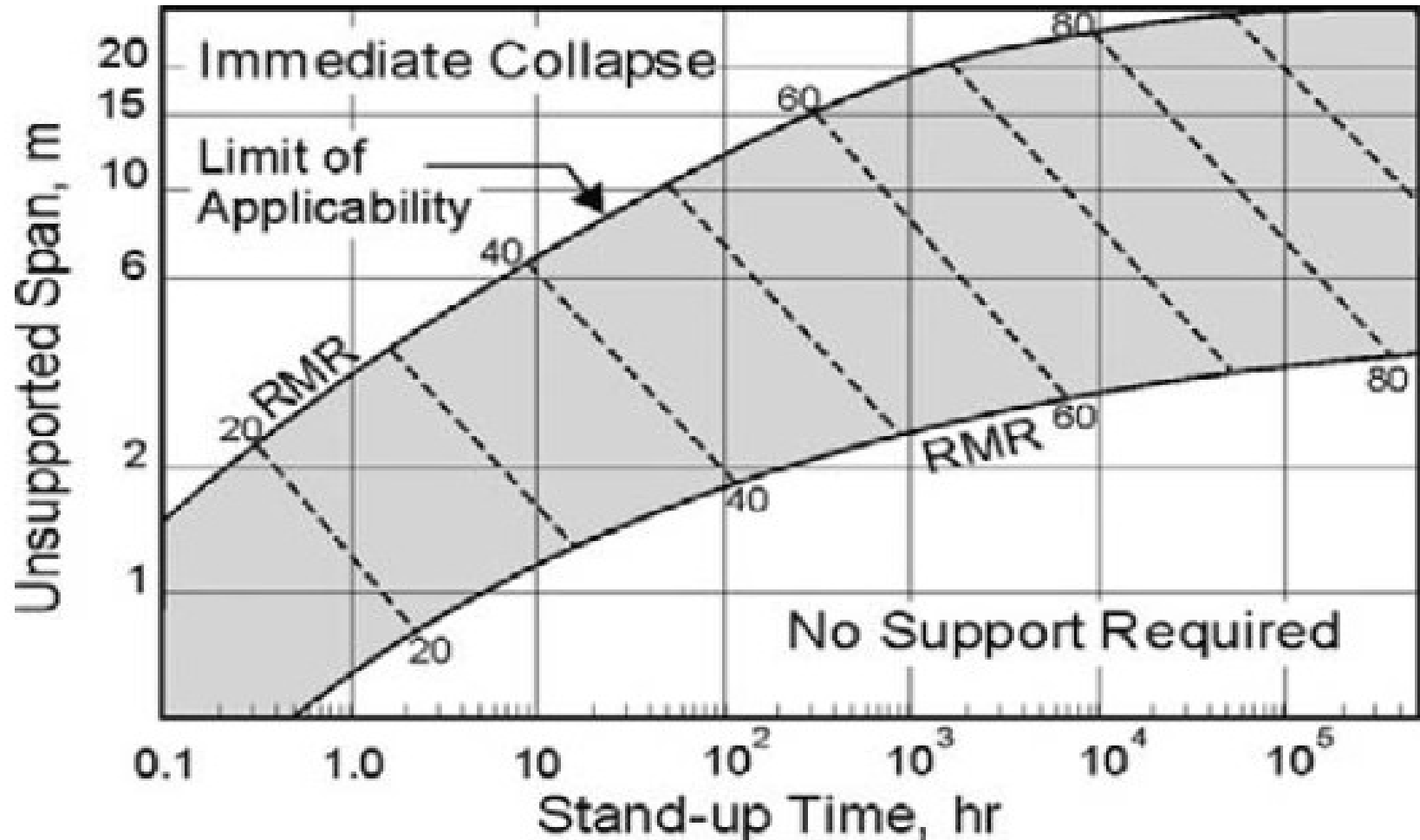
- Further, RMR value is used in the estimation of various design parameters and engineering properties such as cohesion and angle of internal friction for rock mass, allowable bearing pressure, average stand-up time and slope angle for safe cut.

score	100-81	80-61	60-41	40-21	<20
Rock class	I	II	III	IV	V
Classification	Very good	Good	Fair	Poor	Very Poor
Cohesion of rock mass (MPa)	>0.4	0.4-0.3	0.3-0.2	0.2-0.1	<0.1
Angle of internal friction of rock mass	>45°	45°-35°	35°-25°	25°-15°	<15°
Average stand-up time	20 years for 15m span	1 year for 10m span	1 week for 5m span	10 hours for 2.5m span	30 minutes for 1m span
Allowable bearing pressure (T/m ²)	600-440	440-280	280-135	135-45	45-30
Safe cut slope	>70°	65°	55°	45°	<40°

Average Stand-up Time for an Arched Roof

- The stand-up time depends upon an effective (unsupported) span of the opening, which is defined as the width of the opening or the distance between the tunnel face and the last support (whichever is smaller).
- For arched openings the stand-up time would be significantly higher than for a flat roof. Controlled blasting further increases the stand-up time as damage to the rock mass is decreased. For tunnels with an arched roof the stand-up time is related to the rock mass class in Table 1 (Figure 1).
- Do not unnecessarily delay supporting the roof in a rock mass with high stand-up time as this may lead to deterioration in the rock mass, which ultimately reduces the stand-up time.
- Lauffer (1988) observed that the stand-up time improves by one class of RMR value in excavations by TBM.

Average Stand-up Time for an Arched Roof



Guidelines for excavation and support of 10 m span horseshoe shaped rock tunnels constructed using drill and blast method at a depth of < 900 m, in accordance with the RMR system (after Bieniawski, 1989)

Rock mass class	Excavation	Supports		
		<i>Rock bolts (20 mm diameter, fully grouted)</i>	<i>Conventional shotcrete</i>	<i>Steel sets</i>
Very good rock RMR = 81–100	Full face; 3 m advance	Generally, no support required except for occasional spot bolting		
Good rock RMR = 61–80	Full face; 1.0–1.5 m advance; complete support 20 m from face	Locally, bolts in crown 3 m long, spaced 2.5 m, with occasional wire mesh	50 mm in crown where required	None
Fair rock RMR = 41–60	Heading and bench; 1.5–3 m advance in heading; commence support after each blast; complete support 10 m from face	Systematic bolts 4 m long, spaced 1.5–2 m in crown and walls with wire mesh in crown	50–100 mm in crown and 30 mm in sides	None
Poor rock RMR = 21–40	Top heading and bench; 1.0–1.5 m advance in top heading; install support concurrently with excavation 10 m from face	Systematic bolts 4–5 m long, spaced 1–1.5 m in crown and wall with wire mesh	100–150 mm in crown and 100 mm in sides	Light to medium ribs spaced 1.5 m where required
Very poor rock RMR <20	Multiple drifts; 0.5–1.5 m advance in top heading; install support concurrently with excavation; shotcrete as soon as possible after blasting	Systematic bolts 5–6 m long, spaced 1–1.5 m in crown and walls with wire mesh; bolt invert	150–200 mm in crown, 150 mm in sides, and 50 mm on face	Medium to heavy ribs spaced 0.75 m with steel lagging and forepoling if required; close invert

Shape: Horseshoe; width: 10 m; vertical stress <25 MPa; construction: drilling and blasting.

Source: Bieniawski, 1984.

Direct relations between rock mass classification and properties of rock mass (Aydan, Ulusay, & Tokashiki, 2014)

Property	Empirical relation	Proposed by
Deformation modulus, E_m	$E_m = 2RMR - 100$ (GPa) (for $RMR > 50$)	Bieniawski (1978)
	$E_m = 10^{((RMR-10)/40)}$ (GPa)	Serafim and Pereira (1983)
	$E_m = e^{(4.407+0.081 \cdot RMR)}$ (GPa)	Jasarevic and Kovacevic (1996)
	$E_m = 0.0097 RMR^{3.54}$ (MPa)	Aydan et al. (1997)
	$E_m = 25 \log Q$ (GPa)	Grimstad and Barton (1993)
	$E_m = (1 - \frac{D}{2}) \sqrt{\frac{\sigma_{ci}}{100}} 10^{((GSI-10)/40)}$ (GPa) (for $\sigma_{ci} < 100$ MPa)	Hoek et al. (2002)
	$E_m = 100 \frac{(1-0.5D)}{1+e^{\frac{25+250-GSI}{11}}}$ (GPa)	Hoek and Diederichs (2006)

Direct relations between rock mass classification and properties of rock mass (Aydan, Ulusay, & Tokashiki, 2014)

Property	Empirical relation	Proposed by
Deformation modulus, E_m	$E_m = 0.135 \left[E_i \left(1 + \frac{1}{WD} - \frac{RQD}{100} \right) \right]^{1.1811}$ (GPa)	Kayabasi et al. (2003)
	$E_m = 5.6 R_{Mi}^{0.3}$ (GPa) (for $R_{Mi} > 0.1$)	Palmstrom (1996)
	$E_m = 0.1 \left(\frac{RMR}{10} \right)^3$	Mitri et al. (1994)
	$E_m = 7(\pm 3) \sqrt{10^{(RMR-44)/21}}$ (GPa)	Diederichs and Kaiser (1999)
	$E_m = 10 Q^{1/3}$ (GPa)	Barton (1995)
	$E_m = 10 \left(Q \frac{\sigma_{ci}}{100} \right)^{1/3}$ (GPa)	Barton (2002)
	$E_m = 10^{\left(\frac{GSI-10}{40} \right)} \sqrt{\frac{\sigma_{ci}}{100}}$ (GPa)	Hoek and Brown (1997)
	$E_m = 0.0876 RMR$ (GPa) (for $RMR > 50$)	Galera et al. (2005)
	$E_m = 0.0876 RMR + 1.056(RMR - 50) + 0.015 (RMR - 50)^2$ (GPa) (for $RMR \leq 50$)	Galera et al. (2005)

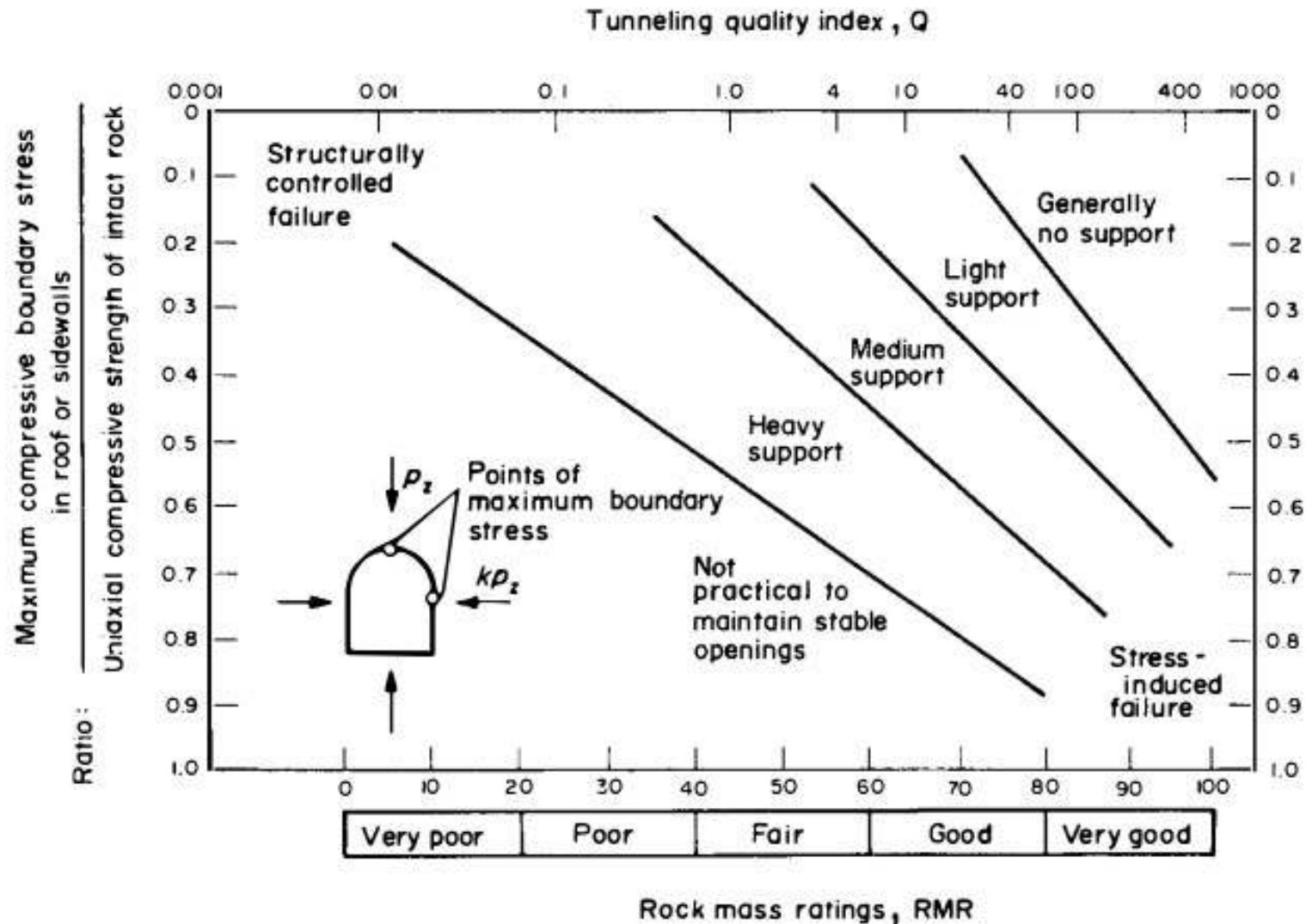
Direct relations between rock mass classification and properties of rock mass (Aydan, Ulusay, & Tokashiki, 2014)

Property	Empirical relation	Proposed by
Uniaxial compressive strength, σ_{cm} (MPa)	$\sigma_{cm} = 0.0016 RMR^{25}$	Aydan et al. (1997)
	$\sigma_{cm} = 5 \gamma \left(Q \frac{\sigma_{cl}}{100} \right)^{1/3}$	Barton (2002)
Friction angle, ϕ_m (°)	$\phi_m = 20 + 0.5 RMR$	<i>Aydan and Kawamoto (2001)</i>
	$\phi_m = 20 \sigma_{cm}^{0.25}$	<i>Aydan et al. (1993)</i>
	$\phi_m = \tan^{-1} \left(\frac{J_r}{J_a} \times \frac{J_w}{1} \right)$	Barton (2002)

Direct relations between rock mass classification and properties of rock mass (Aydan, Ulusay, & Tokashiki, 2002)

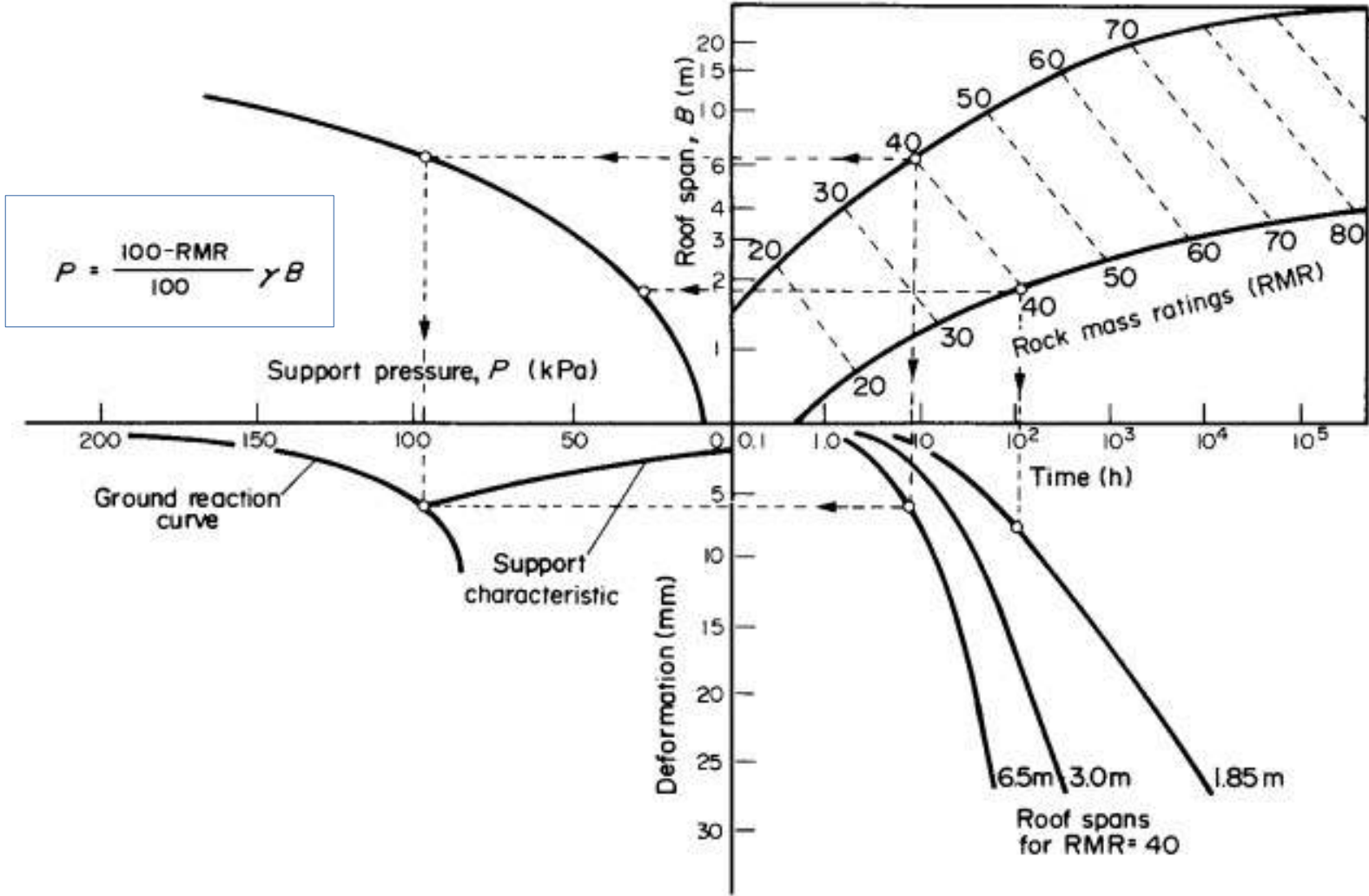
Property	Empirical relation	Proposed by
Cohesion, C_m (MPa)	$C_m = \frac{\sigma_{cm}}{2} \frac{1 - \sin \phi_m}{\cos \phi_m}$	Aydan and Kawamoto (2001)
	$C_m = \left(\frac{RQD}{J_n} \times \frac{1}{SRF} \times \frac{\sigma_{ci}}{100} \right)$	Barton (2002)
Poisson's ratio, ν_m	$\nu_m = 0.25(1 + e^{-\sigma_{cm}/4})$	Aydan et al. (1993)
	$\nu_m = 0.5 - 0.2 \frac{RMR}{RMR + 0.2(100 - RMR)}$	Tokashiki and Aydan (2010)

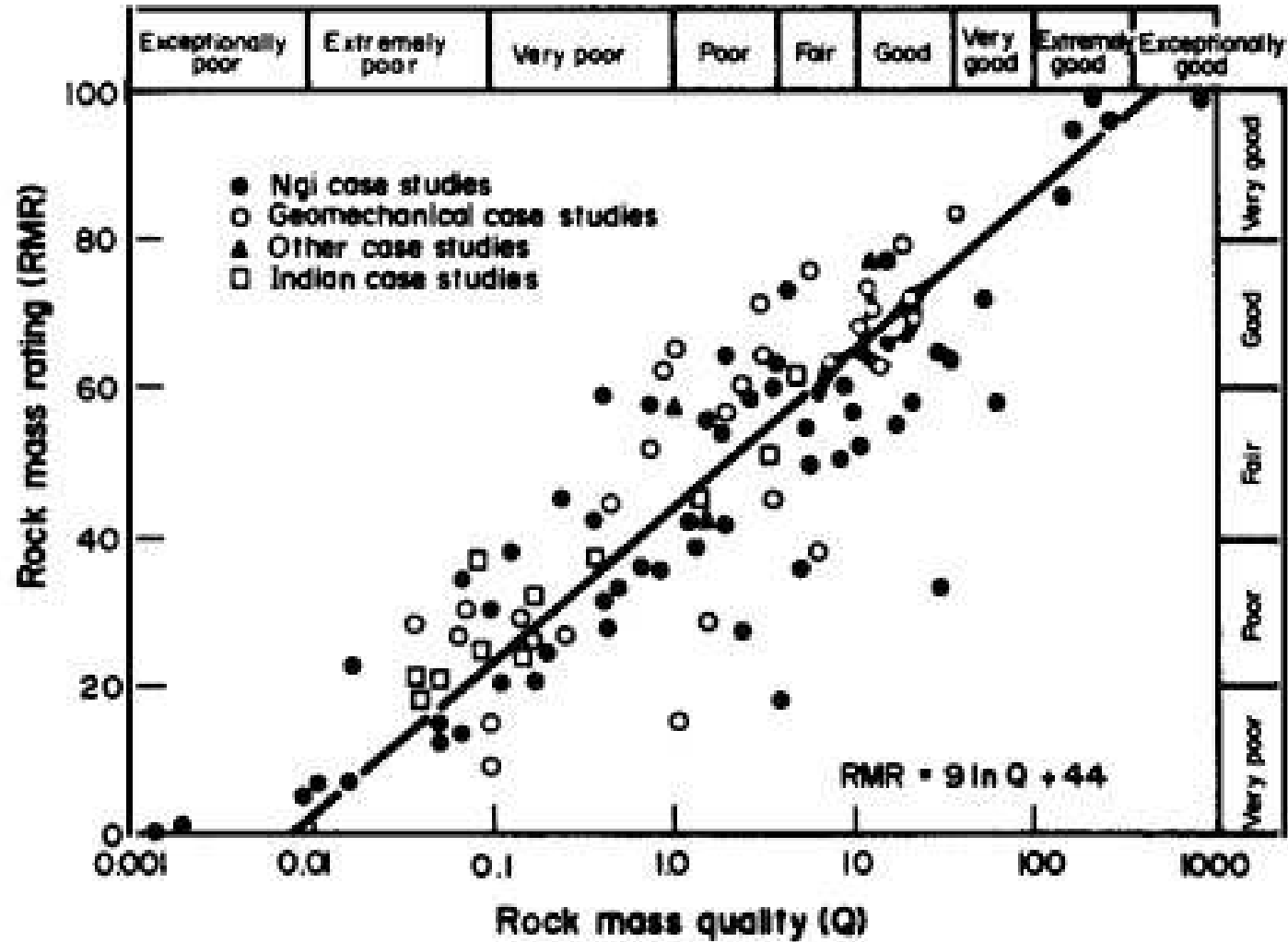
E_m deformation modulus of rock mass, E_i Young's modulus of intact rock, RMR rock mass rating, Q rock mass quality, GSI Geological Strength Index, D Disturbance factor, σ_{ci} uniaxial compressive strength of intact rock, σ_{cm} uniaxial compressive strength of rock mass, RQD Rock Quality Designation, RMi Rock Mass Index, WD weathering degree, ϕ_m friction angle of rock mass, c_m cohesion of rock mass, ν_m Poisson's ratio of rock mass, J_n joint set rating, J_r joint roughness rating, J_w joint water rating, J_a joint alteration rating, SRF stress reduction factor, γ rock density (t/m^3)



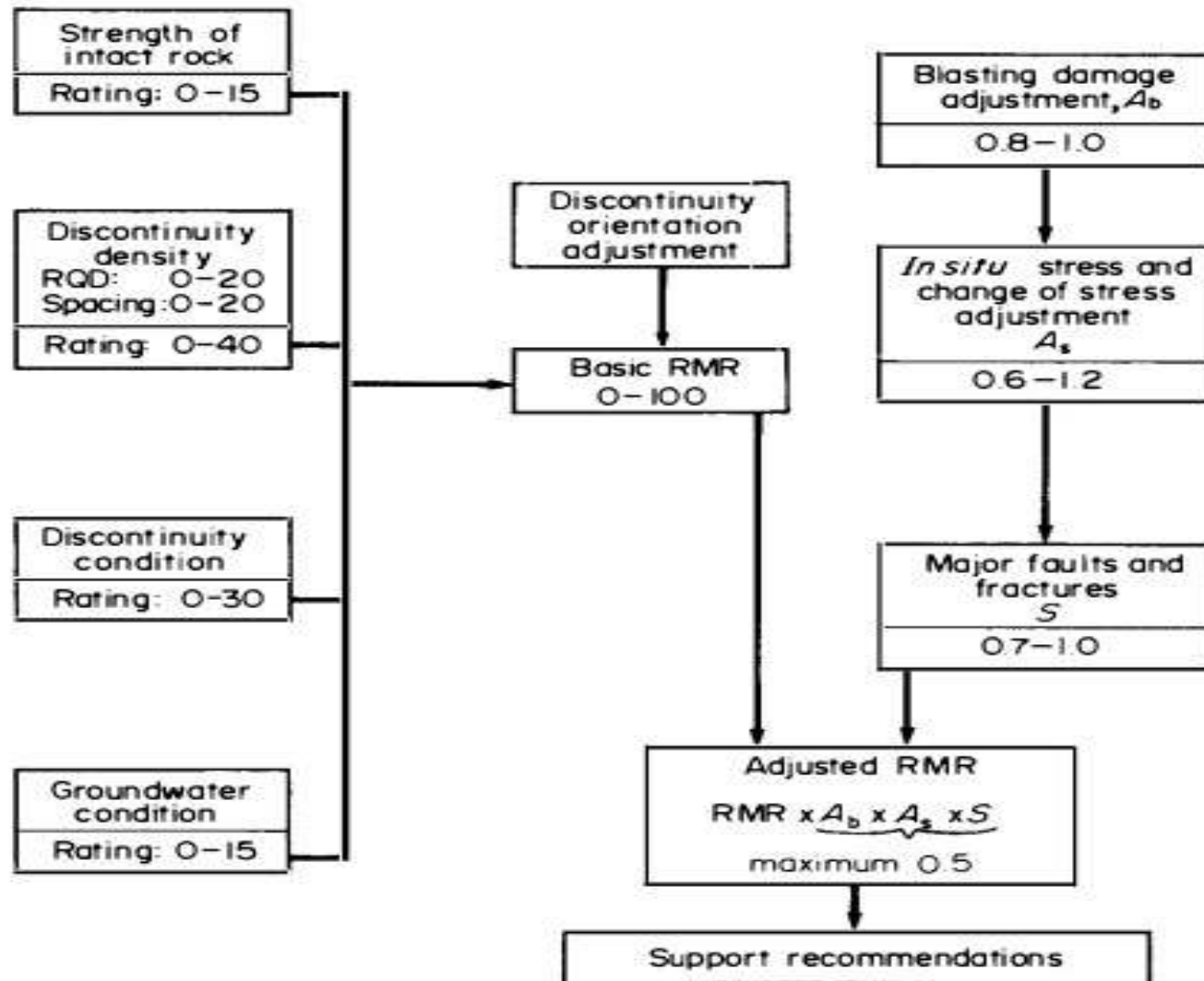
Approximate
support
guidelines
(after Hoek,
1981)

Integration of Rock Mass Rating with support characteristics and roof deformation in coal mines





Correlation between the RMR and the Q index (after Jethwa)



Adjustments to the RMR system for mining applications

Barton's Q-System

$$Q = \frac{RQD}{J_n} \times \frac{J_r}{J_a} \times \frac{J_w}{SRF}$$

Where;

Q = Rock Quality

Designation,

J_n = Number of joint set,

J_r = Joint roughness number,

J_a = Joint alteration number,

J_w = Joint water reduction

factor, and

SRF = Stress reduction

factor.

The rock quality can range from $Q = 0.001$ to $Q = 1000$ on a logarithmic rock mass quality scale.

$$RQD = \frac{\text{Sum of core Pieces} > 10 \text{ cm}}{\text{Total drill run}} \times 100$$

Joint Set

Number of Joint Sets	<i>Joint set No. J_n</i>
Massive/Intact, no or few joints	0.5– 1.0
One joint set	2
one joint set plus random	3
Two joint sets	4
Two joint sets plus random	6
Three joint sets	9
Three joint sets plus random	12
Four or more joint sets, random, heavily jointed, ‘suger cube’, etc.	15
Crushed rock, earthlike	20

Note:

For intersections use $J_n = 3J_n$

For portals use $J_n = 2J_n$

Joint Roughness Number J_r

Description of joint surface roughness	Discontinuous	Undulating	Planar
Rough	4.0	3.0	1.5
Smooth	3.0	2.0	1.0
Slickensided	2.0	1.5	0.5
Planes containing gouge thick enough to prevent rock wall contact	1.5	1.0	1.0

Note:

Add 1.0 to J_r if the mean spacing of the relevant joint set is greater than 3.0 m

Joint Alteration Number J_a

(a) Rock wall contact	Ja	Φt (approx)
A. Tightly healed, hard, nonsoftening, impermeable filling i.e., quartz or epidote.	0.75	25-35°
B. Unaltered joint walls, surface staining only.	1.0	
C. Slightly altered joint walls. Nonsoftening mineral coatings, sandy particles, clay-free disintegrated rock, etc.	2.0	25-30°
D. silty or sandy clay coatings, small clay fraction (nonsoftening)	3.0	20-25°
E. Softening or low-friction clay mineral coatings, i.e., kaolinite, mica Also chlorite, talc, gypsum, and graphite, etc and small quantities of swelling clays(discontinuous coatings, 1-2 mm or less in thickness.	4.0	8-16°
(b) Rock wall contact before 10 cm shear		
F. Sandy particles, clay-free disintegrated rock, etc.	4.0	25-30°
G. Strongly over-consolidated, non-softening clay mineral fillings (continuous. < 5 mm in thickness)	6.0	16-24°
H. Medium or low over-consolidation, softening clay mineral fillings. (continuous, < 5 mm in thickness)	8.0	12-16°
J. Swelling clay fillings, i.e. montmorillonite (continuous, < mm in thickness) value of Ja depends on percentage of swelling clay sized particles, and access to water ect.	8.0-12.0	6-12°
(c) No rock wall contact when sheared.		
K. Zones or bands of disintegrated or crushed rock and clay (see G.,H., J. for description of clay condition)	6-8. or 8-12	6-24°
L. Zones or bands of silty or sandy clay, small clay fraction (nonsoftening)	5.0	
M. Thick, continuous zones or bands of clay (see G.,H.,J. for description of clay condition)	10-13 or 13-20	6-24°
Note: (i) Values of Φ, are intended as an approximate guide to the mineralogical properties of the alteration products. If present.		

Joint water Reduction Factor J_w

	J_w	Water Pressure (kg/cm ²)	
A. Dry excavations or minor inflow, i.e., <5 l/min locally	1.0	<1	Note: (i) Factors (C to F) are crude estimates. Increase J_w if drainage measures are installed. (ii) Special problems caused by ice formation are not considered.
B. Medium inflow or pressure occasional outwash of joint fillings	0.66	1.0-2.5	
C. Large inflow or high pressure in competent rock with unfilled joints	0.5	2.5-10.0	
D. Large inflow or high pressure, considerable outwash of joint fillings	0.33	2.5-10.0	
E. Exceptionally high inflow or water pressure at blasting, decaying with time.	0.2-0.1	>10.0	
F. Exceptionally high inflow or water pressure continuing without noticeable decay.	0.1-0.05	>10.0	

Stress Reduction Factor (SRF)

(a) Weakness zones intersecting excavation, which may cause loosening of rock mass when tunnel is excavated.		SRF
A.	Multiple occurrences of weakness zones containing clay or chemically disintegrated rock, very loose surrounding rock (any depth)	10.0
B.	Single-weakness zones containing clay or chemically disintegrated rock (depth of excavation ≤ 50 m)	5.0
C.	Single –weakness zones containing clay or chemically disintegrated rock (depth of excavation > 50 m)	2.5
D.	Multiple- shear zones in competent rock (clay-free), loose surrounding rock (any depth)	7.5
E.	Single-shear zones in competent rock (clay-free) (depth of excavation ≤ 50 m)	5.0
F.	Single- shear zones in competent rock (clay-free) depth of excavation > 50 m)	2.5
G	Loose open joints, heavily jointed or “sugar cube” etc. (any depth)	5.0

Note: Reduce these SRF values by 25-50% if the relevant shear zones only influence but do not intersect the excavation

Stress Reduction Factor (SRF)continued

(b) Competent rock, rock stress problems

		q_c/σ_1	σ_θ/q_c	SRF (old)	SRF (new)
H	Low stress, near surface, open joints	>200	<0.01	2.5	2.5
J	Medium stress, favorable stress condition	200–10	0.01–0.3	1.0	1.0
K	High stress, very tight structure; usually favorable to stability, may be unfavorable to wall stability	10–5	0.3–0.4	0.5–2.0	0.5–2.0
L	Moderate slabbing after >1 hour in massive rock	5–3	0.5–0.65	5–9	5–50
M	Slabbing and rock burst after a few minutes in massive rock	3–2	0.65–1.0	9–15	50–200
N	Heavy rock burst (strain-burst) and immediate dynamic deformations in massive rock	<2	>1	15–20	200–400

For strongly anisotropic virgin stress field (if measured): when $5 < \sigma_1 / \sigma_3 < 10$, reduce q_c to $0.75 q_c$; when $\sigma_1 / \sigma_3 > 10$, reduce q_c to $0.50 q_c$ (where q_c is unconfined compressive strength), σ_1 and σ_3 are major and minor principal stresses, and σ_θ is the maximum tangential stress (estimated from elastic theory). Few case records available where depth of crown below surface is less than span width; suggest SRF increase from 2.5 to 5 for such cases (see H).

Stress Reduction Factor (SRF)continued

Cases L, M, and N are usually most relevant for support design of deep tunnel excavation in hard massive rock masses, with RQD/J n ratios from about 50–200.

(c) Squeezing rock; plastic flow of incompetent rock under the influence of high rock pressures

		SRF (old)	SRF (new)
O	Mild squeezing rock pressure	1–5	5–10
P	Heavy squeezing rock pressure	>5	10–20

(d) Swelling rock; chemical swelling activity depending on presence of water

Q	Mild swelling rock pressure	5–10	
R	Heavy swelling rock pressure	10–15	

For general characterization of rock masses distant from excavation influences, the use of SRF 5, 2.5, 1.0, and 0.5 is recommended as depth increases from, say, 0–5, 5–25, 25–250, >250 m. This will help to adjust Q for some of the effective stress effects, in combination with appropriate characterization values of J w . Correlations with depth-dependent static modulus of deformation and seismic velocity will then follow the practice used when these were developed.

Cases of squeezing rock may occur for depth $H > 350Q^{1/3}$ (Singh & Goel, 2006). Rock mass compressive strength can be estimated from $q_{\text{cmass}} = 7\gamma(Q)^{1/3}$ (MPa); γ is the rock density in t/m^3 , and q_{cmass} = rock mass compressive strength.



3 Joint Sets

Estimated Spacing?

$$J_1 = 1\text{m}; J_2 = 0.5\text{m}; J_3 = 0.25\text{m}$$

$$J_v = 1/1 + 1/0.5 + 1/0.25 = 7$$

$$\text{RQD} = 115 - 3.3 J_v = \mathbf{93\%}$$

Barton says to closet 5%, so 95%

Joint Set Number

Number of Joint Sets	<i>Joint set No. J_n</i>
Massive/Intact, no or few joints	0.5– 1.0
One joint set	2
one joint set plus random	3
Two joint sets	4
Two joint sets plus random	6
Three joint sets 	9
Three joint sets plus random	12
Four or more joint sets, random, heavily jointed, ‘suger cube’, etc.	15
Crushed rock, earthlike	20

Note:

For intersections use $J_n = 3 J_n$

For portals use $J_n = 2J_n$



Joint Roughness Number Jr

Description of joint surface roughness	Discontinuous	Undulating	Planar
Rough	4.0	3.0 	1.5
Smooth	3.0	2.0	1.0
Slickensided	2.0	1.5	0.5
Planes containing gouge thick enough to prevent rock wall contact	1.5	1.0	1.0

Note:

Add 1.0 to J_r if the mean spacing of the relevant joint set is greater than 3.0 m



(a) Rock wall contact	Joint Alteration Number J_a	J_a	Φ_t (approx)
A. Tightly healed, hard, nonsoftening, impermeable filling i.e., quartz or epidote.		0.75	25-35°
B. Unaltered joint walls, surface staining only.		1.0	
C. Slightly altered joint walls. Nonsoftening mineral coatings, sandy particles, clay-free disintegrated rock, etc.		2.0	25-30°
D. silty or sandy clay coatings, small clay fraction (nonsoftening)		3.0	20-25°
E. Softening or low-friction clay mineral coatings, i.e., kaolinite, mica Also chlorite, talc, gypsum, and graphite, etc and small quantities of swelling clays(discontinuous coatings, 1-2 mm or less in thickness.		4.0	8-16°
(b) Rock wall contact before 10 cm shear			
F. Sandy particles, clay-free disintegrated rock, etc.		4.0	25-30°
G. Strongly over-consolidated, non-softening clay mineral fillings (continuous. < 5 mm in thickness)		6.0	16-24°
H. Medium or low over-consolidation, softening clay mineral fillings. (continuous, < 5 mm in thickness)		8.0	12-16°
J. Swelling clay fillings, i.e. montmorillonite (continuous, < mm in thickness) value of J_a depends on percentage of swelling clay sized particles, and access to water ect.		8.0-12.0	6-12°
(c) No rock wall contact when sheared.			
K. Zones or bands of disintegrated or crushed rock and clay (see G.,H., J. for description of clay condition)		6-8. or 8-12	6-24°
L. Zones or bands of silty or sandy clay, small clay fraction (nonsoftening)		5.0	
M. Thick, continuous zones or bands of clay (see G.,H.,J. for description of clay condition)		10-13 or 13-20	6-24°
Note: (i) Values of Φ , are intended as an approximate guide to the mineralogical properties of the alteration products. If present.			

Joint water Reduction Factor J_w

	J_w	Water Pressure (kg/cm ²)	
A. Dry excavations or minor inflow, i.e., <5 l/min locally 	1.0	<1	Note: (i) Factors (C-F are crude estimates. Increase J_w if drainage measures are installed. (ii) Special problems caused by ice formation are not considered.
B. Medium inflow or pressure occasional outwash of joint fillings	0.66	1.0-2.5	
C. Large inflow or high pressure in competent rock with unfilled joints	0.5	2.5-10.0	
D. Large inflow or high pressure, considerable outwash of joint fillings	0.33	2.5-10.0	
E. Exceptionally high inflow or water pressure at blasting, decaying with time.	0.2-0.1	>10.0	
F. Exceptionally high inflow or water pressure continuing without noticeable decay.	0.1-0.05	>10.0	

Stress Reduction Factor (SRF)continued

(b) Competent rock, rock stress problems

		q_c/σ_1	σ_θ/q_c	SRF (old)	SRF (new)
H	Low stress, near surface, open joints	>200	<0.01	2.5	2.5
J	Medium stress, favorable stress condition 	200–10	0.01–0.3	1.0	1.0
K	High stress, very tight structure; usually favorable to stability, may be unfavorable to wall stability	10–5	0.3–0.1	5–2.0	0.5–2.0
L	Moderate slabbing after >1 hour in massive rock	5–			
M	Slabbing and rock burst after a few minutes in massive rock				
N	Heavy rock burst (strain-burst) and immediate dynamic deformations in massive rock	<2	>1		200–400

UCS = q_c = 200 MPa
 σ_1 = 6 MPa (tunnel at 150 m)
(q_c/σ_1) = 33

For strongly anisotropic virgin stress field (if measured): when $5 < \sigma_1/\sigma_3 < 10$, reduce q_c to $0.75 q_c$; when $\sigma_1/\sigma_3 > 10$, reduce q_c to $0.50 q_c$ (where q_c is unconfined compressive strength), σ_1 and σ_3 are major and minor principal stresses, and σ_θ is the maximum tangential stress (estimated from elastic theory). Few case records available where depth of crown below surface is less than span width; suggest SRF increase from 2.5 to 5 for such cases (see H).

$$Q = 95/9 \times 1.5/1 \times 1/1 \\ = 15.8$$

Equivalent RMR, from Bieniawski:

$$\text{RMR} = 9\ln Q + 44 = 69 \quad (\text{natural log})$$

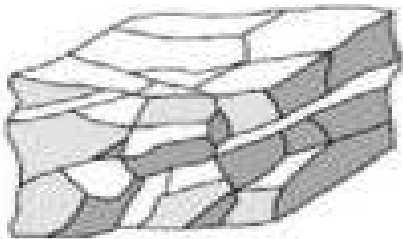
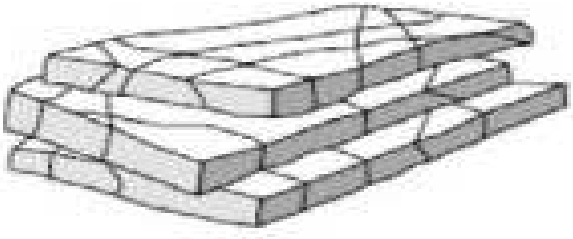
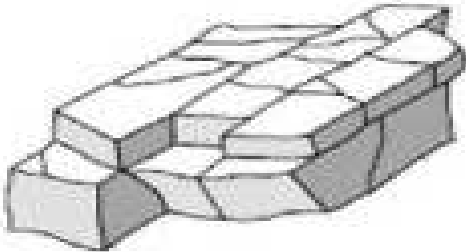
or, from Barton:

$$\text{RMR} = 15\log Q = 50 = 68$$

Rock mass quality (Q) may be considered a function of only three parameters, which are approximate measures of

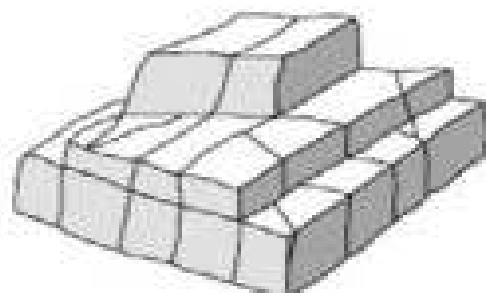
a. Block size (RQD/J _n):	It represents overall structure of rock mass (Table 4.5 next slide)
b. Inter-block shear strength (J _r /J _a):	• It has been found that $\tan^{-1} (J_r / J_a)$ is a fair approximation of the actual peak sliding angle of friction along the clay-coated joints (Table 8.7). • This has been later modified by Barton (2008) as given in Eq. (8.16)
c. Active stress (J _w /SRF):	It is an empirical factor describing the active effective stress

TABLE 4.5 Terms to Describe the Main Rock Mass Structures and Block Shapes

S. No.	Terms	Figure	Description
1	Polyhedral blocks		Irregular discontinuities without arrangement into distinct sets, and of small persistence
2	Tabular blocks		One dominant set of parallel discontinuities, for example, bedding planes, with other noncontinuous joints; thickness of blocks much less than length or width
3	Prismatic blocks		Two dominant sets of discontinuities, approximately orthogonal and parallel, with a third irregular set; thickness of blocks much less than length or width

4

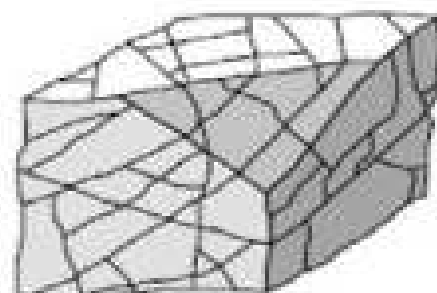
Equidimensional
blocks



Three dominant sets
of discontinuities,
approximately orthogonal,
with occasional irregular
joints, giving equi-
dimensional blocks

5

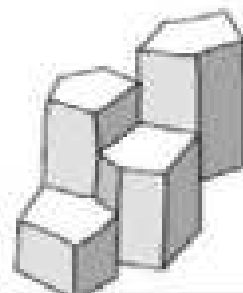
Rhombohedral
blocks



Three (or more) dominant,
mutually oblique, sets of
joints, giving oblique-
shaped, equidimensional
blocks

6

Columnar blocks



Several, usually more than
three, sets of discontinuities;
parallel joints usually
crossed by irregular joints;
length much greater than
other dimensions

It has been found that $\tan^{-1} (J_r / J_a)$ is a fair approximation of the actual peak sliding angle of friction along the clay-coated joints

TABLE 8.7 Estimation of Angle of Internal Friction from the Parameters J_r and J_a

Description	J_r	$\tan^{-1}(J_r/J_a)$				
(a) Rock wall contact		(Thin coatings)				
		$J_a = 0.75$	1.0	2.0	3.0	4.0
A. Discontinuous joints	4.0	79°	76°	63°	53°	45°
B. Rough, undulating	3.0	76°	72°	56°	45°	37°
C. Smooth, undulating	2.0	69°	63°	45°	34°	27°
D. Slickensided, undulating	1.5	63°	56°	37°	27°	21°
E. Rough, planar	1.5	63°	56°	37°	27°	21°
F. Smooth, planar	1.0	53°	45°	27°	18°	14°
G. Slickensided, planar	0.5	34°	27°	14°	9.5°	7.1°

(b) Rock wall contact when sheared

	<i>(Thin filling)</i>				
	J_r	$J_a = 4.0$	6	8	12
A. Discontinuous joints	4.0	45°	34°	27°	18°
B. Rough, undulating	3.0	37°	27°	21°	14°
C. Smooth, undulating	2.0	27°	18°	14°	9.5°
D. Slickensided, undulating	1.5	21°	14°	11°	7.1°
E. Rough, planar	1.5	21°	14°	11°	7.1°
F. Smooth, planar	1.0	14°	9.5°	7.1°	4.7°
G. Slickensided, planar	0.5	7°	4.7°	3.6°	2.4°

(c) No rock wall contact when sheared

	<i>(Thick filling)</i>				
	J_r	$J_a = 5$	6	8	12
Nominal roughness of discontinuity rock walls	1.0	11.3°	9.5°	7.1°	4.8°
	J_r	$J_a = 13$	16	20	—
	1.0	4.4°	3.6°	2.9°	—

Source: Barton, 2002.

Barton (2008) suggested the following correlations to obtain the cohesive strength (c_p) and angle of internal friction or frictional strength (ϕ_p) of the rock mass.

$$c_p = \frac{\text{RQD}}{J_n} \times \frac{1}{\text{SRF}} \times \frac{q_c}{100} \text{ MPa}$$

$$1.124\phi_p = \tan^{-1} \left(\frac{J_r}{J_a} \times J_w \right), \text{ degrees}$$

Eqn, 8.16

When uncertainty of Parameters exists then use the Weighted Average Method

TABLE 8.8 Weighted Average Method of Obtaining Q-Value

Parameter of Q	Poorest value (10%)	Most typical value (60%)	Maximum value (30%)	Weighted average
RQD	25	65	85	67
J_n	12	9	—	9.42
J_r	1.5	3	4	2.05
J_a	4	2	1	1.9
J_w	0.66	1	1	0.966
SRF	7.5	5	2.5	4.5

The rock mass quality (Q) is a very sensitive index and its value varies from 0.001 to 1000. Use of the Q-system is specifically recommended for tunnels and caverns with an arched roof. On the basis of the Q-value, the rock masses are classified into nine categories

TABLE 8.9 Classification of Rock Mass Based on Q-Values

Q	Group	Classification
0.001–0.01		Exceptionally poor
0.01–0.1	3	Extremely poor
0.1–1		Very poor
1–4	2	Poor
4–10		Fair
10–40		Good
40–100	1	Very good
100–400		Extremely good
400–1000		Exceptionally good

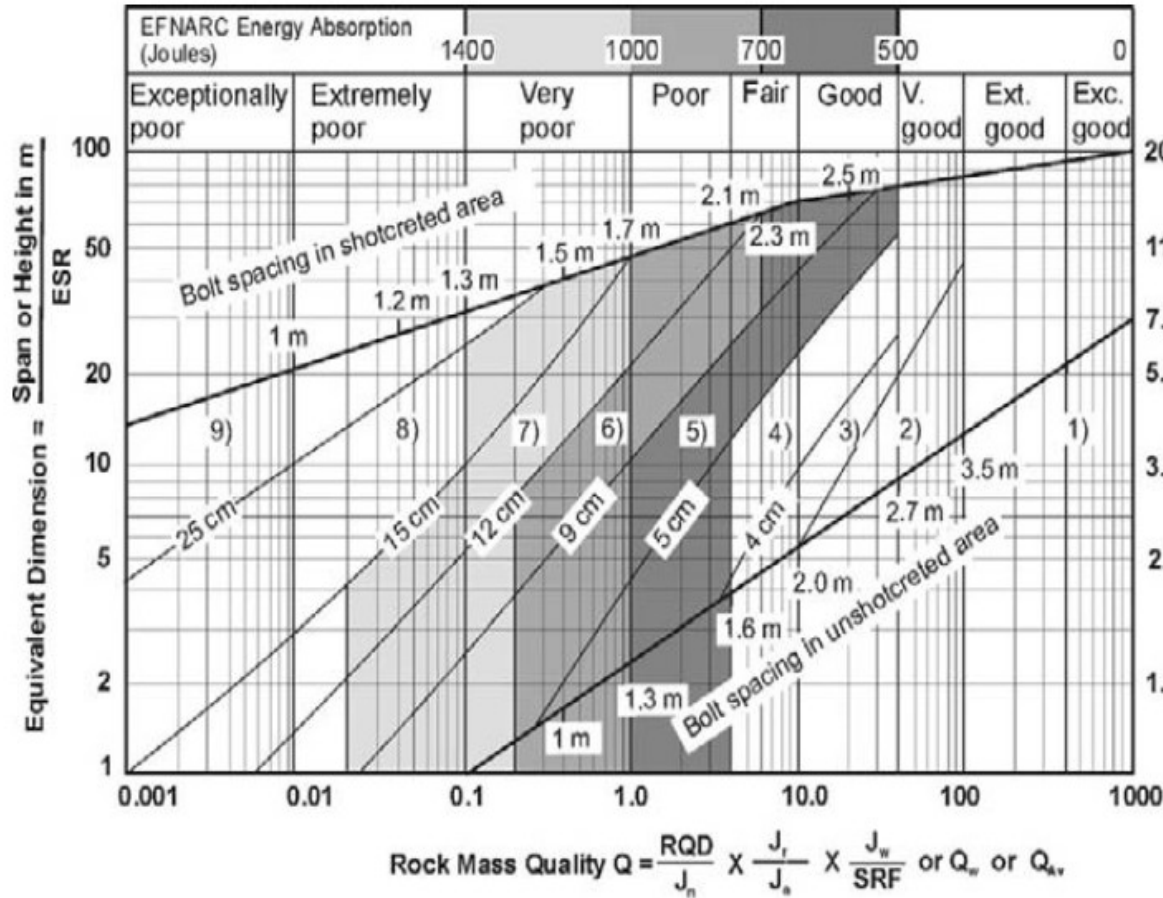
The rock bolt length (L) can be estimated from the excavation span (B) and the ESR value:

$$L = 2 + \frac{0.15 B}{ESR}$$

Equivalent Dimensions can be obtained using following equation

Type of excavation		ESR
A	Temporary mine openings etc.	2–5
B	Permanent mine openings, water tunnels for hydro power (excluding high pressure penstocks), pilot tunnels, drifts and headings for large openings, surge chambers.	1.6–2.0
C	Storage caverns, water treatment plants, minor road and railway tunnels, access tunnels.	1.2–1.3
D	Power stations, major road and railway tunnels, civil defence chambers, portal, intersections.	0.9–1.1
E	Underground nuclear power stations, railway stations, sports and public facilities, factories, major gas pipeline tunnel.	0.5–0.8

DESIGN OF SUPPORTS

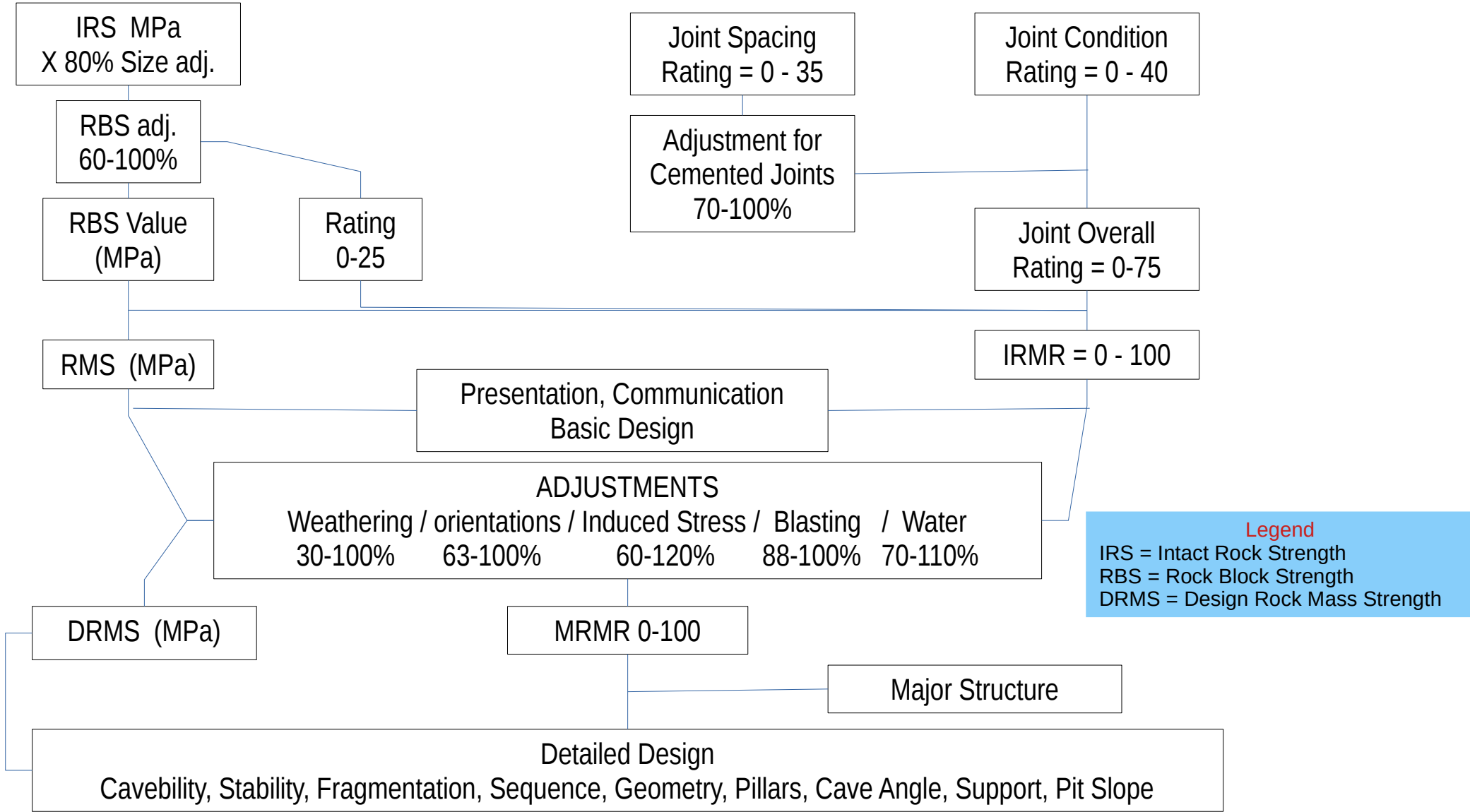


REINFORCEMENT CATEGORIES

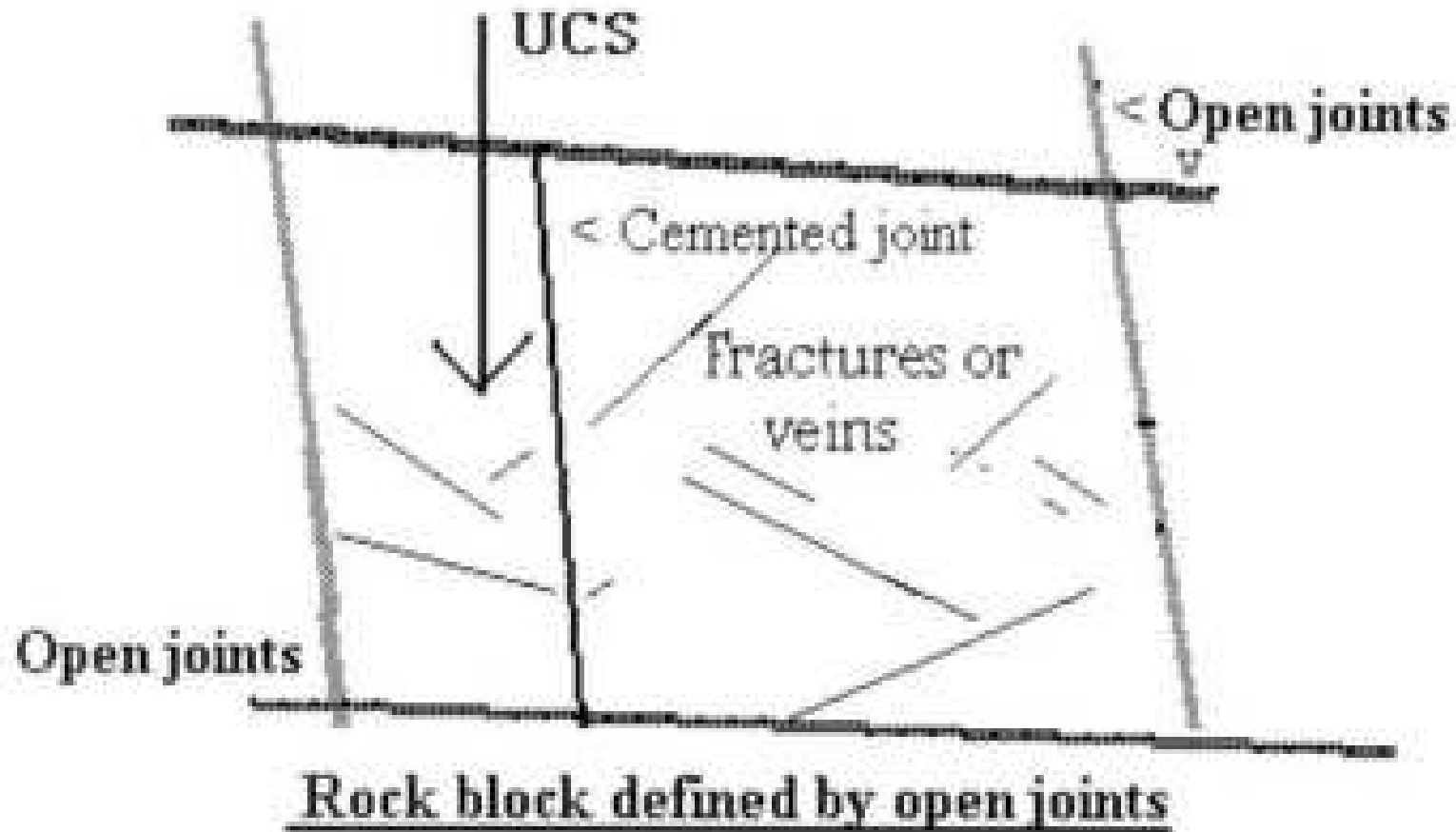
- 1) Unsupported
- 2) Spot bolting, sb
- 3) Systematic bolting, B
- 4) Systematic bolting (and unreinforced shotcrete, 4 to 10 cm, B(+S))
- 5) Fiber reinforced shotcrete and bolting, 5 to 9 cm, S(fr)+B
- 6) Fiber reinforced shotcrete and bolting, 9 to 12 cm, S(fr)+B
- 7) Fiber reinforced shotcrete and bolting, 12 to 15 cm, S(fr)+B
- 8) Fiber reinforced shotcrete > 15 cm, reinforced ribs of shotcrete and bolting, S(fr), RRS+B
- 9) Cast concrete lining, CCA

Grimstad and Barton (1993) chart for the design of support including the required energy absorption capacity of SFRS suggested by Papworth (2002).

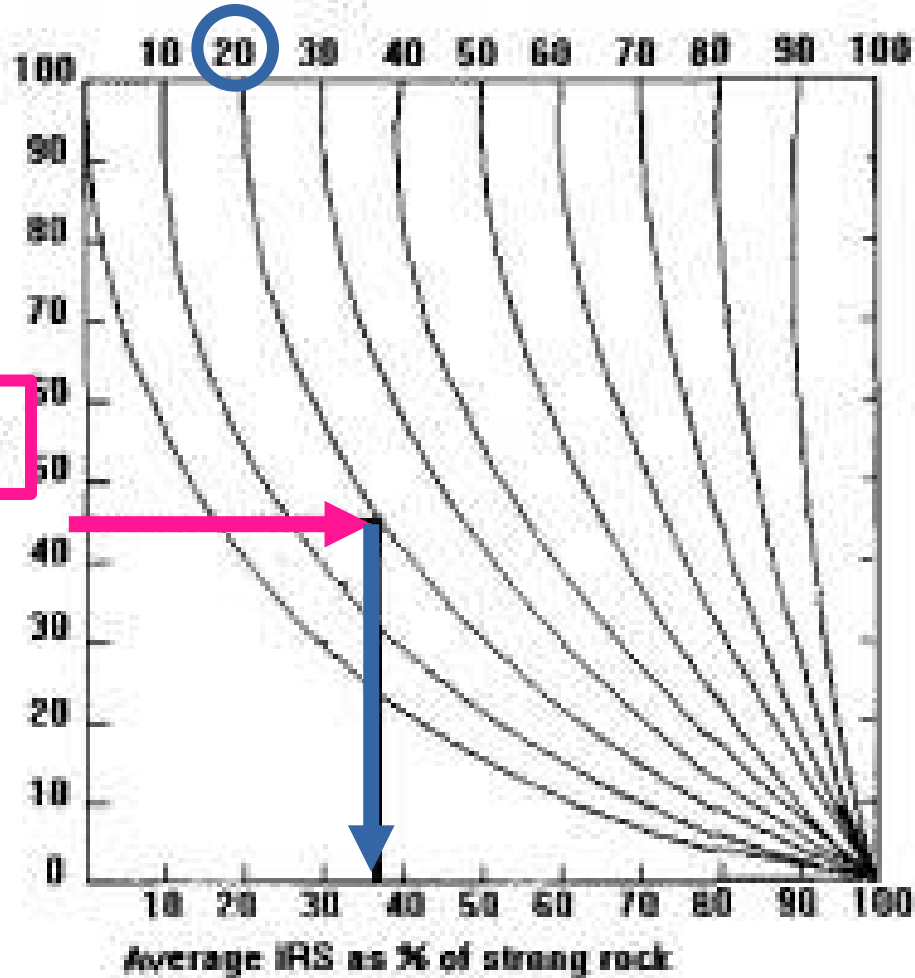
Laubscher's MRMR



Definition of the structural terms



Select curve using weak rock UCS as % of strong rock UCS



Nomogram for determining the “corrected” IRS value

- The **unconfined compressive strength (UCS)** is the value derived from testing cores and is the value assigned to the IRS.
- The intact rock specimen may be homogeneous or have intercalations of weaker material.
- In the latter case**, the procedure shown in Figure should be adopted. Care must be taken in determining this value as the cores that are selected often represent the stronger material in the rock mass.
- Example:** Let the UCS values for the strong and weak rock be **100 (MPa) and 20 MPa**, respectively.
- It is estimated that 45% is made up of weak rock.** Using Figure, one locates this value on the Y-axis, moves horizontally to the curve representing the strength of the weak rock, and then drops down to the horizontal axis. In this case, the appropriate **“corrected” IRS is 37 MPa.**

Rock Block Strength: To obtain the RBS from the “corrected IRS, various factors are applied depending upon whether the rock blocks are homogeneous or contain fractures and veins.

Homogeneous Rock Blocks-If the rock block **does not contain fractures or veins**, the RBS is the IRS value reduced to 80% to adjust for the small-to-large specimen effect.

Rock Blocks with Fractures and Veins-Fractures and veins reduce the strength of the rock block in terms of the number and frictional properties of the features.

The Moh’s hardness number is used to define the frictional properties of the vein and fracture filling.

The standard hardness table is used only up to 5, because values greater than 5 are not likely to be significant.

Open fractures or veins would be given a value of 1. The vein and fracture filling must be weaker than the host rock.

Mineral Hardness Scales	
Mineral	Mohs
Talc	1
Gypsum	2
Calcite	3
Fluorite	4
Apatite	5
Orthoclase	6
Quartz	7
Topaz	8
Corundum	9
Diamond	10

example:

IRS = 100 MPa,

Gypsum veins: Moh's hardness = 2

Fracture frequency per m = 8.0

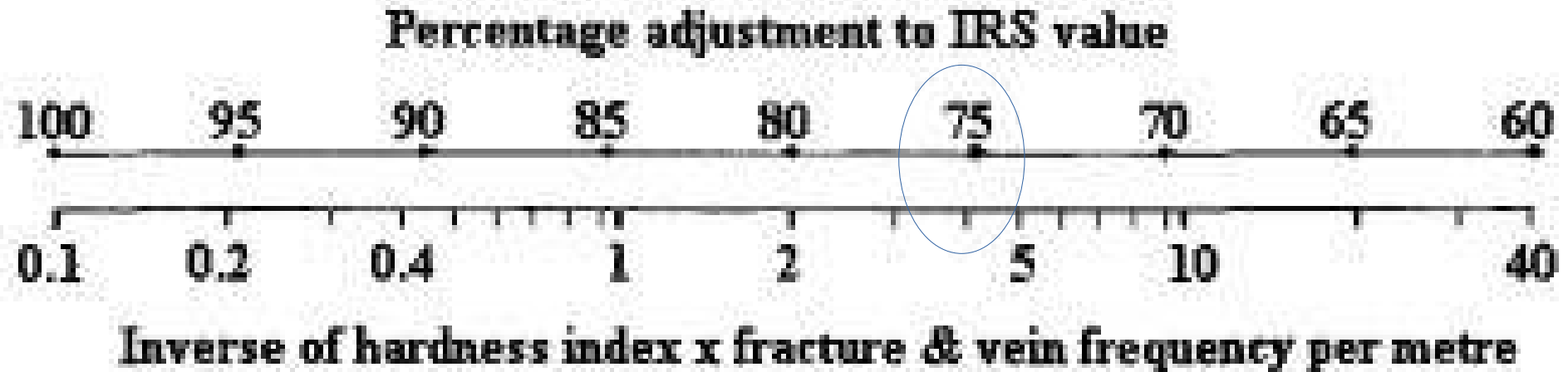
The product of the inverse of hardness and the fracture frequency is:

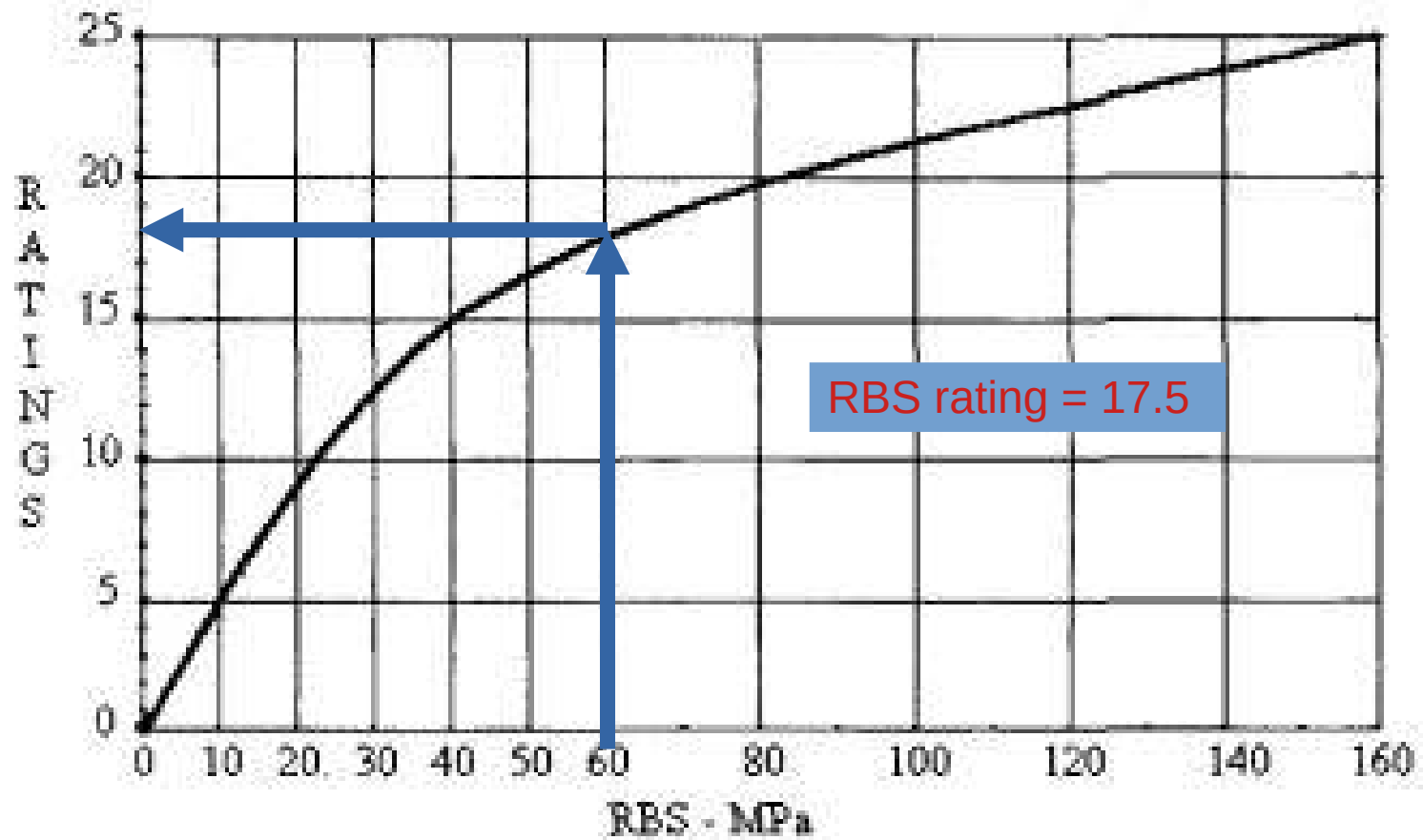
Inverse of hardness index x fracture frequency = $0.5 \times 8 = 4.0$

Using Figure below, one finds that the adjustment is 0.75.

$$\text{RBS} = 100 \times 0.8 \times 0.75 = 60 \text{ MPa.}$$

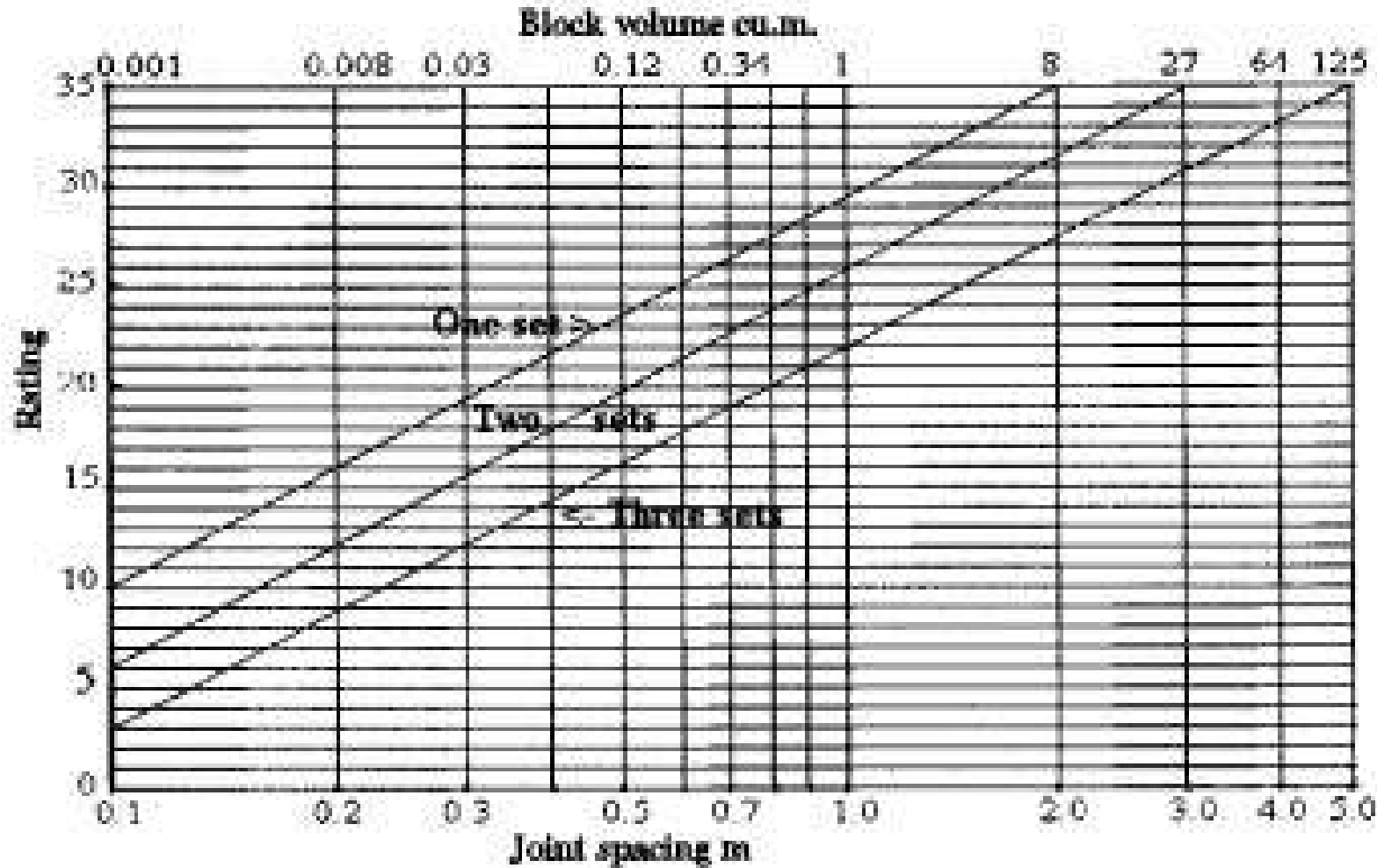
To obtain the RBS,
the corrected IRS is
adjusted by the size
factor of 80% and
then by the fracture
frequency and
hardness
adjustment.





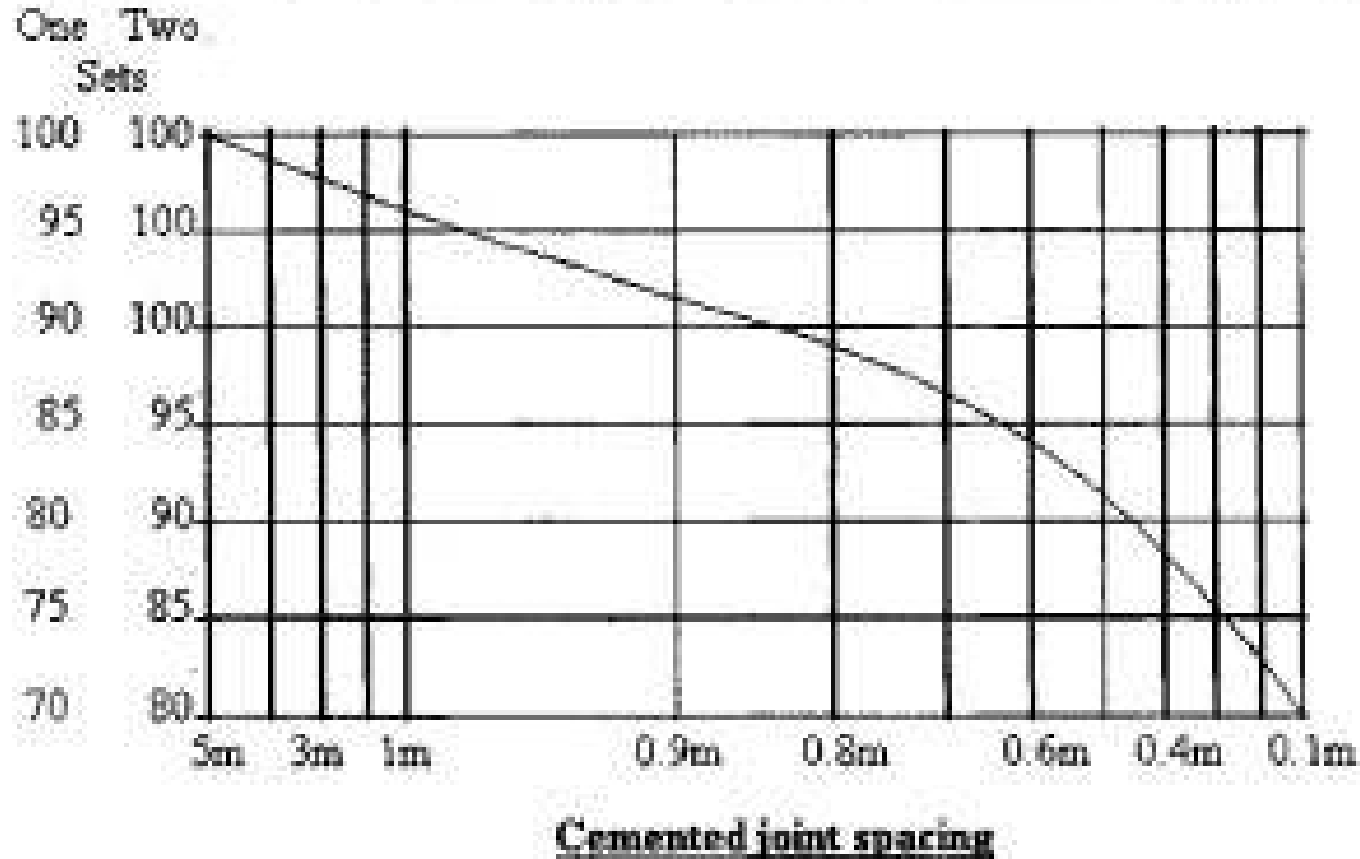
Rock block strength rating as a function of rock block strength

joint spacing ratings



Adjustment factor for cemented joints

% Adjustment to the open joint spacing rating for one or two filled joints in a three joint set



For example, if the rating for two open joints at 0.5-m spacing were 23,

an additional cemented joint with a spacing of 0.85 m would have an adjustment of 90%.

The final rating would be 21, which is equivalent to a three-joint set with an average spacing of 0.65 m.

TABLE 57.1 Joint condition adjustments

A. Large-scale joint expression		Adjustment % of 40
Wavy—multidirectional		100
Wavy—unidirectional		95
Curved		90
Straight, slight undulation		85
B. Small-scale joint expression (200 mm × 200 mm)		
Rough stepped/irregular		95
Smooth stepped		90
Slickensided stepped		85
Rough undulating		80
Smooth undulating		75
Slickensided undulating		70
Rough planar		65
Smooth planar		60
Polished		55
C. Joint wall alteration weaker than sidewall and filling		75
D. Gouge		
Thickness < amplitudes		60
Thickness > amplitudes		30
E. Cemented/filled joints—cement weaker than wall rock. The percentage in the column is the adjustment to obtain the cemented filled-joint condition rating		

Hardness

Adjustment:

5

95%

4

90%

3

85%

2

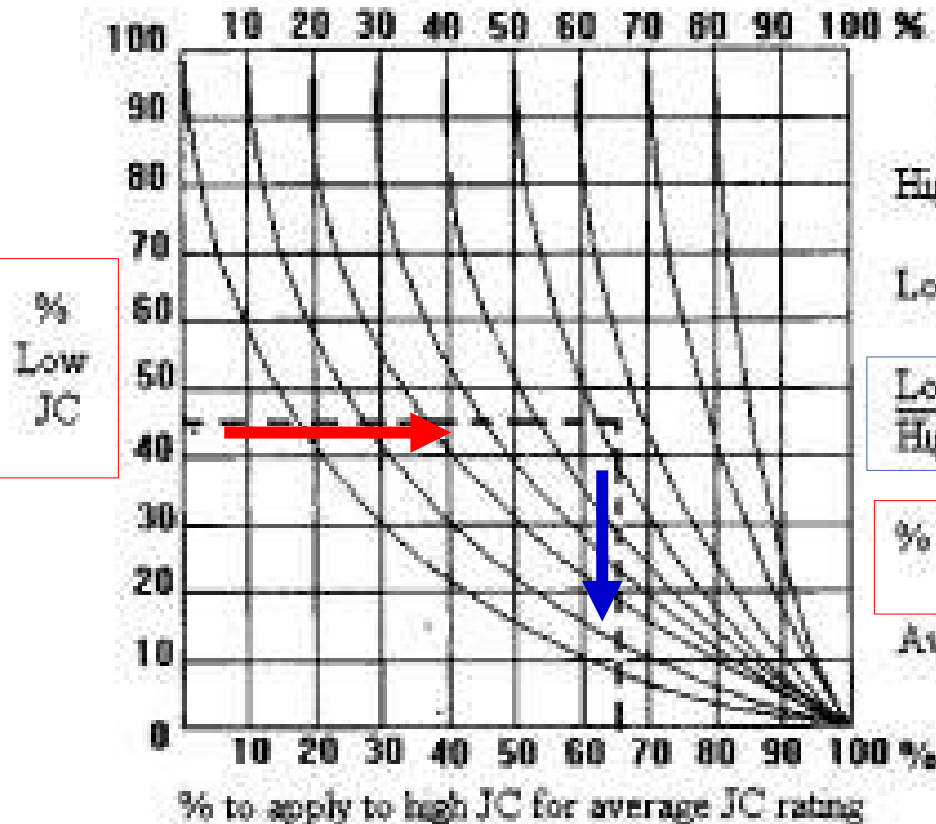
80%

1

75%

Chart for considering multiple joint sets

Select curve using lowest joint condition rating as % of highest joint condition rating



EXAMPLE:

High JC rating = 38

Low JC rating = 20

$$\frac{\text{Low JC rating}}{\text{High JC rating}} \times 100 = 53\%$$

$$\% \text{ low JC} = \frac{\text{Low JC joints (3)}}{\text{High JC joints (7)}} = 43\%$$

$$\text{Av. JC rating} = 65\% \times 38 = \underline{25}$$

ROCK-MASS VALUES

Rock-Mass Rating

The RMR is defined as:

$$\text{RMR} = \text{RBS rating} + \text{Overall joint rating}$$

Rock-Mass Strength

$$\text{RMS} = (A - B)/70 (100 - \text{RBS rating}) \times C$$

where:

A = RMR value

B = RBS rating

C = RBS value in MPa

MRMR ADJUSTMENT PROCEDURE

- The RMR rating is multiplied by an adjustment factor to give the MRMR rating.
- For example, a bad blasting adjustment would apply in a low-stress area, but in a high-stress area, the damage from the stresses would exceed that of the blasting, and the only adjustment would be the mining induced stress.
- The MRMR for a **cavability assessment** would **not have blasting as an adjustment**, **nor would it have weathering** unless the weathering effects were so rapid as to exceed the rate of cave propagation as a result of the structural and stress effects.
- The joint-orientation and mining-induced stress adjustments tend to complement each other. **The object of the adjustments is for the geologist, rock mechanics engineer, and planning engineer to adjust the RMR so the MRMR is a realistic number, reflecting the RMS for that particular mining situation.**

Adjustments for weathering

It does not refer to the existing weathered state of the rock,

TABLE 57.2 Adjustments for weathering

Description	Potential weathering and % adjustments				
	6 months	1 year	2 years	3 years	4+ years
Fresh	100	100	100	100	100
Slightly	88	90	92	94	96
Moderately	82	84	86	88	90
Highly	70	72	74	76	78
Completely	54	56	58	60	62
Residual soil	30	32	34	36	38

Certain rock types weather readily, and this must be taken into consideration in terms of life and size of opening and the support Design.

In the case of the fast-weathering kimberlites, for example, the rock surface needs to be sealed.

The weathering adjustment refers to the anticipated change in RMS as the weathering process alters the exposed surfaces and joint fillings.

Joint-Orientation Adjustment

TABLE 57.3 Joint-adjustment factors

Number of joints defining the block	Number of faces inclined from vertical	Orientation % adjustments for ranges in joint condition		
		0–15	16–30	31–40
3	3	70	80	95
	2	80	90	95
4	4	70	80	90
	3	75	80	95
	2	85	90	95
5	5	70	75	80
	4	75	80	85
	3	80	85	90
	2	85	90	95
	1	90	95	

- attitude of the joints, with respect to the vertical axis of the block,
- the frictional properties of the joints, and
- whether the bases of the rock blocks are exposed,

have a considerable influence on stability; and the RMR rating must be adjusted accordingly

- The adjustment for the **orientation of shear zones** at an angle to the development is:

$$0^{\circ}-15^{\circ} = 76\%$$

$$16^{\circ}-45^{\circ} = 84\%$$

$$46^{\circ}-75^{\circ} = 92\%$$

- Advancing tunnel faces in the direction of the dip is preferable to advancing against the dip, as it is easier to support blocks with joints dipping in the direction of advance.
- **An adjustment of 90% should be made when the advance is into the dip of a set (or sets) of joints.**

Adjustments to Mining Induced Stresses

The orientation, magnitude, and ratio of the field stresses should be known either from stress measurements and/or stress analyses.

*A compressive stress at a large angle to structures will increase the stability of the rock mass and inhibit caving. In this case, the adjustment factor is **120%**. This was the situation in a caving operation where the back was stable and caving only occurred when adjacent mining removed the high horizontal stress. Stresses at a low angle will result in shear failure and have an adjustment of **70%**.*

Verbatim from original paper

- The adjustment for high stresses that cause rock failure can be as low as 60%.
- A classic example of this was on a mine where the RMR was 60 in the low-stress area, but the same rock mass in a high-stress area was classified as having an RMR of 40. The 40 is not the RMR, but the MRMR, and the adjustment in this case is $40/60 = 67\%$.

TABLE 57.4 Blasting adjustment factors

Technique	Adjustment (%)
Boring	100
Smooth-wall blasting	97
Good conventional blasting	94
Poor blasting	80

TABLE 57.5 Water adjustment factors

Molst	Moderate pressure—1–5 MPa 25–125 l/m	High pressure — >5 MPa >125 l/m
95% to 90%	90% to 80%	80% to 70%

DESIGN RATINGS AND STRENGTHS

Design Rating MRMR

The MRMR is used for design. This is the RMR value as adjusted for weathering, orientation, induced stress, blasting, and water.

$$\mathbf{MRMR = RMR \times \text{adjustment factors}}$$

Design Rock Mass Strength

The design rock mass strength (DRMS) is the RMS reduced by the same adjustment factor relating the RMR to the MRMR. In the case where:

$$\text{RMS} = 40 \text{ MPa}$$

$$\text{RMR} = 50$$

$$\text{MRMR} = 40,$$

then the DRMS would be:

$$\text{DRMS} = \text{RMS} \times \text{MRMR/RMR} = 40/50 \times 40 = 32 \text{ MPa.}$$

PRESENTATION

The RMR data **should be plotted on plans and sections**. The range of 0-100 covers all variations in jointed rock masses from very poor to very good. The classification is divided into **five classes of 20 rating with A and B subdivisions**. A colour scheme is used to denote the classes on the plan with **full colour for the A subdivision** and **cross-hatched for the B subdivision**. The colours are:

Class 1	Class 2	Class 3	Class 4	Class 5
(100–81)	(80–61)	(60–41)	(40–21)	(20–0)
Blue	Green	Yellow	Red	Brown

SCALE EFFECTS

- when examining mass-mining methods. For example, the stability/cavability of a deposit cannot be based on the MRMR from a drift assessment alone, as widely spaced major structures play a significant role.
- Structures at a spacing of 10 m would have a marginal effect on the overall IRMR value obtained from drift mapping, but would have a large effect on the cavability of an orebody by providing planes along which displacements can occur.

The MRMR value gives a hydraulic radius (HR) required for assuring cavability. This figure should be adjusted for the influence of major structures by using the following procedure to obtain an "influence" number.

Rankings

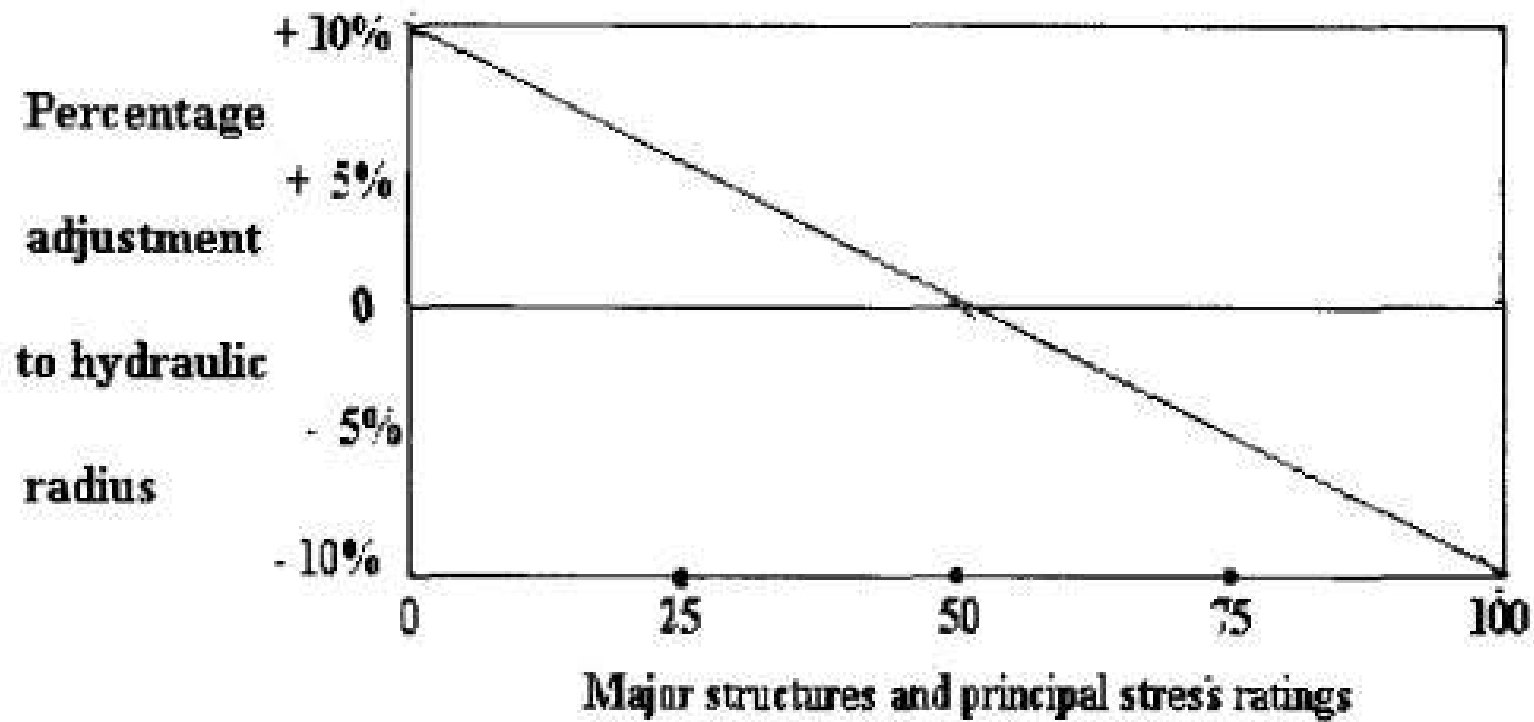
- A **Dip:** $0^{\circ}-20^{\circ} = 6$, $21^{\circ}-40^{\circ} = 4$, $31^{\circ}-40^{\circ} = 2$, $41^{\circ}-60^{\circ} = 1$, $>61^{\circ} = 0$
- B **Spacing:** $0-9\text{ m} = 6$, $10-15\text{ m} = 4$, $16-21\text{ m} = 3$, $22-27\text{ m} = 1$, $>27\text{ m} = 0$
- C **Joint Condition:** $0-10 = 6$, $10-15 = 4$, $15-20 = 2$, $20-25 = 1$, $>25 = 0$
- D **Stress/structure orientation:** $0^{\circ}-20^{\circ} = 7$, $21^{\circ}-30^{\circ} = 9$, $31^{\circ}-40^{\circ} = 6$, $41^{\circ}-50^{\circ} = 3$, $51^{\circ}-60^{\circ} = 2$, $61^{\circ}-70^{\circ} = 1$, $>71 = 0$
- E **Distance of major structures from undercut boundaries:** $0-9\text{ m} = 12$, $10-20\text{ m} = 8$, $21-30\text{ m} = 2$, $>31\text{ m} = 0$
- F **Stress values—Sigma 1 as % of RMS:** $>100\% = 14$, $80\%-99\% = 12$, $60\%-79\% = 8$, $40\%-59\% = 4$, $20\%-39\% = 2$, $<20\% = 0$

The rankings are plotted in Table

TABLE 57.6 Form for determining the major structure influence number

Major structures	A	B	C	D	E	F	Total
Set 1							
Set 2							
Set 3							
Total							

Adjustment factor for major structures



Applications of MRMR

Table 10 Support Guide for Tunnels using MRMR

MRMR	RMR									
	1A	1B	2A	2B	3A	3B	4A	4B	5A	5B
	← rock reinforcement-plastic deformation →									
1A										
1B										
2A	a									
2B	a	a								
3A	b	b	a	a						
3B	b	b	b	b	b	d				
4A	r	r	c	c	c	d	d			
4B				d	e	f	f	c + l		
5A						f/p	h + f/p	h + f/l	h + f/l	
5B							h + f/p	f/o	t	t

Alphabets described in next slide

(i) Rock reinforcement

a - local bolting at joint intersections

b - bolts at 1 m spacing

c - b and straps and mesh if rock is finely jointed

d - b and mesh/steel fiber-reinforced shotcrete, bolts as lateral restraint

e - d and straps in contact with or shotcreted in

f - e and cable bolts as reinforcing and lateral restraint

g - f and pinning

h - spiling

i - grouting

(ii) Rigid lining

j timber

-

k - rigid steel sets

l - massive concrete

m - k and concrete (all low deformation)

n - structurally reinforced concrete.

(iii) Yielding lining (a repair technique of high deformation)

o - yielding steel arches; and

p - yielding steel arches set in concrete/shotcrete.

(iv)

Fill

q - fill.

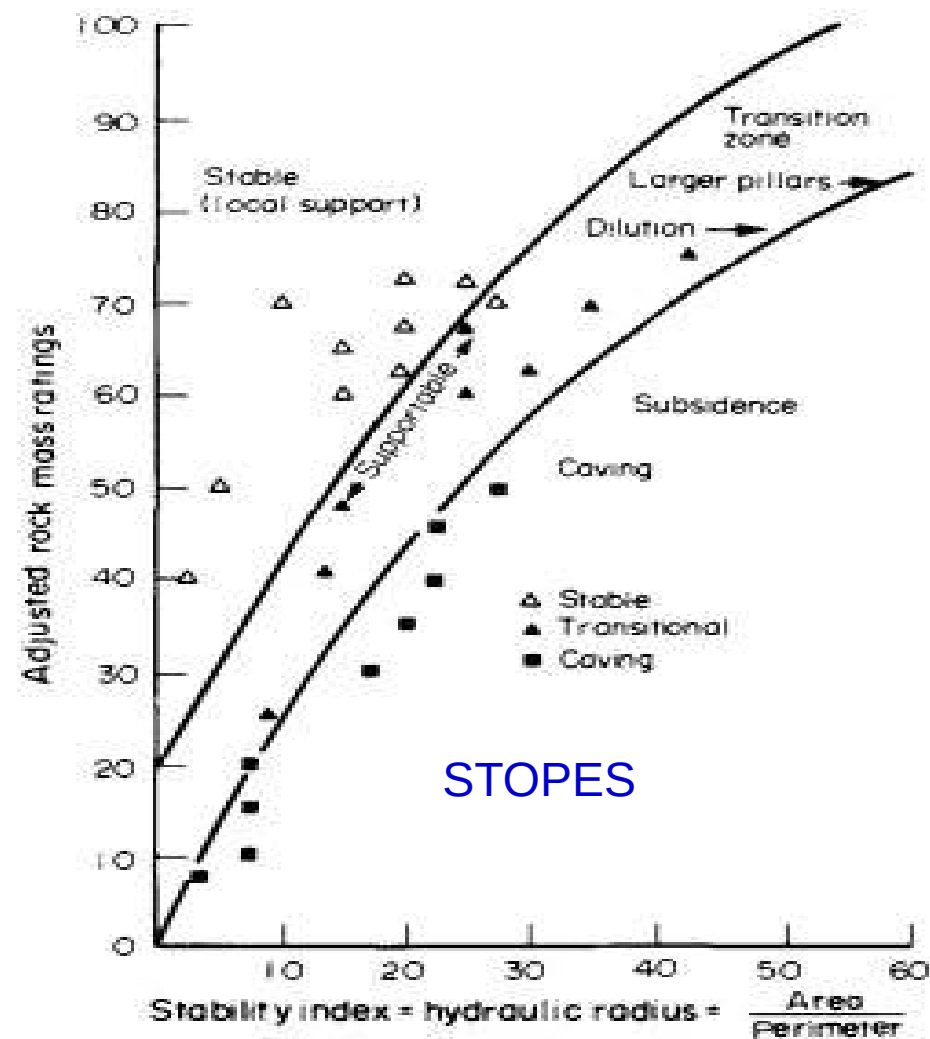
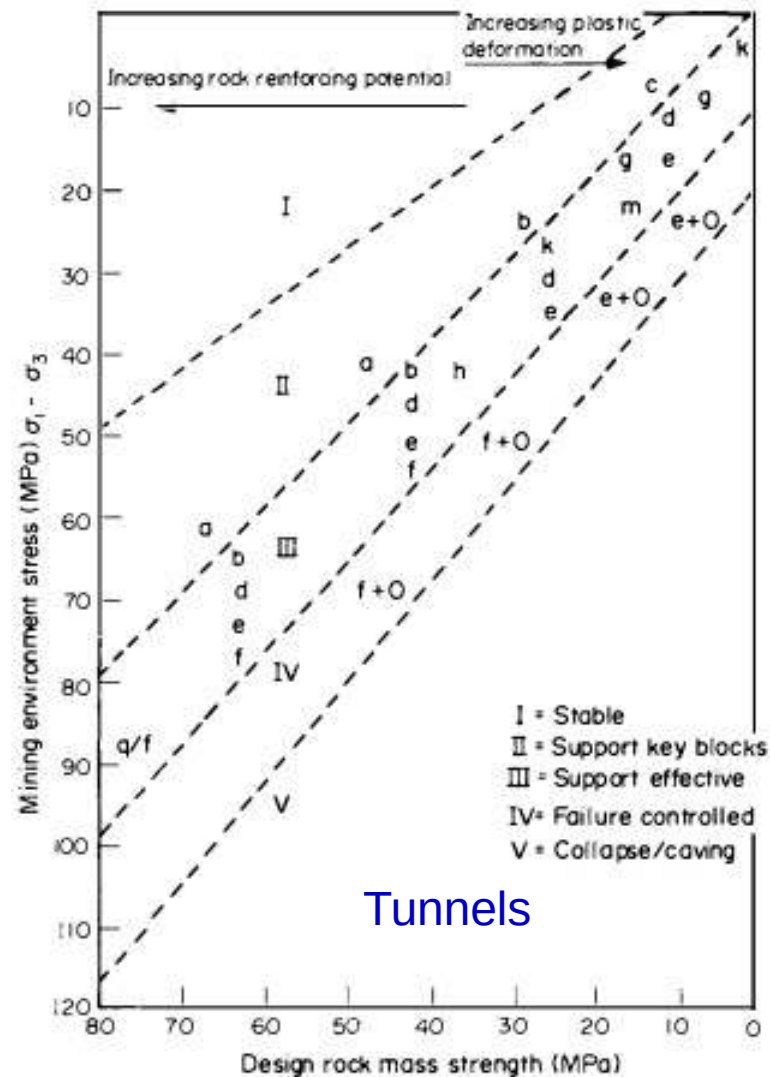
(v) Spalling control

r - bolts and rope laced mesh.

(vi) Rock replacement

s - replace rock with stronger material

t - avoid development if possible



Hoek's
GSI

Geological Strength Index (GSI)

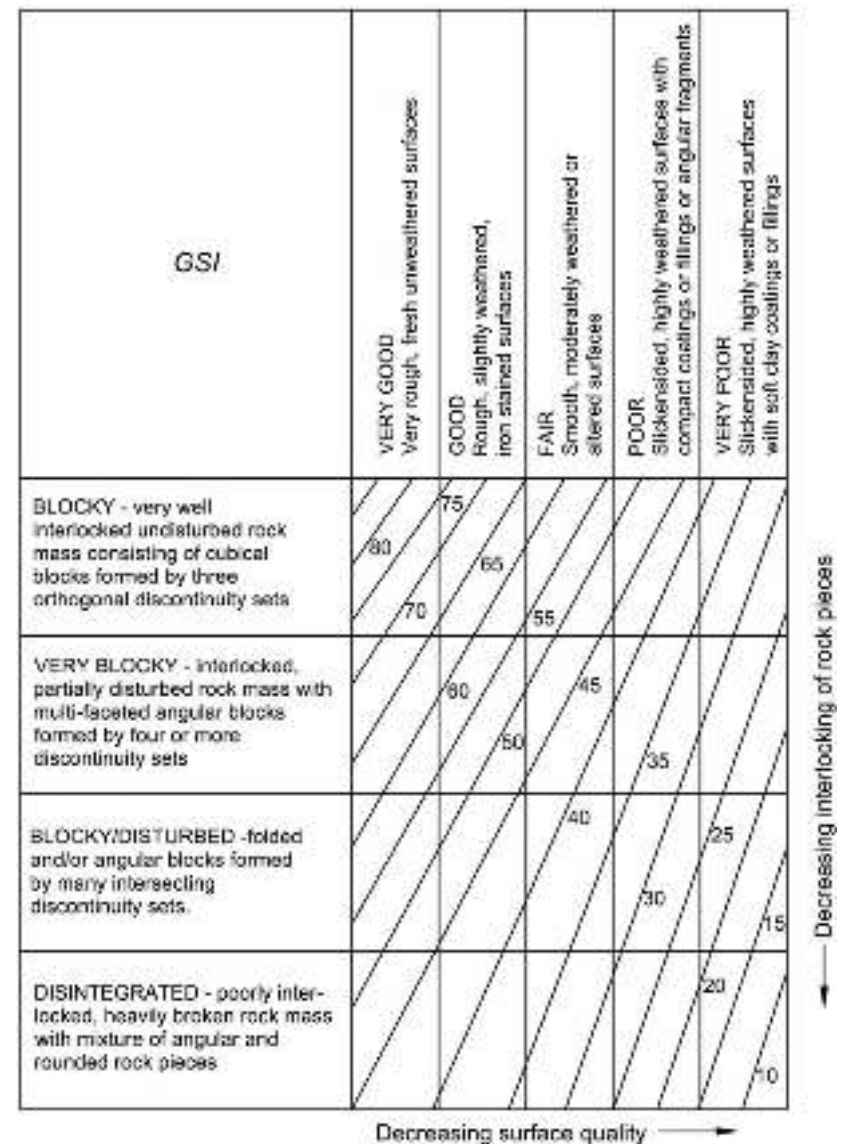
- Hoek et al. (1995) proposed the **GSI chart** for classification of rock mass.
- GSI chart is more suitable for rock mass classification as it overcomes the difficulties that are encountered during the **evaluation of weak rock mass** in RMR (Hoek and Brown, 1997).
- It is broadly used for quick and qualitative assessment of rock mass based on two parameters, **(i) the structure of the rock mass and (ii) the condition of the discontinuities..**

Geological Strength Index (GSI)

- Initially, the chart had four categories of rock mass structure and five categories of surface conditions.
- Later, it was modified by Hoek et al. (1998) and an additional category of foliated/laminated rock type was added into the rock mass structure.
- After a year, one more category was introduced by Hoek (1999) to accommodate massive or intact rocks.

GSI Chart

- It is one of the early chart provided by Hoek and Brown (2002).
- The chart have four categories of rock mass structure namely *blocky*, *very-blocky*, *blocky-disturbed* and *disintegrated*.
- The five categories of surface conditions are *very-good*, *good*, *fair*, *poor* and *very-poor*.



Geological Strength Index (GSI)

- The main job of GSI is to **calculate the rock mass strength** using Hoek-Brown failure criteria (Hoek et al., 2002) and **modulus of deformation** (Hoek and Diederichs, 2006).
- The relationships between GSI, RMR and Q was given by Hoek et al. (1995) as shown below.
- **$GSI = RMR_{89} - 5; \quad [for\ GSI \geq 18\ or\ RMR \geq 23]$**
- **$GSI = 9 \ln Q + 44 \quad [for\ GSI \geq 18] \quad (For\ poor\ rock\ masses)$**

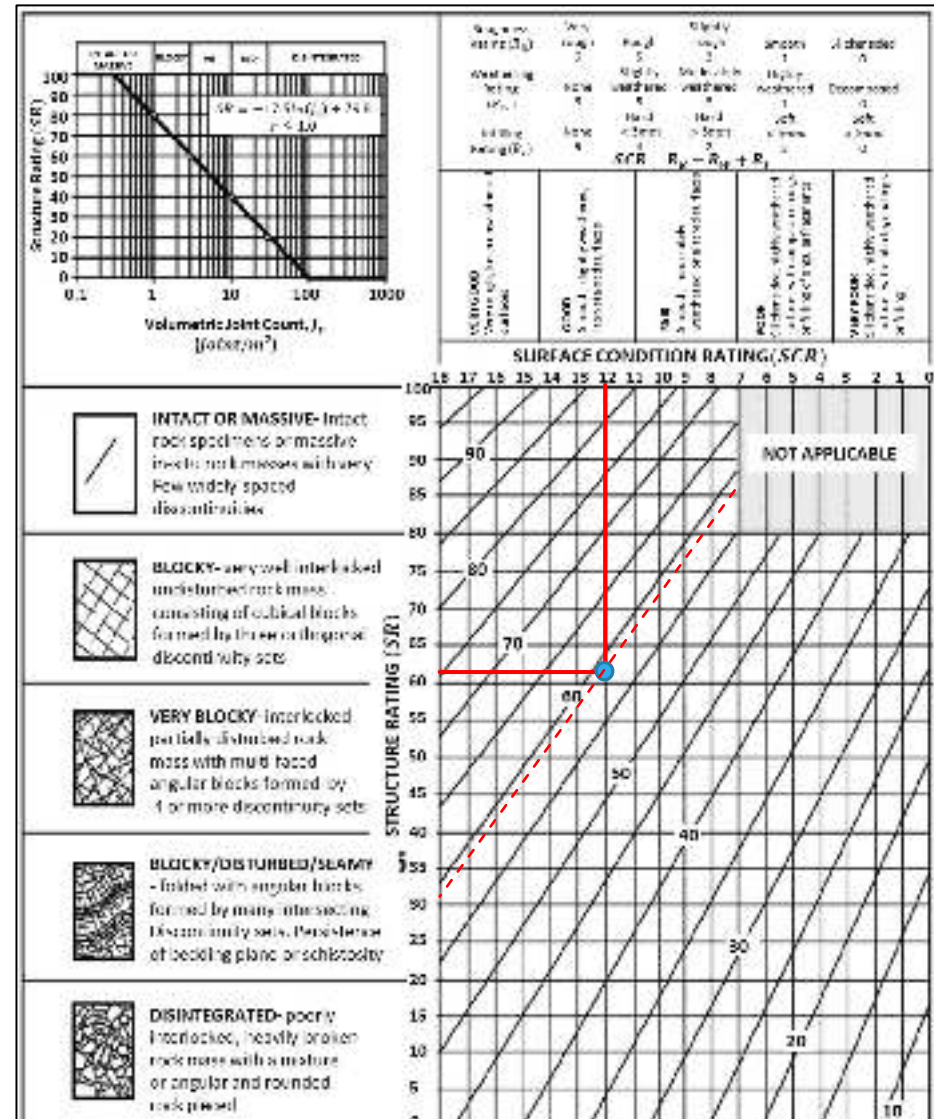
Geological Strength Index (GSI)

- Later, GSI chart was slightly modified by Marinos and Hoek (2000) that had six classes of rock structure namely intact or massive, blocky, very blocky, blocky/folded, crushed/disintegrated and laminated/sheared using five categories of surface conditions of discontinuities namely very-good, good, fair, poor and very-poor.
- The purpose of Marinos and Hoek (2000) chart is to provide a range of GSI based on visual inspection by an expert.
- GSI chart was further simplified by Somnez and Ulusey (2002), Cai et al. (2004), Russo (2009) and Hoek et al. (2013) to obtain values based on field measurements in case of difficulty in using the chart provided by Marinos and Hoek (2000).

GSI Chart

by Somnez and Ulusey (2002):

- and calculated on the basis of formula given in quantified GSI chart (2002).
- The slope shows ‘Blocky’ rock mass with a ‘good’ surface condition.



Based on this research on the nonlinear Griffith failure criterion, Hoek and Brown (1980) proposed the following empirical equation to fit the results of a wide range of triaxial tests on intact rock samples:

$$\sigma_1 = \sigma_3 + \sigma_{ci} \sqrt{m_i \frac{\sigma_3}{\sigma_{ci}} + 1} \quad (1)$$

The generalized Hoek & Brown criterion for the estimation of rock mass strength. introduced by Hoek (1994)

$$\sigma_1 = \sigma_3 + \sigma_{ci} \left(m_b \frac{\sigma_3}{\sigma_{ci}} + s \right)^a \quad (3)$$

where m_b , s , and a are the rock mass material constants, given by

$$m_b = m_i \exp[(GSI - 100)/(28 - 14D)] \quad (4)$$

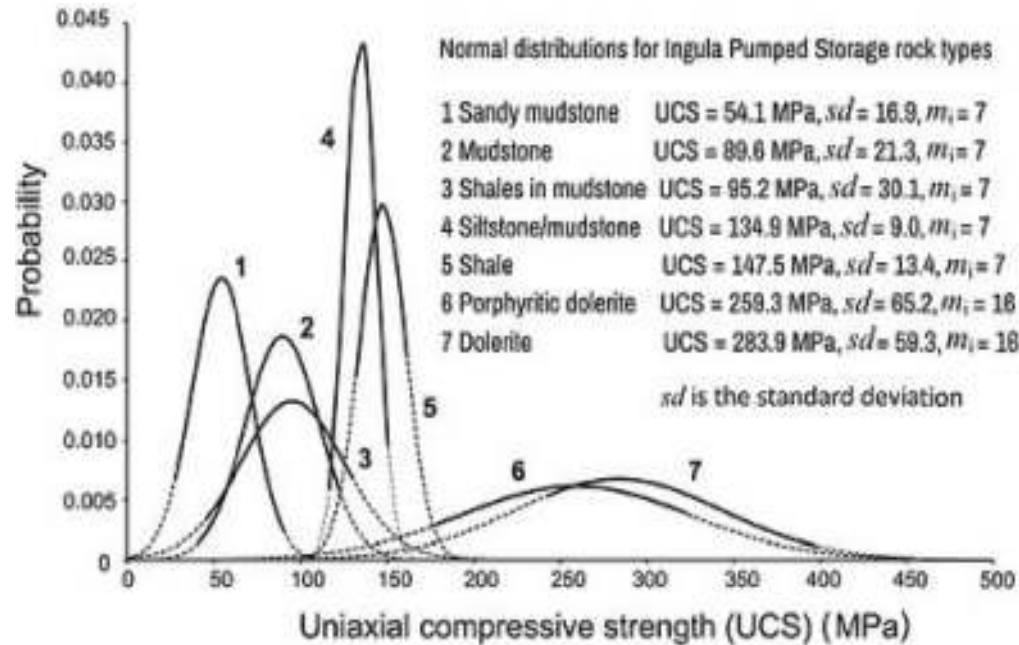
$$s = \exp[(GSI - 100)/(9 - 3D)] \quad (5)$$

$$a = 1/2 + 1/6 \left(e^{-GSI/15} - e^{-20/3} \right) \quad (6)$$

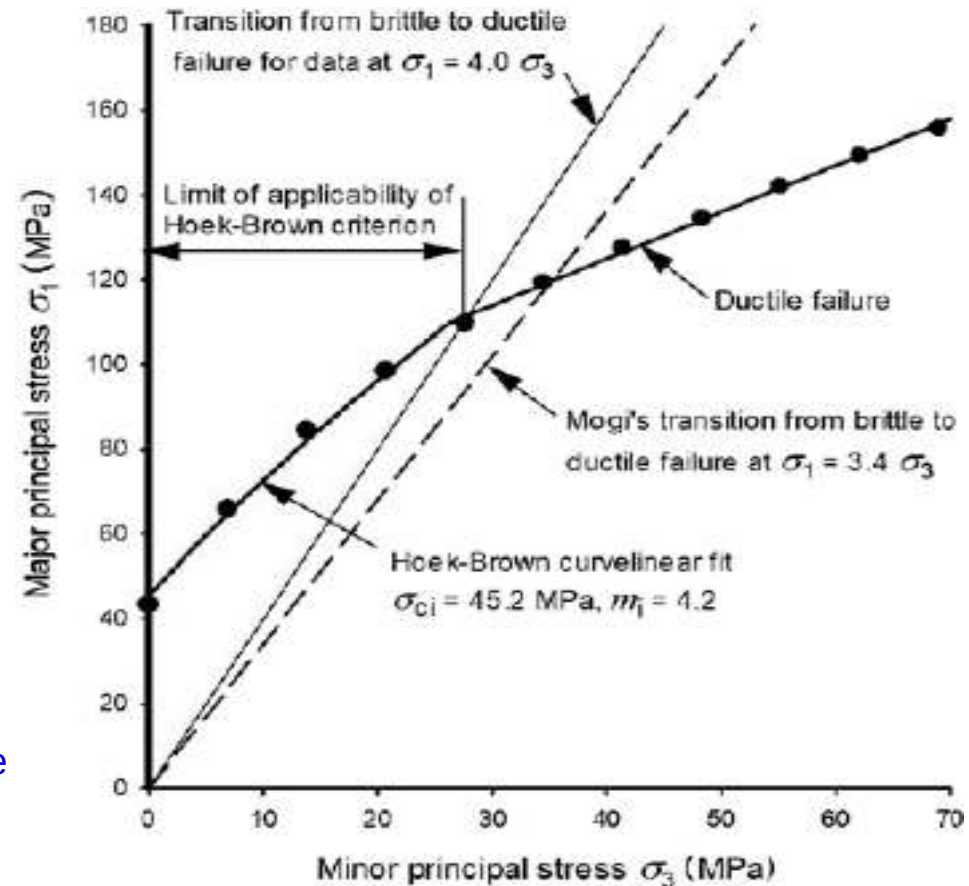
D is a factor which depends upon the degree of disturbance to which the rock mass has been subjected to blast damage and stress relaxation

Strength of intact rock

The constants m_b , s , and 'a' define the shape of the curvilinear failure plot



Mogi (1966) investigated the transition from shear to ductile failure in a wide range of rock types and found that the average transition is defined by $\sigma_1 = 3.4 \sigma_3$. This is a useful guide for the maximum confining pressure for triaxial testing of intact



- Tensile failure ($\sigma_3 < 0$) is not dealt with by the Hoek & Brown criterion. However, tensile failure is an important factor in some rock engineering problems.
- as proposed by Fairhurst (1964), can be generalized in terms of the ratio of compressive to tensile strength, $\sigma_{ci} / |\sigma_t|$, as follows

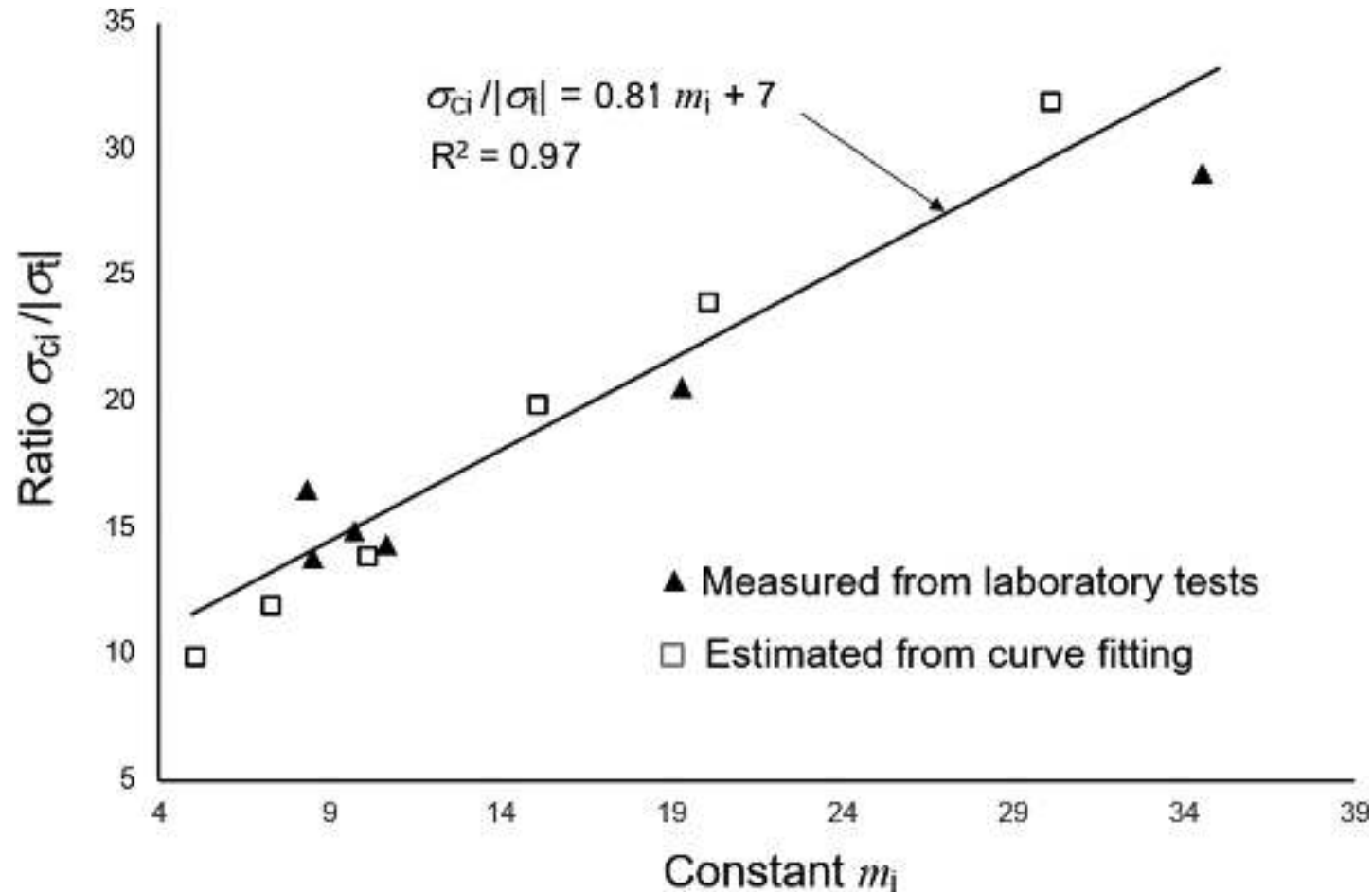
- (1) If $w(w - 2)\sigma_3 + \sigma_1 \leq 0$, failure occurs when $\sigma_3 = \sigma_t$;
- (2) If $w(w - 2)\sigma_3 + \sigma_1 > 0$, failure occurs when

$$\sigma_1 = \frac{(2\sigma_3 - A\sigma_t) + \sqrt{(A\sigma_t - 2\sigma_3)^2 - 4(\sigma_3^2 + A\sigma_t\sigma_3 + 2AB\sigma_t^2)}}{2} \quad (7)$$

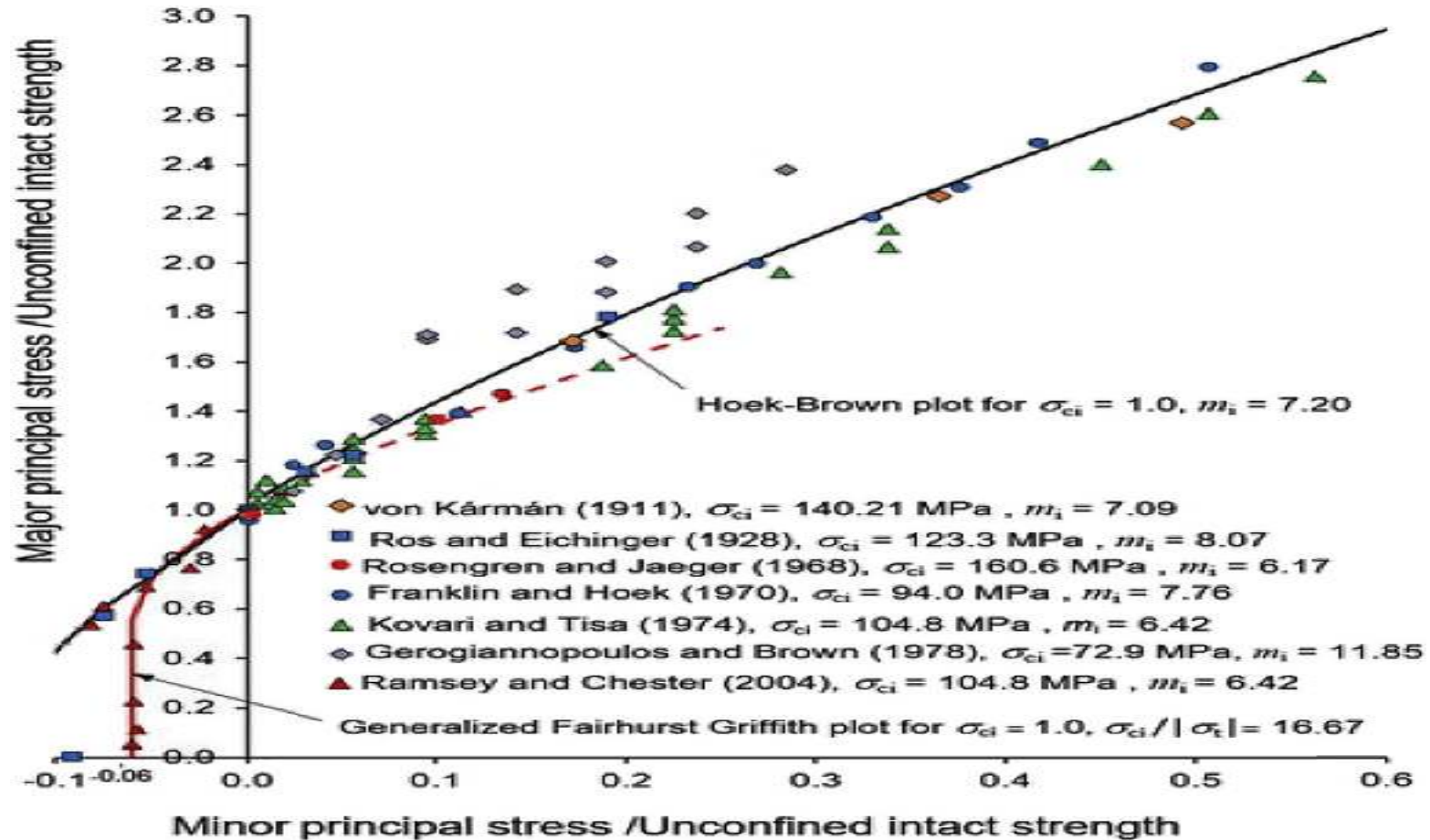
where

$$A = 2(w - 1)^2, B = \left(\frac{w - 1}{2}\right)^2 - 1, w = \sqrt{\frac{\sigma_{ci}}{|\sigma_t|} + 1}$$

As an interim measure, the following approximate relationship between the compressive to tensile strength ratio, $\sigma_{ci} / |\sigma_t|$, and the Hoek & Brown parameter m_i was proposed:



Hoek and Martin (2014) proposed that, for practical rock engineering purposes, a Hoek-Brown failure envelope with a **tensile cut-off**, based on the **generalized Griffith failure criterion theory proposed by Fairhurst (1964)**, can provide an effective solution.



Estimating rock mass deformation modulus

Hoek and Diederichs (2006), using a database of rock mass deformation modulus measurements from projects in China (including Taiwan), proposed the following equation for estimating rock mass modulus

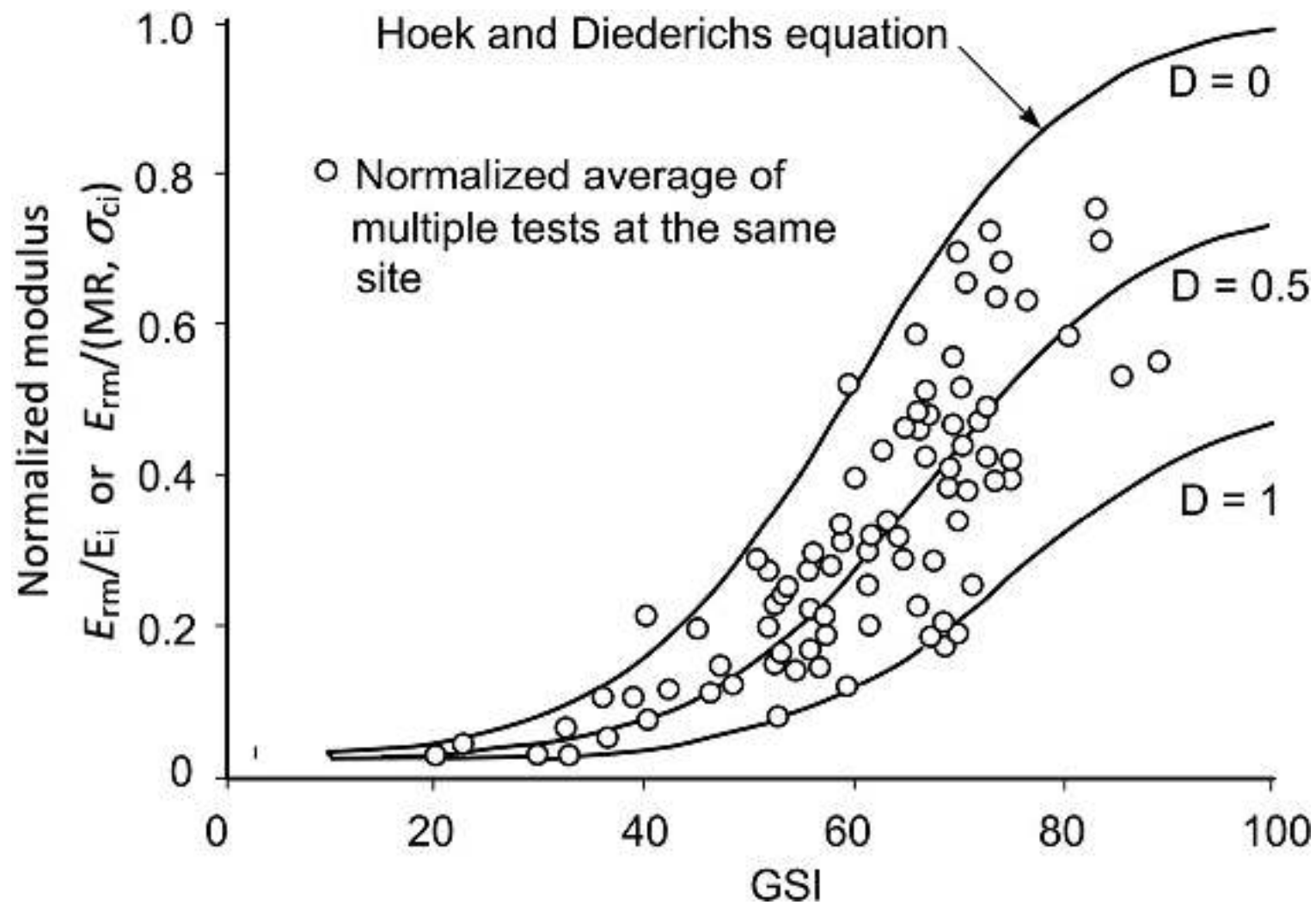
$$E_{rm} = E_i \left\{ 0.02 + \frac{1 - D/2}{1 + \exp[(60 + 15D - GSI)/11]} \right\} \quad (9)$$

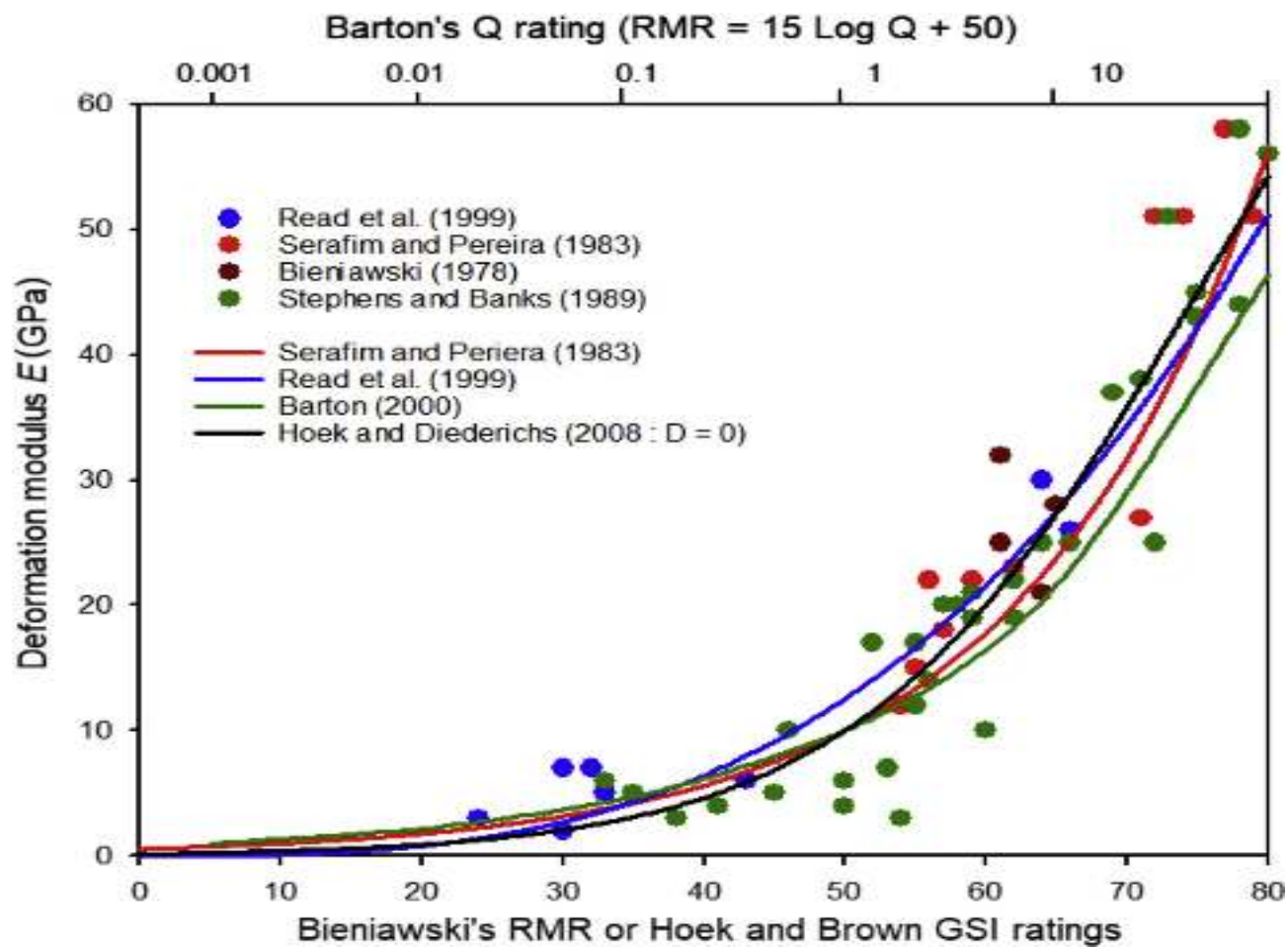
where E_i is the intact rock deformation modulus (MPa).

Estimating rock mass deformation modulus

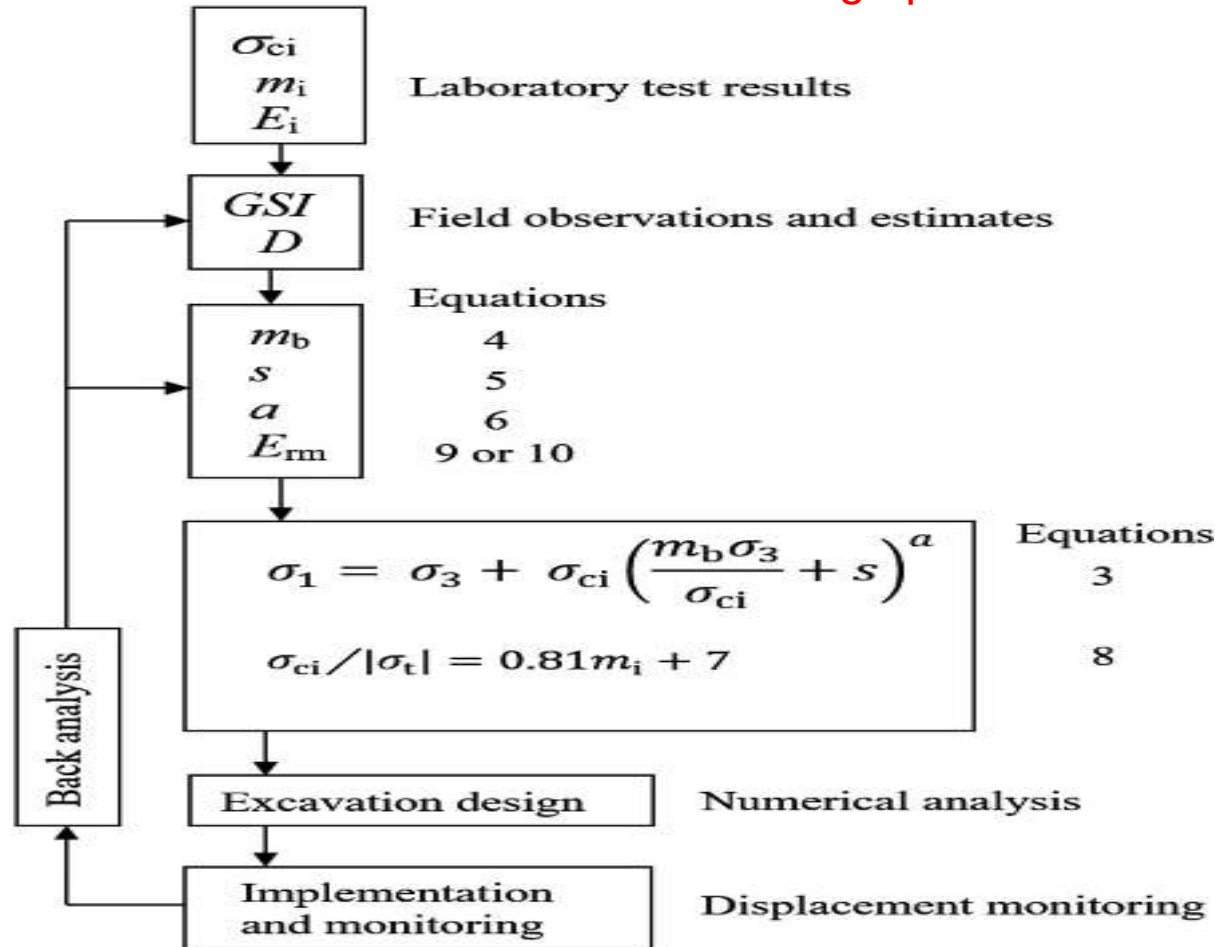
when the laboratory measured values for E_i are not available, the rock mass reduction values (MR) proposed by Deere (1968) can be used for estimating the intact rock modulus. When no information on the intact rock deformation modulus is available, the following alternative equation for estimating the rock mass modulus E_{rm} (MPa) was proposed by Hoek and Diederichs (2006)

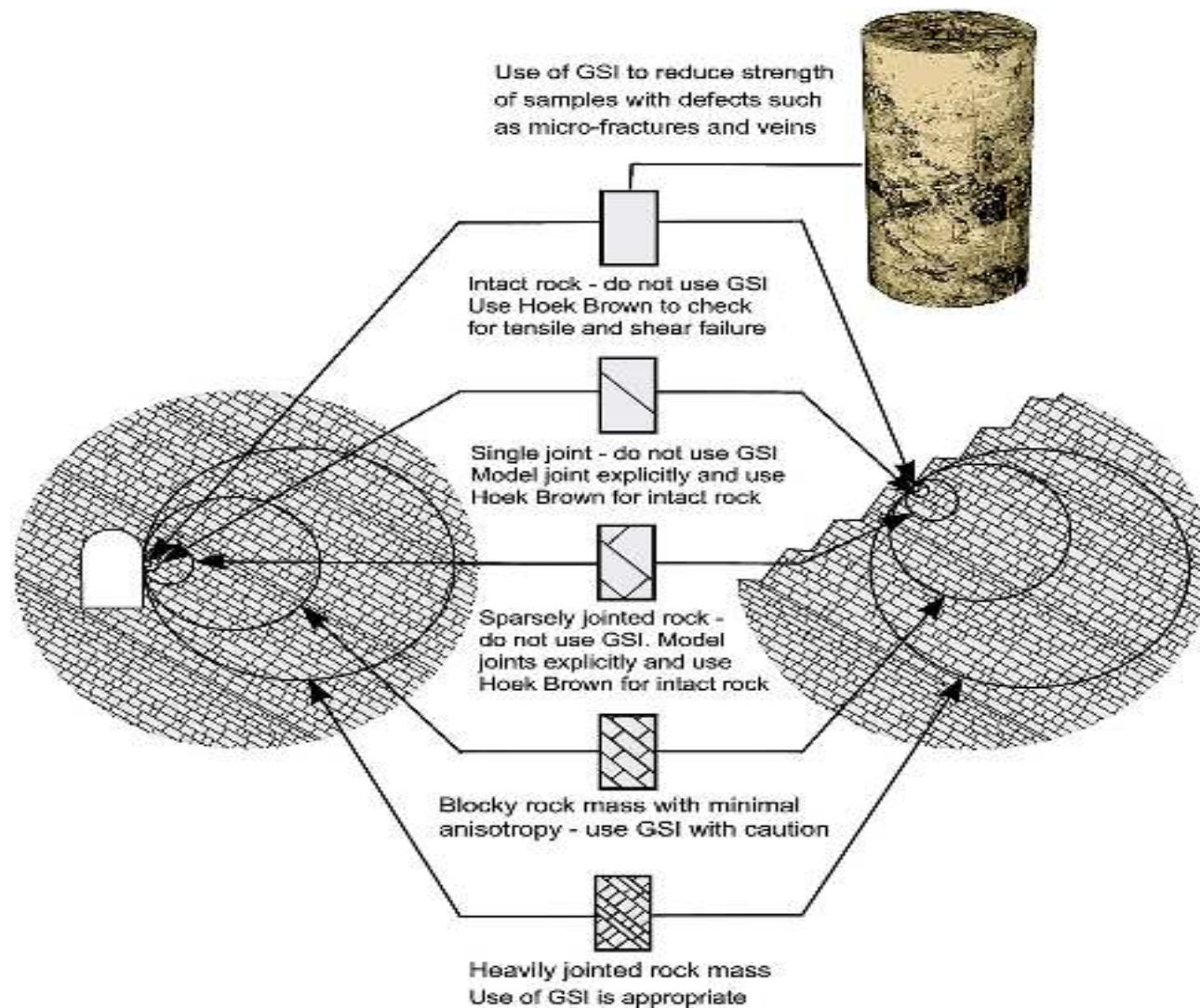
$$E_{rm} = 10^5 \frac{1 - D/2}{1 + \exp[(75 + 25D - GSI)/11]} \quad (10)$$





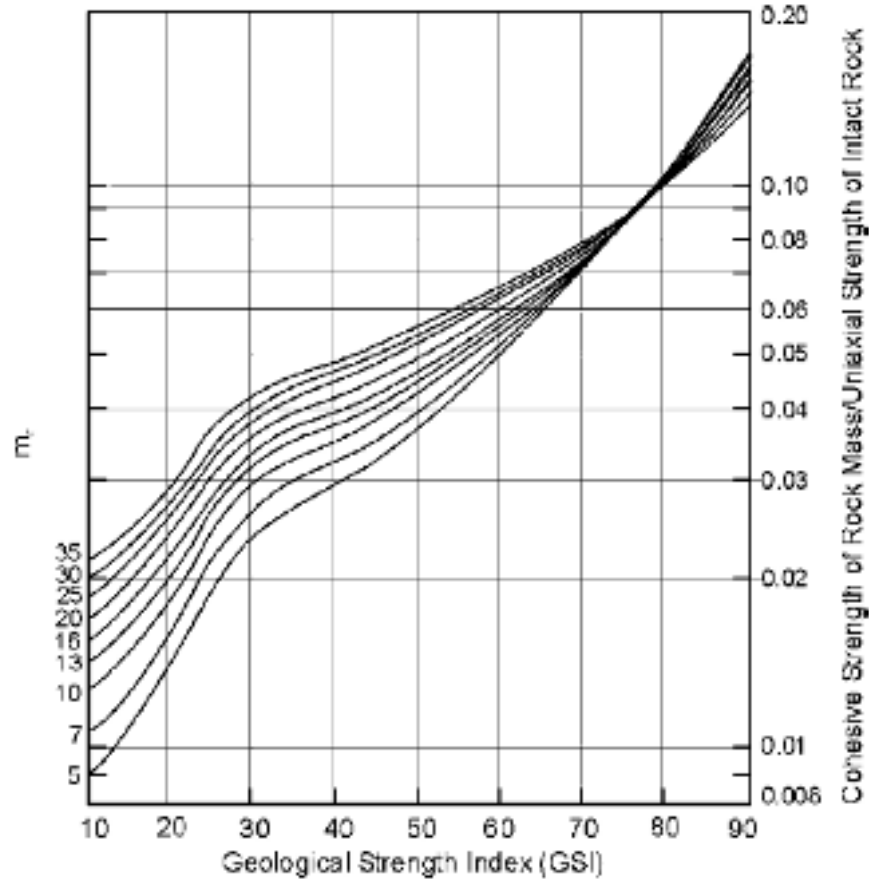
The overall design process



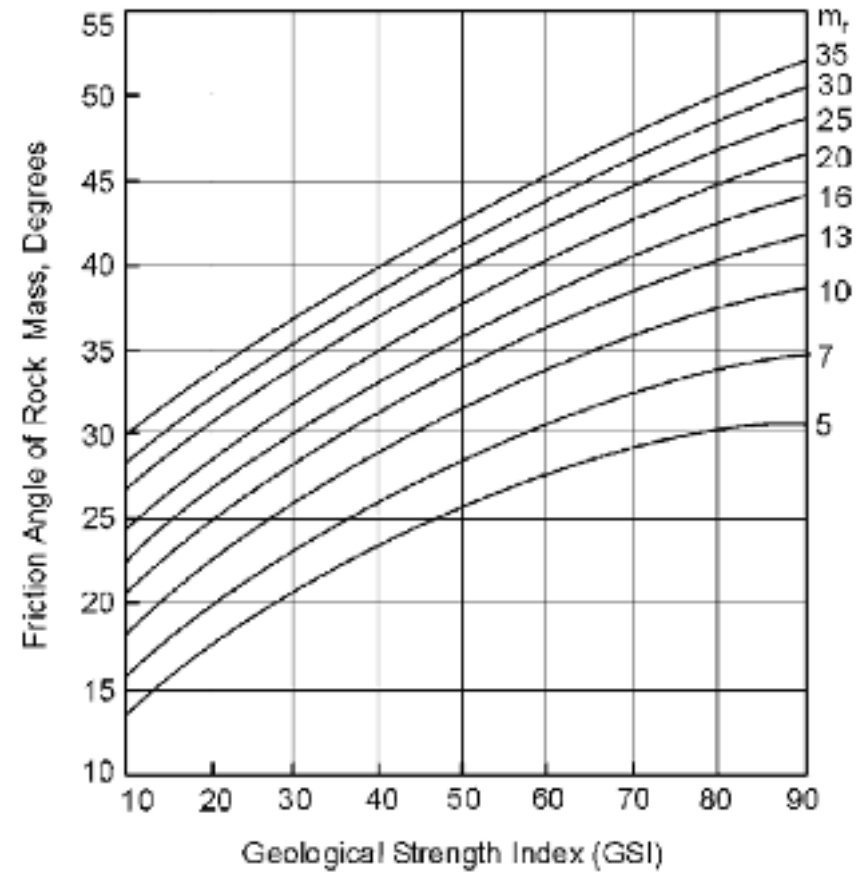


- The influence of size is considered in determining the use of GSI.
- The starting point for this chart is a typical intact rock core, but there is no reason why this starting point should not be the intrablock structure within the core as suggested by Day et al. (2012).
- They emphasized that the reduction of the intact rock strength by this method must be carried out with care to avoid over-penalization of the rock mass strength

MOHR-COULOMB STRENGTH PARAMETERS

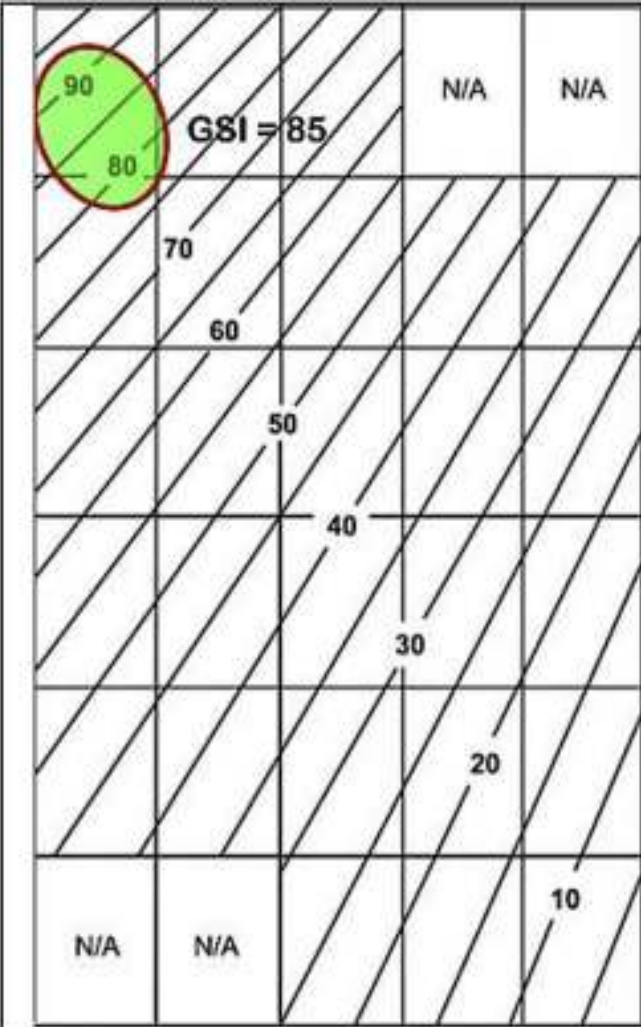


Relationship between ratio of cohesive strength of rock mass to UCS of intact rock (c/q_c) and GSI for different m_r values for $D = 0$. (From Hoek and Brown, 1997)



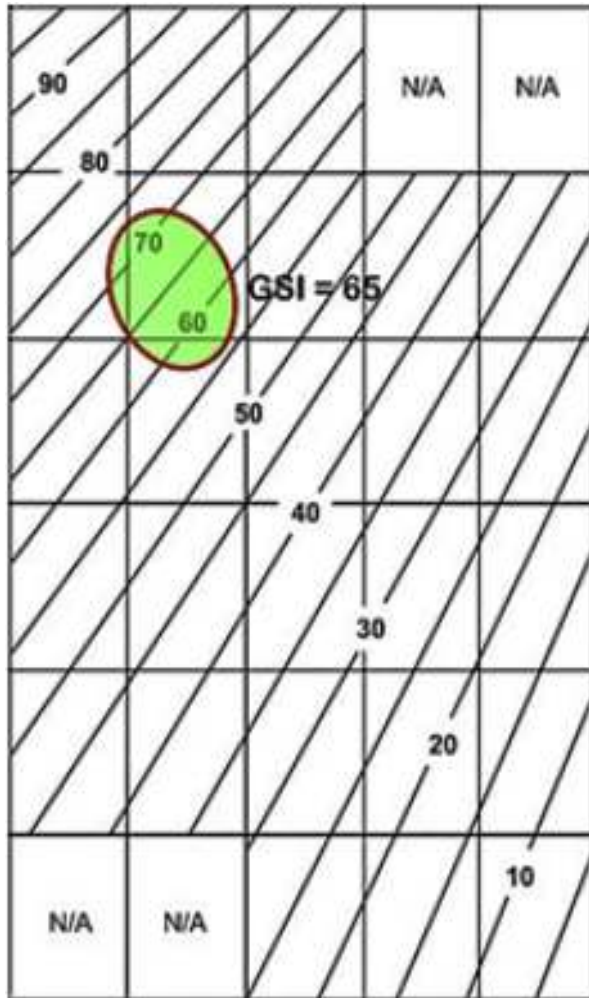
Friction angle (ϕ) of rock mass for $D = 0$ for different GSI and m_r values. (From Hoek and Brown, 1997)

Typical applications of the GSI chart to exposed faces in a range of rock formations.



Spalling in the sidewalls of a mine tunnel in intact hard rock subjected to anisotropic horizontal stresses. GSI is not applicable in the analysis of these stress-induced spalls but it can be used for other applications.

Typical applications of the GSI chart to exposed faces in a range of rock formations.



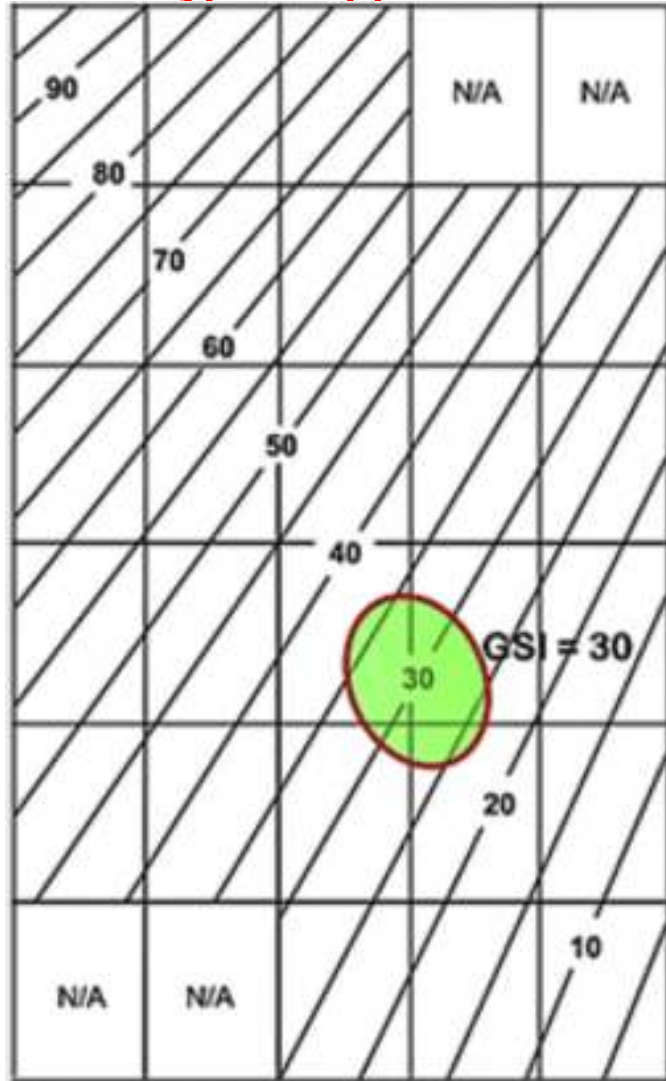
Orthogonal jointing in granitic rock on a dam site. GSI is not applicable on this scale since the stability of the exposed face is controlled by the geometry of intersecting joints. It can be applied to larger scale excavations.

Typical applications of the GSI chart to exposed faces in a range of rock formations.



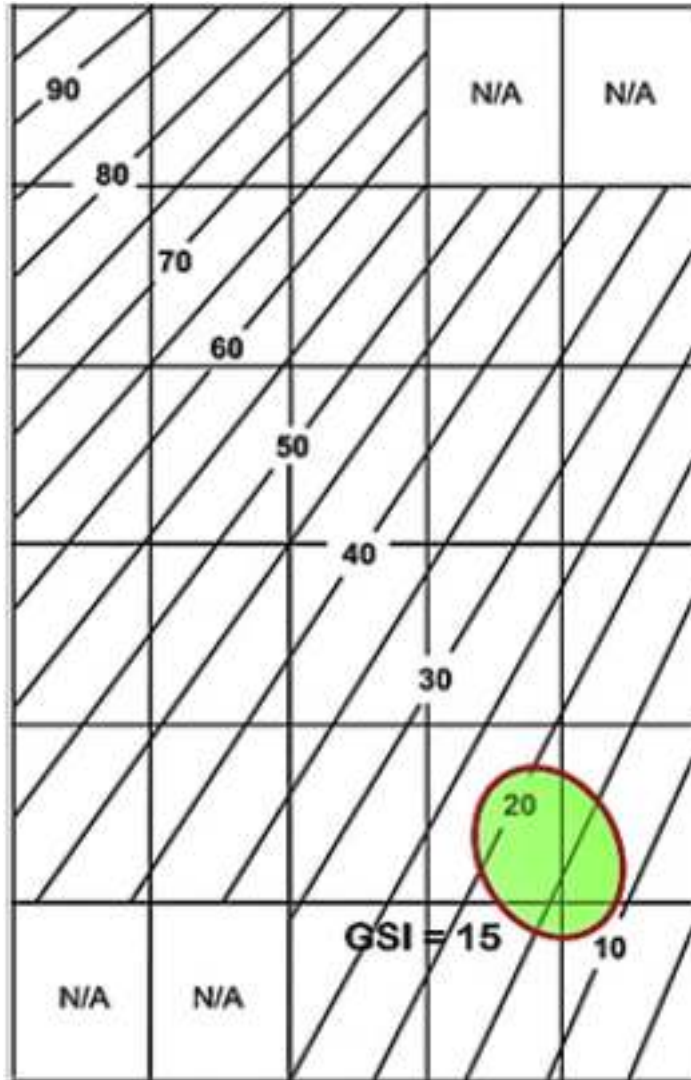
Interlocking angular Andesite blocks defined by several joint sets, exposed in an open pit mine bench. GSI is fully applicable in this situation and on this scale

Typical applications of the GSI chart to exposed faces in a range of rock formations.



Complex folding in a bedded sedimentary deposit. GSI is applicable with care since averaging of the intact properties is required to calculate rock mass properties

Typical applications of the GSI chart to exposed faces in a range of rock formations.



Tectonically deformed sediments with almost complete loss of structural patterns. Care is required in using GSI in this type of rock mass. Use the GSI charts by Marinos et al. (2005) and Marinos (2017).

Palmstrom's RMi

Rock Mass Index (RMi)

- Palmstrom (1995) proposed a rock mass index (RMi) to characterize rock mass strength as a construction material.
- The presence of various defects (discontinuities) in a rock mass that tend to reduce its inherent strength are taken care of in rock mass index (RMi), which is expressed as:

$$RMi = q_c \cdot J_p$$

where,

q_c = uniaxial compressive strength (UCS) of the intact rock material in MPa.

J_p = jointing parameter composed of mainly four jointing characteristics, namely, block volume or density of joints, joint roughness, joint alteration, and joint size.

Selection of Parameters used in RMi:

- study of more than 15 different classification systems have been used by Palmstrom (1995) when selecting the following input parameters for RMi:
 1. Size of the blocks delineated by joints—measured as block volume, V_b
 2. Strength of the block material—measured as UCS, q_c
 3. Shear strength of the block faces—characterized by factors for the joint characteristics, jR and jA (Tables 10.1 and 10.3 *see subsequent slides*)
 4. Size and termination of the joints—given as their length and continuity factor, jL (Table 10.2 *see subsequent slides*)

- The UCS of intact rock, q_c , is defined and can be determined within a reasonable accuracy.
- The jointing parameter (J_P), however, is a combined parameter made up of the block volume, V_b , which can be found from field measurements, and the joint condition factor, J_C , which is the result of three independent joint parameters (roughness, alteration, and size),
 - It has been shown by Palmstrom (2005), the correlation between the block volume (V_b) and the volumetric joint count (J_v) {see the RQD slides} is

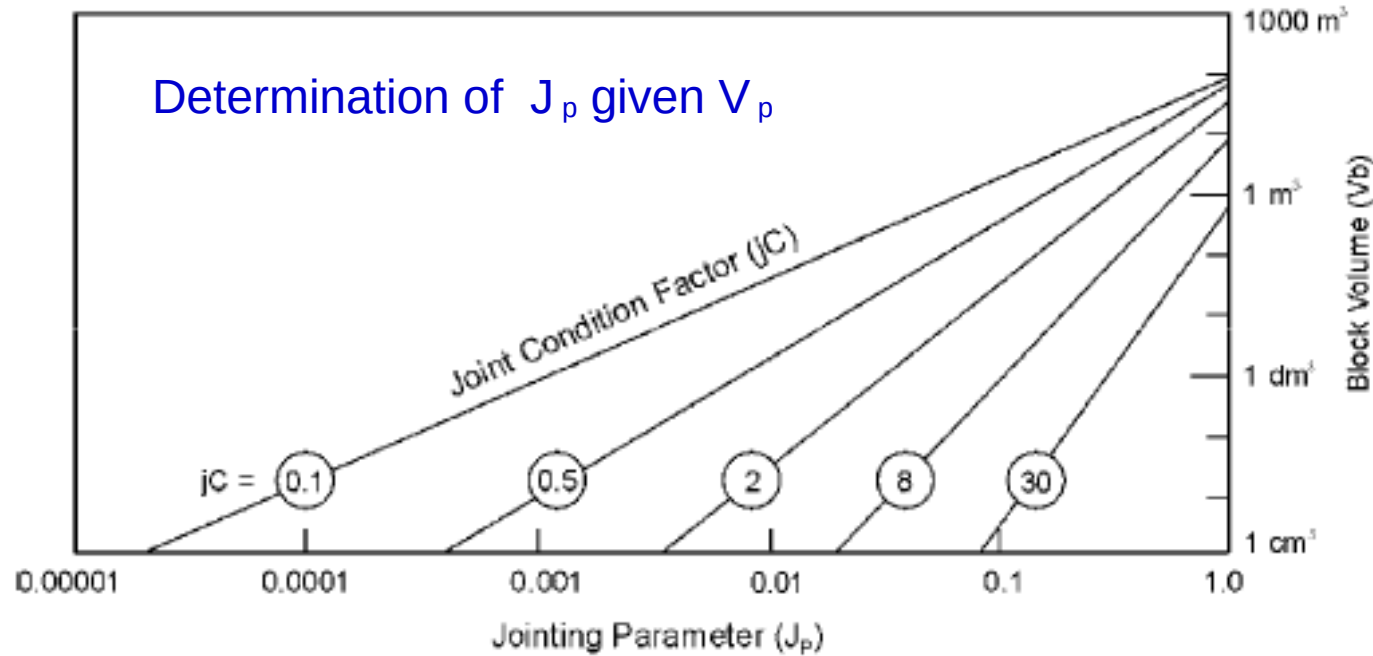
$$V_b = \beta (J_v)^{-3}$$

- where β is the block shape factor, having the following characterization:
 - For equi dimensional (cubical or compact) blocks $\beta = 27$
 - For slightly long (prismatic) and for slightly flat (tabular) blocks $\beta = 28-32$
 - For moderately long and for moderately flat blocks $\beta = 33-59$
 - For long and for flat blocks $\beta = 60-200$
 - For very long and for very flat blocks $\beta > 200$.
 - A common value for $\beta = 36$.

Palmstrom (2005) has shown that the block shape factor (β) may crudely be estimated from

$$\beta = 20 + 7(a_3)/(a_1)$$

where a_1 and a_3 are the shortest and longest dimensions of the block.



$$J_p = 0.2 (jC)^{0.5} \cdot (V_b)^D$$

V_b is given in m^3

$$D = 0.37 \cdot jC^{-0.2}$$

$$jC = jL(jR/jA)$$

Please refer following tables to calculate jC

TABLE 10.2 The Joint Length and Continuity Factor (jL)

TABLE 10.1 The Joint Roughness (jR) Found from Smoothness and Waviness

TABLE 10.3 Characterization and Rating of the Joint Alteration Factor (jA)

TABLE 10.2 The Joint Length and Continuity Factor (jL)

Joint length (m)	Term	Type	jL	
			<i>Continuous joints</i>	<i>Discontinuous joints**</i>
<0.5	Very short	Bedding/foliation parting	3	6
0.1–1.0	Short/small	Joint	2	4
1–10	Medium	Joint	1	2
10–30	Long/large	Joint	0.75	1.5
>30	Very long/ large	Filled joint, seam or shear*	0.5	1

*Often a singularity (special feature), and should in these cases be treated separately.

**Discontinuous joints end in massive rock mass.

Source: Palmstrom, 1996, 2000.

TABLE 10.1 The Joint Roughness Found from Smoothness and Waviness

Small-scale smoothness* of joint surface	Large-scale waviness of joint plane				
	Planar	Slightly undulating	Undulating	Strongly undulating	Stepped or interlocking
(The ratings in bold are similar to Jr in the Q-system)					
Very rough	2	3	4	6	6
Rough	1.5	2	3	4.5	6
Smooth	1	1.5	2	3	4
Polished or slickensided*	0.5	1	1.5	2	3

For filled joints: $jR = 1$; for irregular joints a rating of $jR = 6$ is suggested

**For slickensided surfaces the ratings given cover possible movement along the lineation. (For movements across lineation, a rough or very rough rating should be applied for the surface.)*

Source: Palmstrom, 2000.

TABLE 10.3 Characterization and Rating of the Joint Alteration Factor

Term	Description	jA
<i>A. Contact between rock wall surfaces</i>		
<i>Clean joints</i>		
Healed or welded joints	Softening, impermeable filling (quartz, epidote, etc.)	0.75
Fresh joint walls	No coating or filling on joint surface, except from staining (rust)	1
<i>Alteration of joint wall</i>		
i. 1 grade more altered	The joint surface exhibits one class higher alteration than the rock	2
ii. 2 grade more altered	The joint surface shows two classes higher alteration than the rock	4
<i>Coating or thin filling</i>		
Sand, silt, calcite, etc.	Coating of friction materials without clay	3
Clay, chlorite, talc, etc.	Coating of softening and cohesive minerals	4

B. Filled joints with partial or no contact between the joint wall surfaces

Type of filling material	Description	Partial wall contact (thin filling <5 mm*)	No wall contact (thick filling or gouge)
Sand, silt, calcite, etc. (non-softening)	Filling of friction material without clay	4	8
Compacted clay materials	"Hard" filling of softening and cohesive materials	6	6–10
Soft clay materials	Medium to low over-consolidation of filling	8	12
Swelling clay materials	Filling material exhibits clear swelling properties	8–12	13–20

*Based on joint thickness division in the RMR system (Bieniawski, 1973).

Source: Palmstrom, 1996, 2000.

Flow Chart for Estimating Rock Mass Index

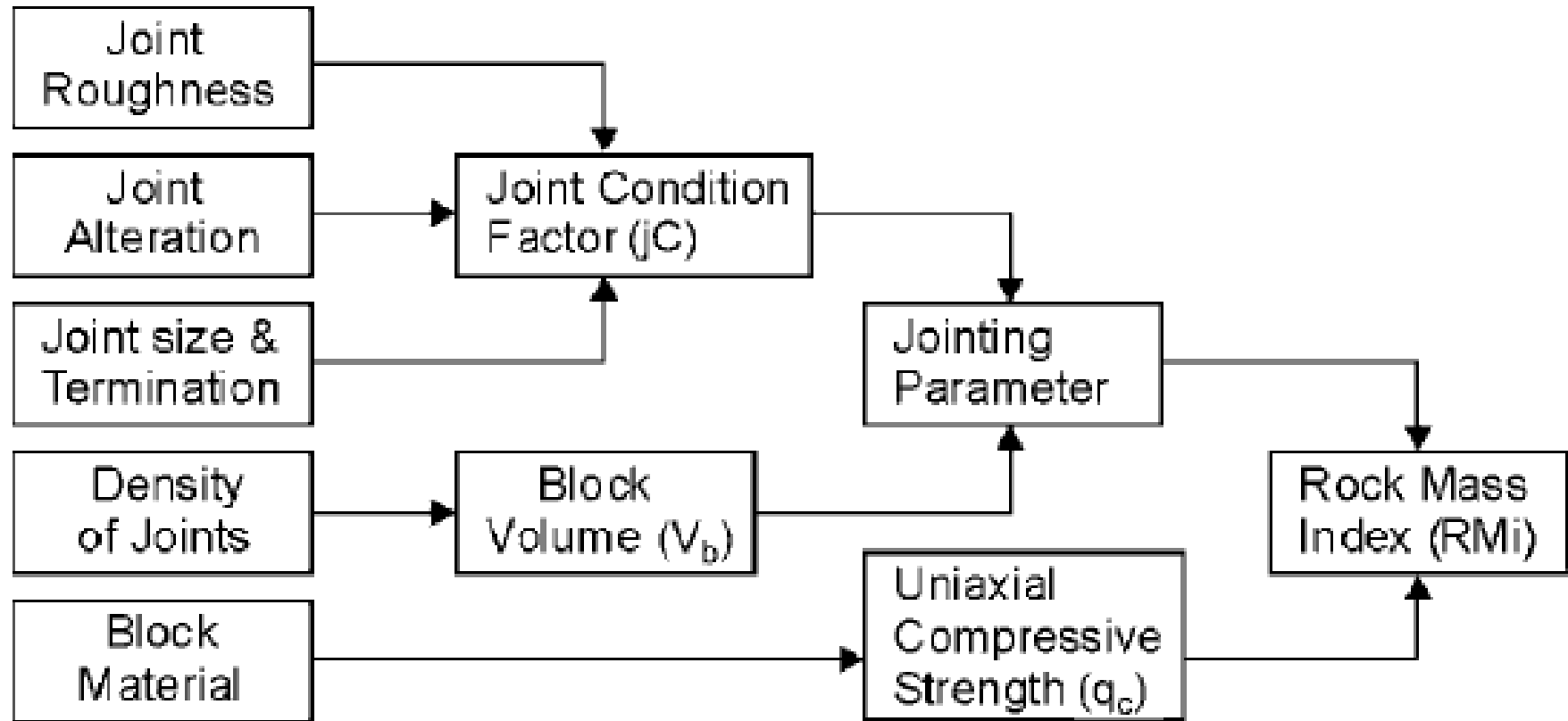


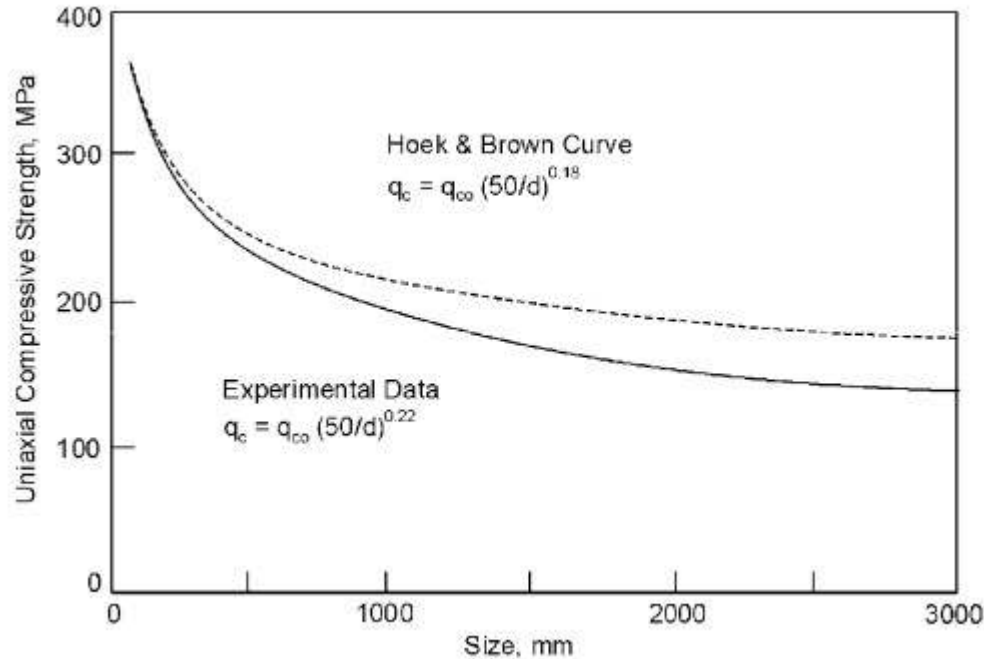
TABLE 10.4 Classification of R_{Mi}

Term		
<i>For R_{Mi}</i>	<i>Related to rock mass strength</i>	R _{Mi} value
Extremely low	Extremely weak	<0.001
Very low	Very weak	0.001–0.01
Low	Weak	0.01–0.1
Moderate	Medium	0.1–1.0
High	Strong	1.0–10.0
Very high	Very strong	10–100
Extremely high	Extremely strong	> 100

Source: Palmstrom, 1996.

Applications

SCALE EFFECT



Empirical equations for scale effect of uniaxial compressive strength. (From Barton, 1990, based on data from Hoek and Brown, 1980, and Wagner, 1987)

$$q_c = q_{co} (50/d)^{0.2} = q_{co} (0.05/Db)^{0.2} = q_{co} \cdot f$$

q_{co} is the UCS for a 50 mm sample size

$$f = (0.05/Db)^{0.2}$$

$$Db = (V_b)^{0.33}$$

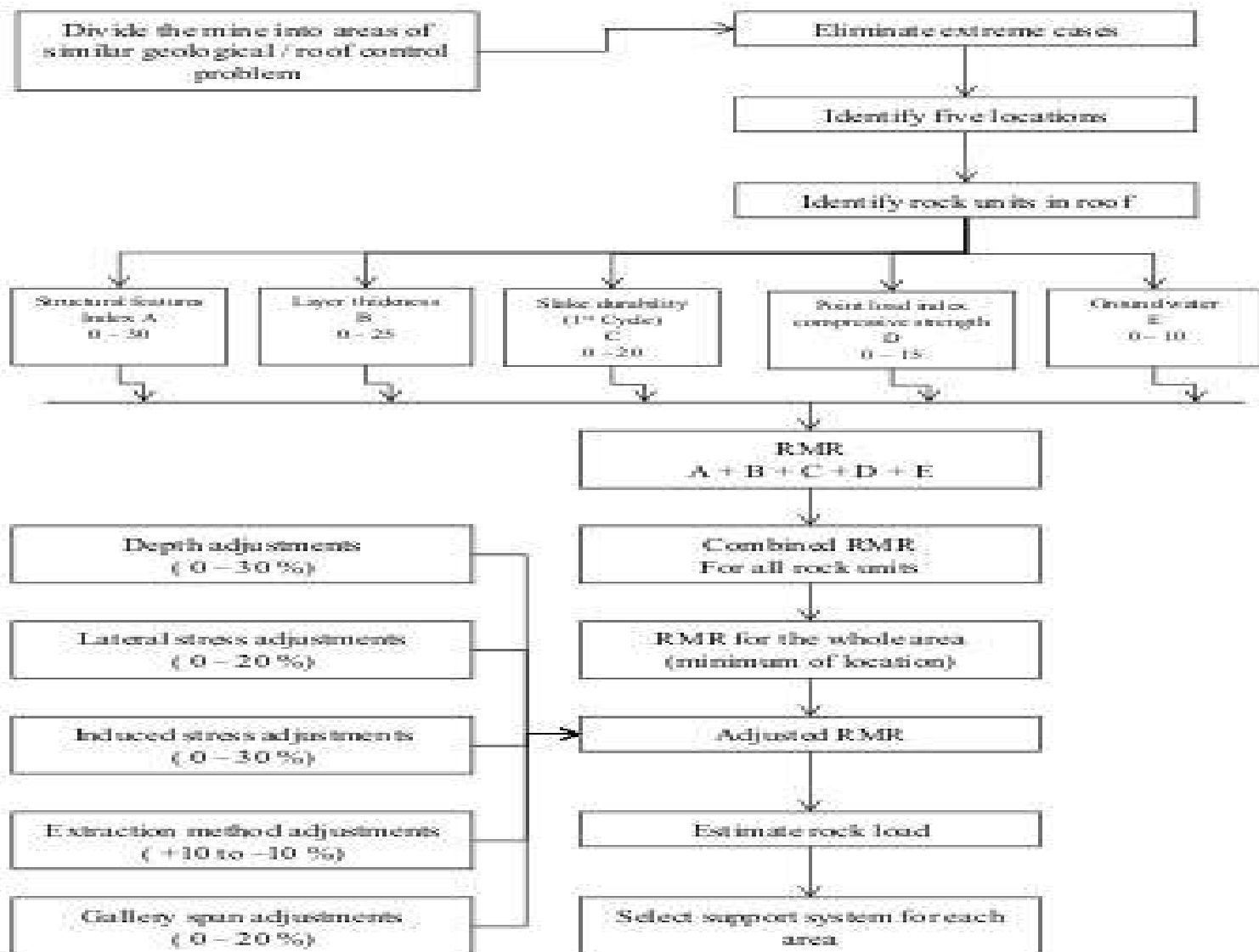
Applications

Cai and Kaiser (2006), using surface fitting techniques, suggested the following equation to calculate GSI from J_c and V_b :

$$GSI(V_b, J_c) = \frac{26.5 + 8.79 \ln J_c + 0.9 \ln V_b}{1 + 0.0151 \ln J_c - 0.0253 \ln V_b}$$

CMRI-ISM

Rock mass classification



1. Layering

- Spacing between the bedding planes or planes of discontinuities should be measured using borehole staratascope in a 2.0 m long drill hole made in the roof.
- Alternately, all bedding planes or fissile (weak) planes within the roof strata can be measured in any roof exposure like a roof fall area, shaft section or cross measure drift.
- Core drilling should be attempted wherever feasible, and the core log can be used to evaluate RQD and layer thickness.
- Average of five values should be taken and layer thickness should be expressed in cm.

Table 1: Structural Features

Presence of major faults		Rating
	net displacement > 10 m	15
	displacement 2 – 10 m	8
	displacement < 2m	5
Presence of minor faults/slips		
	spacing < 5 m	
	orientation unfavourable	10
	orientation not unfavourable	5
	spacing > 5 m	
	orientation unfavourable	7
	orientation not unfavourable	3

Table 1: Structural Features ...contd

Occurrence of joints and cleats			
		Orientation unfavourable	Orientation favourable
a) Minimum spacing <30 cm			
	single set	6	4
	two sets	7	6
	more than two sets	8	8
b) Minimum spacing > 30 cm			
	single set	5	2
	two sets	6	4
	more than two sets	6	6
Sedimentary features			
	lateral thickness variations	3	
	sandstone channels	6	
	kettlebottoms	4	
	plant impressions	3	
	ball coal	4	
Index for 'structural features' = sum of indices for individual features .			

2. Weatherability

- ISRM standard slake durability test should be conducted on fresh, samples of roof rock collected from the mine to determine the susceptibility of rocks to weathering failure on contact with water or the atmospheric moisture.
- Mean of three such first cycle values should be taken. Core may be broken to obtain the samples.

3. Rock strength

- Point load test is the standard index test for measuring the strength of rocks in the field.
- The load at failure (in kg) divided by the square of the distance between the platens (in cm) gives the point load index (I_s).
- The mean of the highest five values out of at least 10 sample tests should be taken.
- The compressive strength of the rocks can be obtained from the irregular lump point load index for Indian coal measure rocks by the relation: **$C_o = 14 \times I_s$ (in kg/cm^2)**

4. Ground water

- A 2 m long vertical hole should be drilled in the immediate roof and the water seeping through the hole after half an hour should be collected in a measuring cylinder.
 - The average of three values from three different holes should be taken and expressed in ml per minute.
-
- *The ratings for the five parameters are given in Table 2 (next slide).*
 - *Rock Mass Rating (RMR) is the sum of the five parameter ratings.*
 - *If there are more than one rock type in the roof,*
 - *RMR is evaluated separately for each rock type, and the combined RMR is obtained as:*
-
- Combined RMR = Σ (RMR of each bed X bed thickness) / Σ (thickness of each bed)***
- *The RMR so obtained may be adjusted, if necessary, to account for some special situations in the mine like great depth, stresses and method of work.*

Table 2: Classification ratings for parameters

Parameter	Range of values					
1. Layer thickness	(cm) Rating	<2.5 0-5	2.5-7.5 6-12	7.5 – 20 13-20	20- 50 21 -26	>50 27-30
2. Structural features	(index) Rating	> 14 0-4	14-11 5- 10	11-7 11-16	7-4 17-21	4-10 22-25
3. Weatherability	(%) Rating	<60 0-3	60-80 4-8	85 – 97 9-13	97-99 14- 17	>9Q 18-20
4. Strength of the rock	(kg/sq.cm) Rating	< 100 0- 1	100-300 3-6	300-600 7-10	600 – 900 11-13	>900 14- 15
5. Ground water seepage rate	(ml/m in) Rating	>2000 0- 1	2000-200 2-4	200-20 5-7	20-0 8-9	Dry 10
RMR		0-20	20 – 40	40-60	60-80	80-100
CLASS		V	IV	III	II	1
DESCRIPTION		VERY POOR	POOR	FAIR	GOOD	VERY GOOD

Table 3: Adjustment Factors for RMR

Adjustment to RMR	Adjusted RMR	
1. Depth		
Less than 250 m	Nil	RMR x 1.0
250-400m	10% reduction	RMR x 0.9
400- 600 m	20% reduction	RMR x 0.8
More than 600 m	30% reduction	RMR x 0.7
2. Lateral stress		
Assumed stress amount		
Small	10% reduction	RMR x 0.9
Moderate	20% reduction	RMR x 0.8
High	30% reduction	RMR x 0.7

Table 3: Adjustment Factors for RMR ... contd

Adjustment to RMR	Adjusted RMR	
3. Induced stresses		
Stress situation :		
No adjacent working in the same seam	Nil	RMRx1 .0
Extraction area within 20-40 m in the same seam	!0% reduction	RMR x 0.9
Extraction areas within 10-20 m in the same scam	Upto 30% reduction	RMR x (0.7 to 0.8)
Working above with 1 0-20 m parting	10% reduction	RMR x 0.9
Working above with 3 -10 m parting	Upto 30% reduction	RMR x {0.9 to 0.8)
4. Method of excavation		
Continuous miner	10% increase	RMR x 1.1
Undercut and blasting	Nil	RMR x 1.0
Blasting off -the-solid	10% reduction	RMR x 0.9
5. Gallery span		
Less than 4.5 m	Nil	RMR x 1.0
4.5-6.0 m	1 0 – 20% reduction	RMR x (0.8 to 0.9)

The final RMR for the roof is classified as

RMR	Roof Class	Rood Description
0-20	V	Very poor
20-40	IV	Poor
40-60	III	Fair
60-80	II	Good
80-100	I	Very good

An empirical relation obtained between RMR and rock loads is:

$$\text{Likely rock load in tonnes/m}^2 = \text{span in m} \times \text{mean rock density} \times (1.7 - 0.037 \text{ RMR} + 0.0002 \text{ RMR}^2)$$

In the USA, Unal related rock load to RMR by the following expression:

$$P = VB (100 - \text{RMR}) / 100$$

Where

P = rock load in t/m²

B = width of gallery in m

V = density of rock in t/m³

Pre-Mining Assessment of Roof Quality

- The Geo-mechanics classification can be applied even at the exploratory stage of a mine.
- Most of the parameters can be obtained by core drilling from the surface.
- The core pieces of the immediate roof of a seam can be measured to determine RQD and layer thickness. Dry density, strength and weatherability properties can be obtained by laboratory testing of the cores.
- Pumping tests above the seam level would indicate the groundwater character of the roof strata.
- Most of the structural features like jointing and major faults, and many sedimentary features could be identified by a set of two or more drill holes.
- From this information, a rough assessment of the support requirement could be made prior to mining. This should be updated after making exploratory drivages and while actual mining.
- In case there are other working mines in the area, information can be obtained and used to plan initially which may need to be updated depending on the experience.

Table 4: Recommended support system for development workings

Roof description (class)	Boiling systems	Conventional supports
1. Very Poor Roof (Class V)		(a) Yielding steel arches under high stress conditions, (b) Rigid steel arches under moderate stress conditions.
2. Poor Roof (Class IV)	(a) Full column grouted bolts using quick setting grout, supplemented with steel channel/w-strap. For flaky roof, wire mesh in addition. (b) For temporary roadways, roof stitching along with roof bolts and W-straps.	(a) Rigid steel arches for roof with structural disturbances, (b) Steel beam (200 mm x 100 mm) set on brick wall for permanent roadways (c) Steel bars set into pillars or on steel props.
3. Fair Roof (Class III)	(a) Full column grouted bolts using quick setting grout, (b) For temporary roadways, roof stitching supplemented by spot bolting in disturbed areas.	(a) Steel cross bars set into pillars or on steel props for permanent roadways. (b) Steel props for temporary roadways. Can also be used in permanent roadways.
4. Good Roof (Class II)	(a) Full column grouted bolts for permanent roadways, (b) Roof stitching with additional spot bolting in disturbed areas in permanent roadways.	Steel props for permanent roadways.
5. Very Good Roof (Class I)	No support except for spot bolting in disturbed areas.	

N.B.: 1. Grouted bolts also imply the use of cement capsules.

2. 'Quick setting' implies the development of an anchorage capacity of at least 10 kN in 30 minutes and about 50 kN in 2 hours.

Table 5: Recommended support system for freshly exposed roof (upto 10 m from the face)

Roof description (class)	Bolting systems	Conventional supports
1. Very Poor Roof (Class V}		The system of support indicated in Table 4 should be erected closely following the face,
2. Poor Roof (Class IV)	(a) Point anchored (mechanical or grouted with quick setting resin) bolts for immediate tension, to be fully grouted soon afterwards, W- straps where necessary.(b) Full column grouted bolts with quick setting grout.	(a) Cross bars on steel chocks or hydraulic props with minimum 5 tonnes setting load.
3. Fair Roof (Class III)	(a) Combination bolts as at (a) above.(b) Full column grouted bolts with quick setting grout. (c) Recoverable, bolts.	(a) Light duty hydraulic props.(b) Steel props. (c) Triangular chocks at junctions. (d) Safari supports.
4. Good Roof (Class II)	(a) Recoverable bolts.(b) Full column grouted -bolts	(a) Steel props. (b) Safari supports.
5. Very Good Roof (Class 1)	No support except for spot bolting in disturbed areas.	

N.B.:

1. Grouted bolts also imply the use of cement capsules,
2. 'Quick setting' implies the development of an anchorage capacity of at least 10 kN in 30 minutes and about 50 kN in 2 hours.
3. Vertical supports should be set with minimum 5 tonnes setting load to prevent dislodgement due to blasting.
4. 'Very poor' and 'poor' roof should be supported as soon as it has been exposed

Load bearing capacity of common support systems being used in Indian coal mines

Sl. No.	Support type	Load bearing capacity (tonnes)
1	Roof bolt (full column grouted with quick setting cement capsules) (Thermo mechanically treated, TMT, ribbed bolt of 22 mm diameter)	6
2.	Roof bolt (fully grouted with resin capsule) (TMT ribbed bolt of 22 mm diameter)	12
3.	Roof stitching	8
4.	Rope dowel	4
5.	Wooden prop	10
6.	Steel prop	30
7.	Steel chock	30
8.	Wooden chock	20
9.	Pit prop (2.5 m height) and (4.5 m height)	15 & 12
10.	Roof/rope truss	10
11.	Brick walling (40 cm thick)	10
12.	Rigid arches (vertical load) and (side loads)	7 (/m length) & 2(/m length)

To estimate the support load density by different support system used in mine, the equation for applied support load density,

$$ASL = \frac{n_b \cdot C_b + m_s \cdot Q_s}{W \cdot a_s}, \text{tonnes/m}^2$$

where;

n_b = number of bolts in a row

C_b = anchorage strength of each bolt (tonnes)

Q_s = load bearing capacity of additional supports if done (tonnes)

m_s = number of additional support at a spacing of “ a ” if it has been used

W = width of split or slice (m)

a_s = spacing between two consecutive rows (m)

If supports used in the mine are combination of rock bolts, cogs and pit props, then the applied support load density can be estimated from the modified equation given as under

$$ASL = \frac{N_b.C_b}{W.S_b} + \frac{N_c.C_c}{W.S_c} + \frac{N_p.C_p}{W.S_p}, \text{tonnes/m}^2$$

N_b = number of bolts in a row

C_b = load bearing capacity of rock bolt (tonnes)

N_c = number of cogs in a row

C_c = load bearing capacity of cogs (tonnes)

N_p = number of props or pit props or timber props in a row

C_p = load bearing capacity of pit prop or timber prop (tonnes)

W = slice/split width (m)

S_b, S_c and S_p = spacing between two consecutive rows of bolts, cogs and pit props respectively, expressed in metre